climate risk in vulnerable mountain regions of the world

Day 2 (28 June)

Jakob Steiner

Course content

27 June (1400 – 1800)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

28 June (1000 – 1800)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

Science to policy

29 June (1000 – 1800)

Serious game – Course wrap up

Course content

28 June (1000 – 1800)

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Mountain hazards – fast onset

Landslides

Debris flows

(Snow/Ice/Rock Ice-)Avalanches

(Glacial/Landslide dammed) Lake Outburst Floods (GLOFs/LLOFs)

Rockfall

Blizzards

Pluvial floods/intense rainfall

COVID

Mountain hazards – slow onset

(Snow) droughts

Snow to Rain transition

Permafrost creep

Subsidence

Road construction/infrastructure development

Hydropower

Urbanisation

Tourism

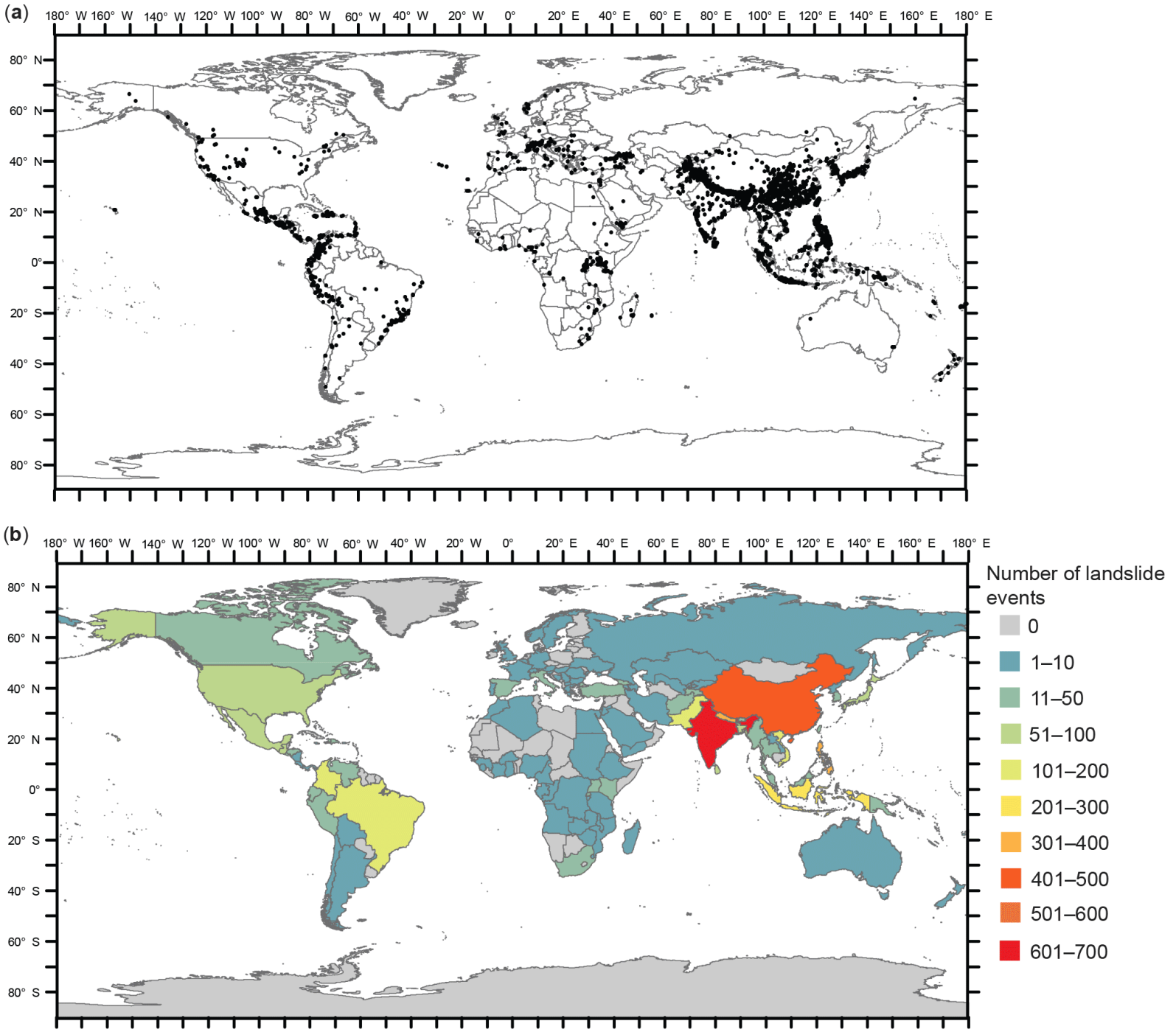
Landslides

2004 – 2016 a total of 4862 (non seismic-)landslides killed 55997 people

Rainfall a dominant driver but no clear signal of climate change

Indication that man made landslides are increasing

Froude and Petley (2018)



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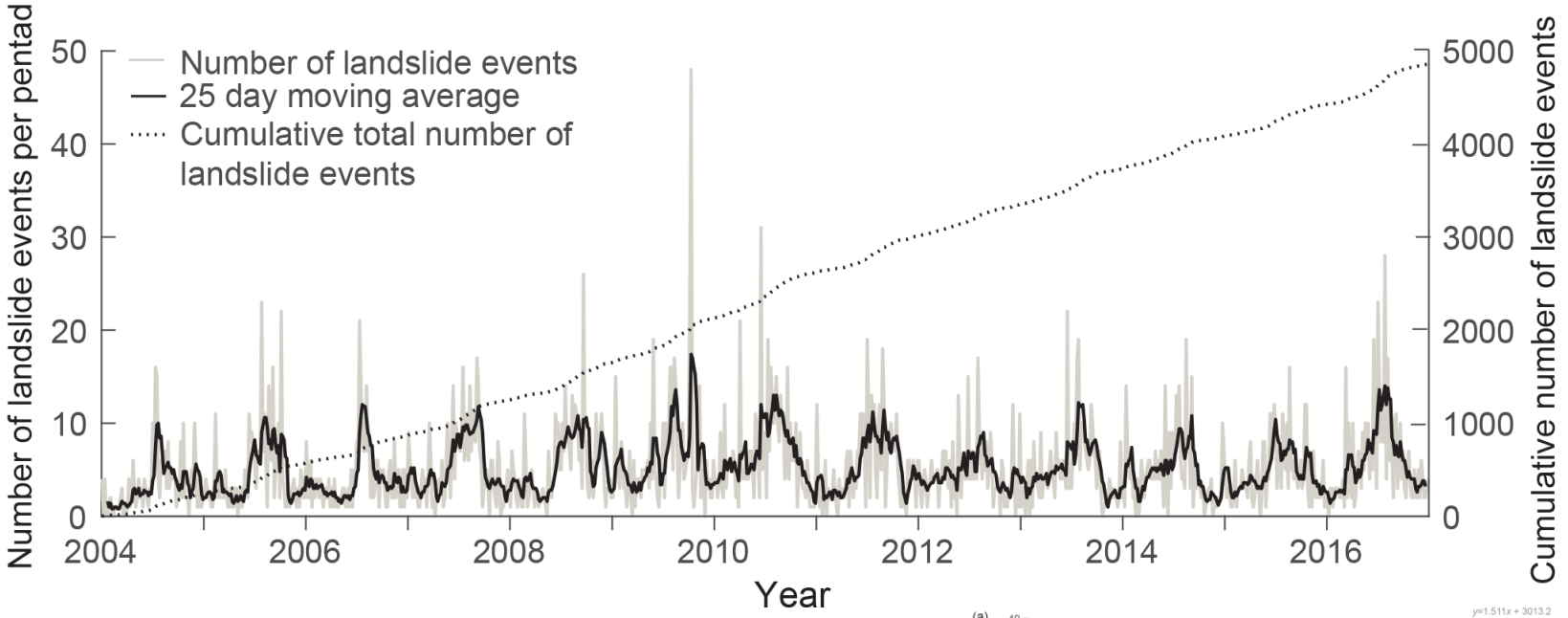

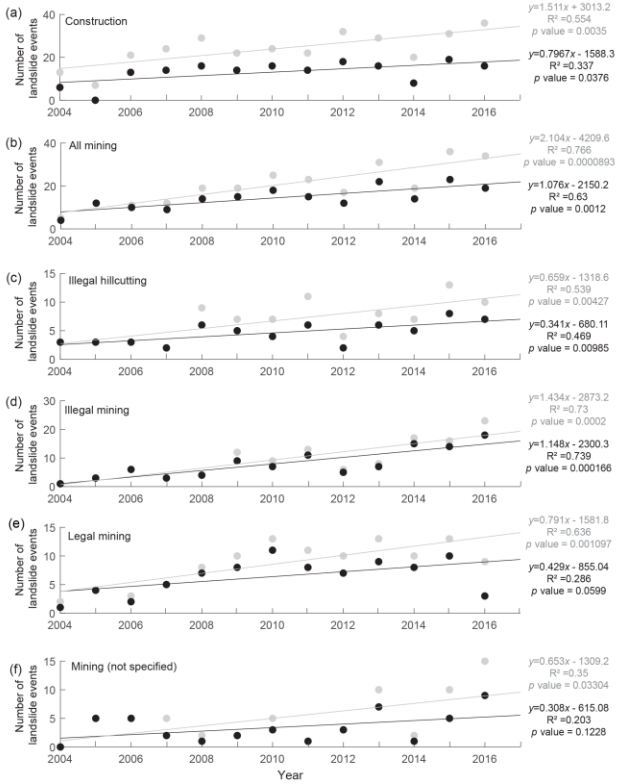

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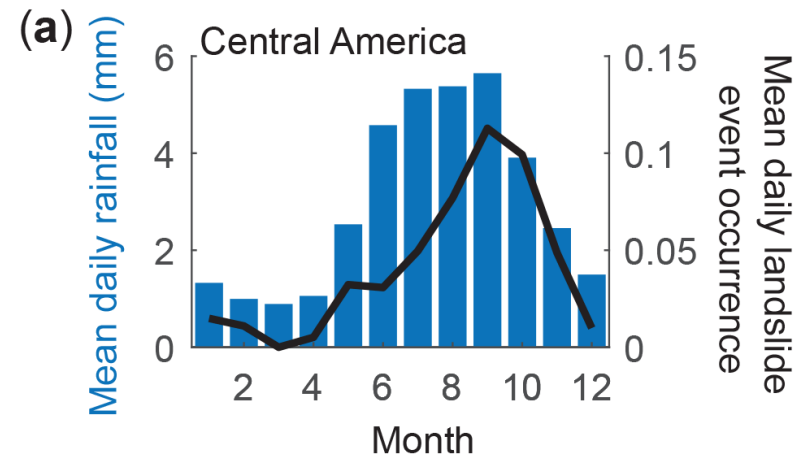
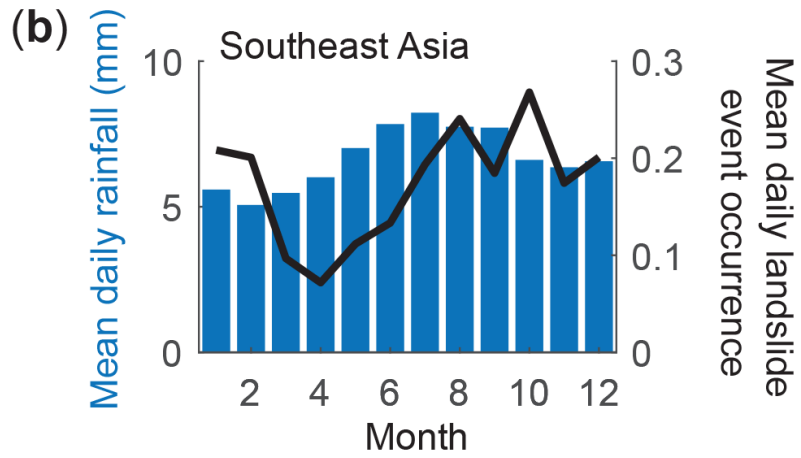
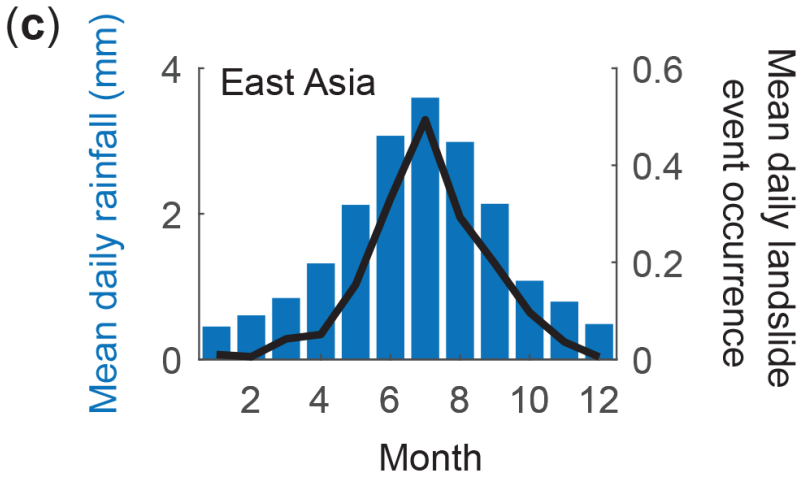
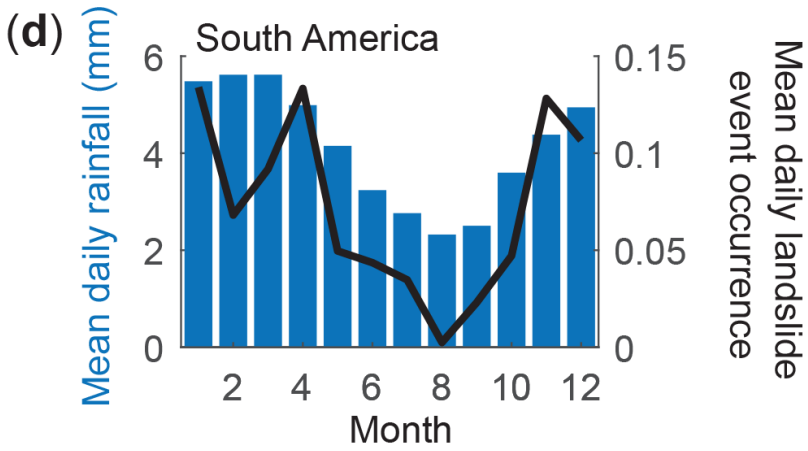
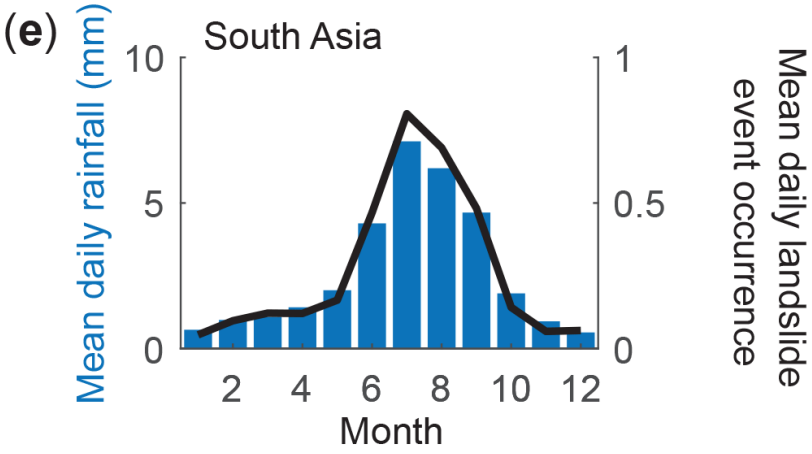


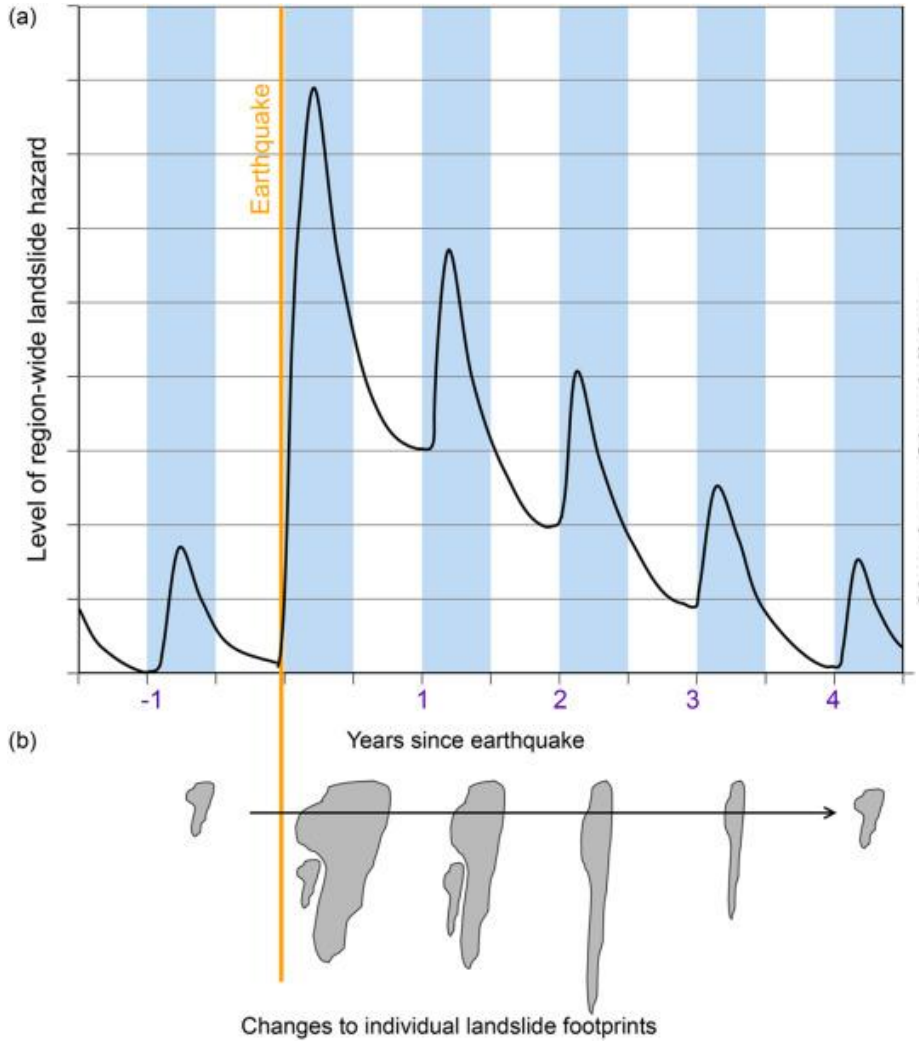


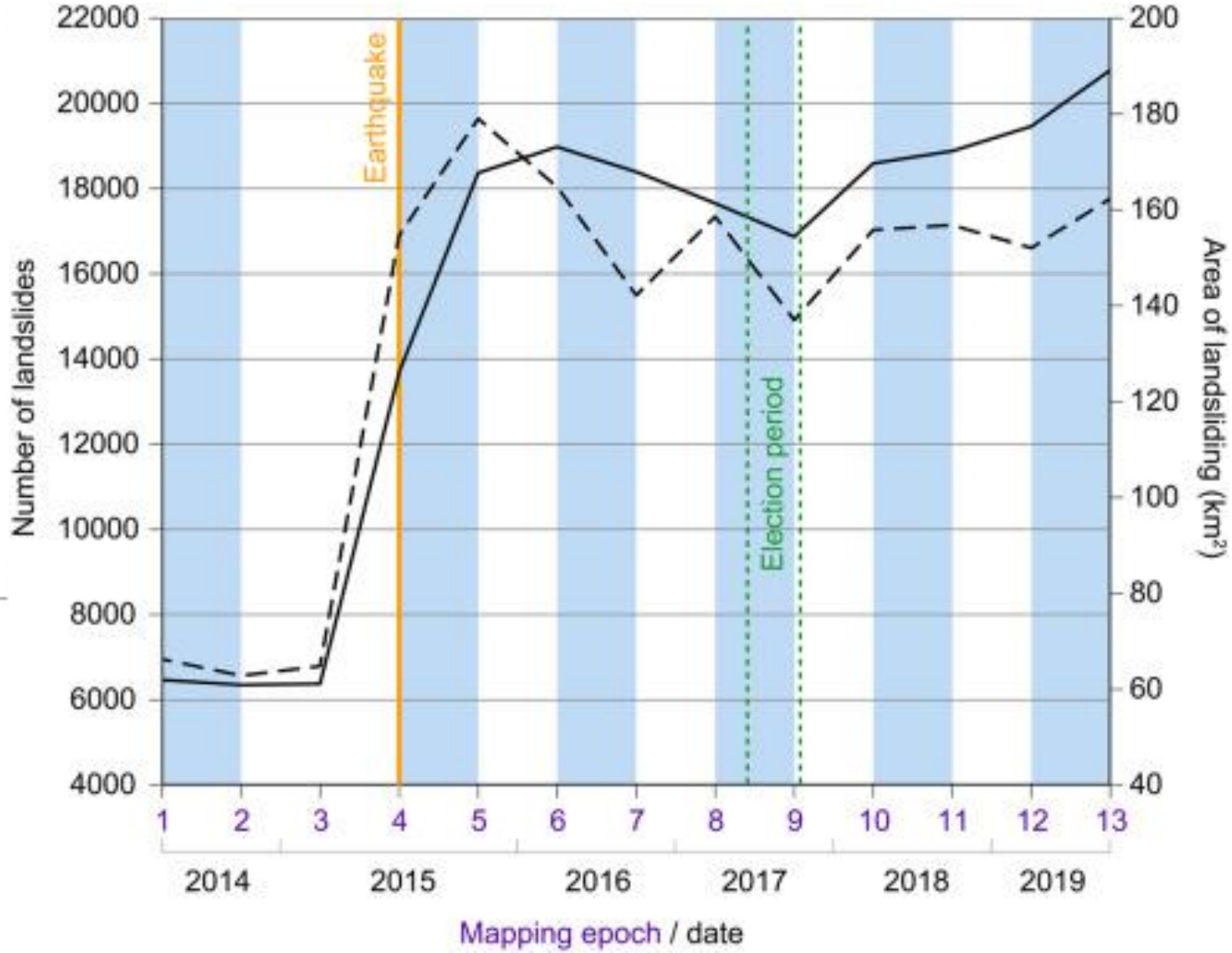






Landslides – expectation vs. reality



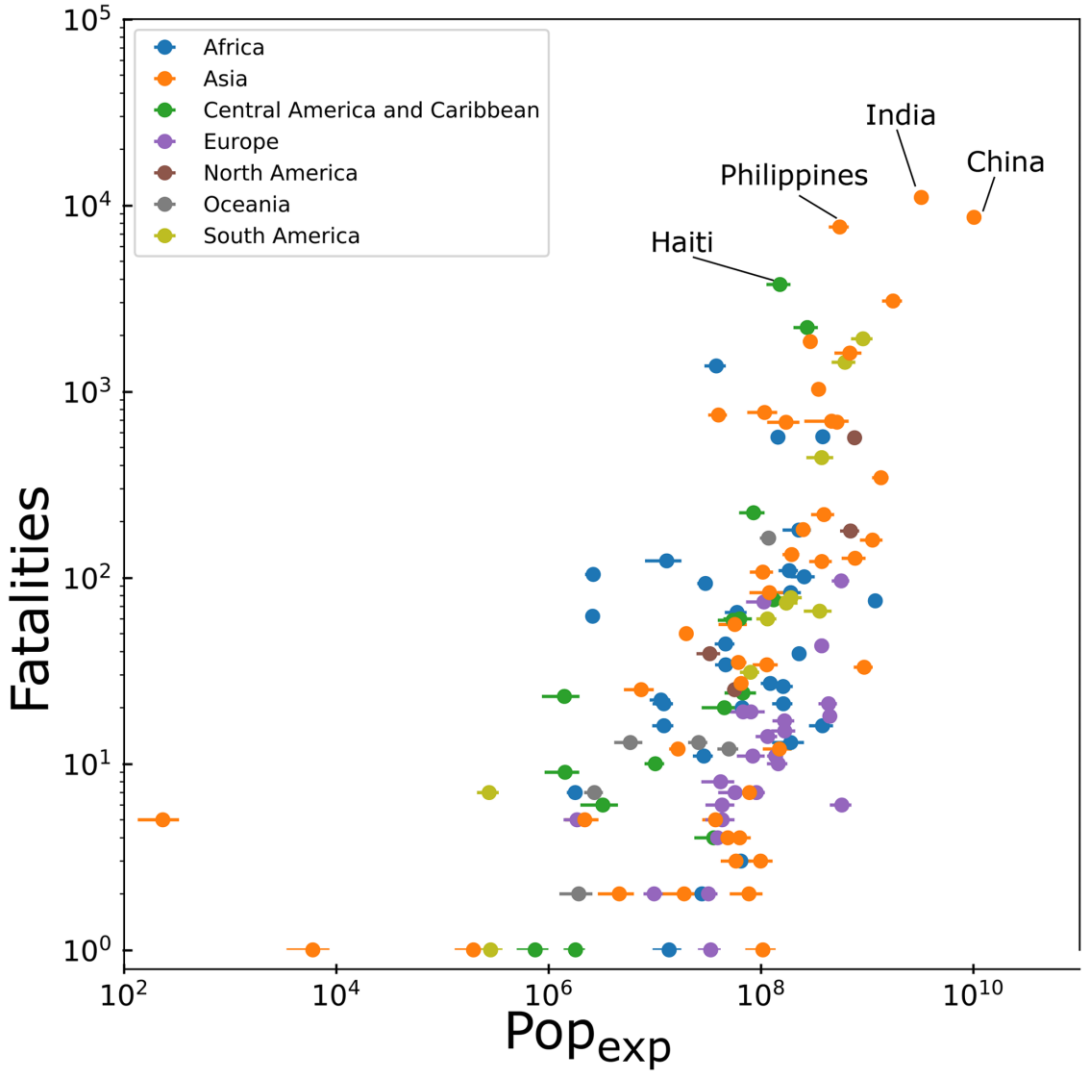


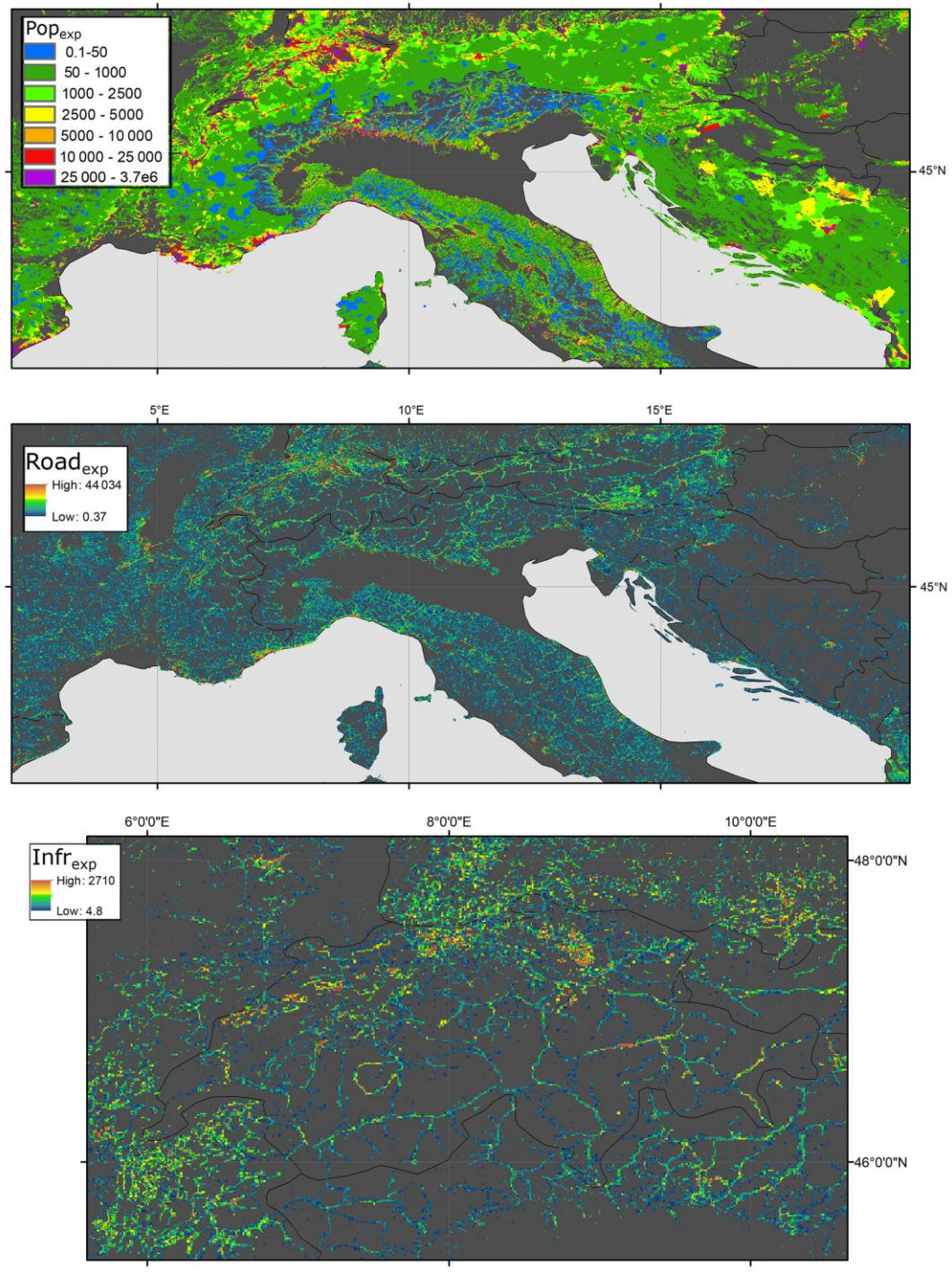
Rosser et al. (2019)

Landslides

Emmer, Steiner and Kalvoda (2024)

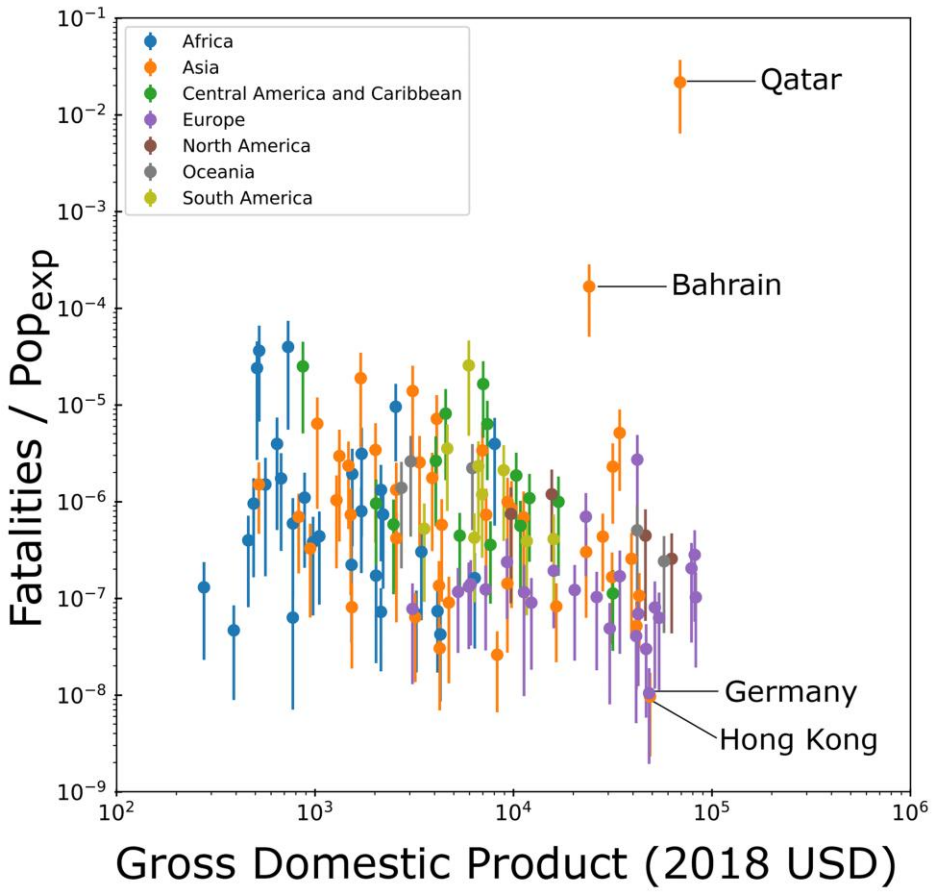
Landslide exposure

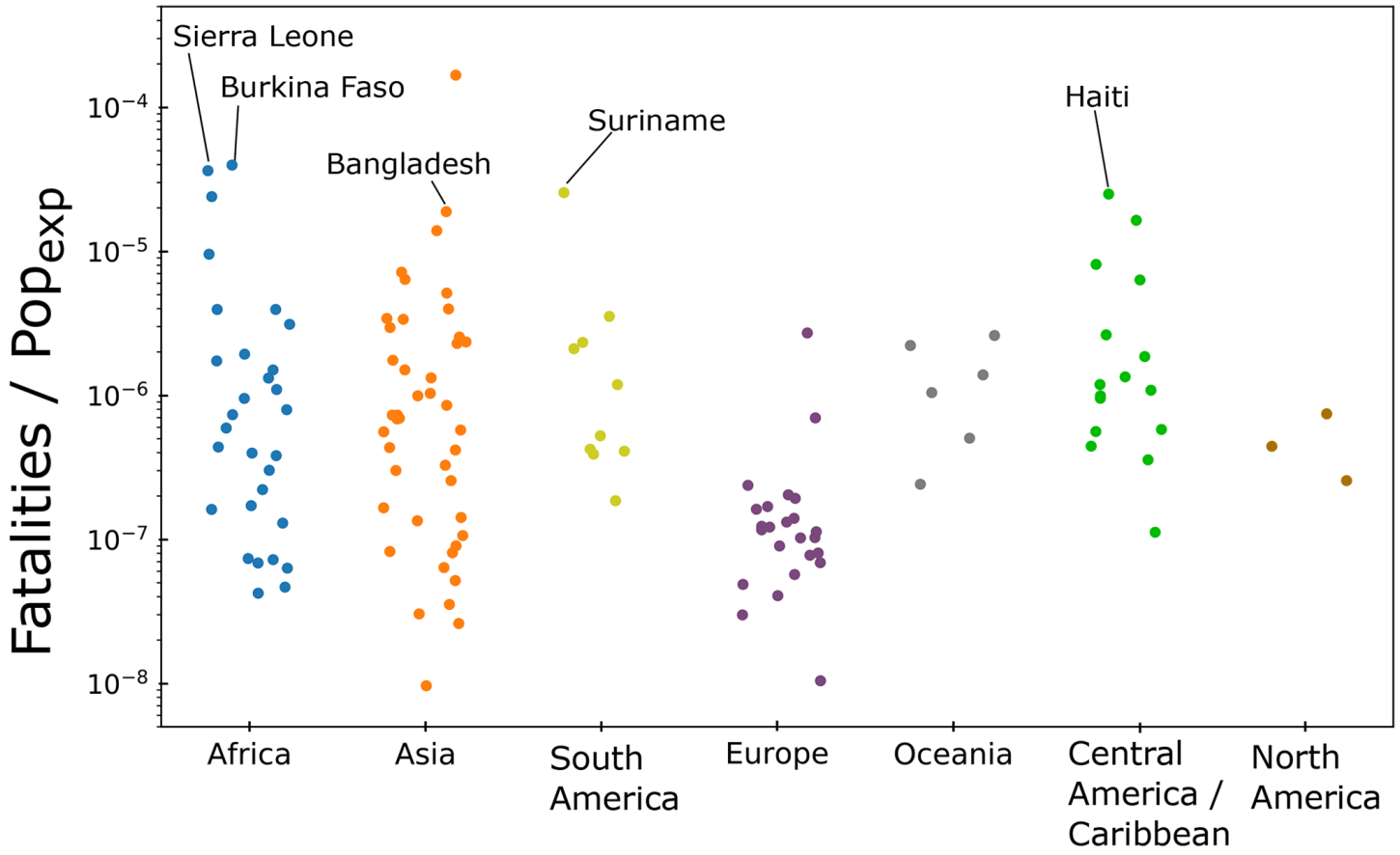




Emberson et al. (2020)

Landslide exposure





Emberson et al. (2020)

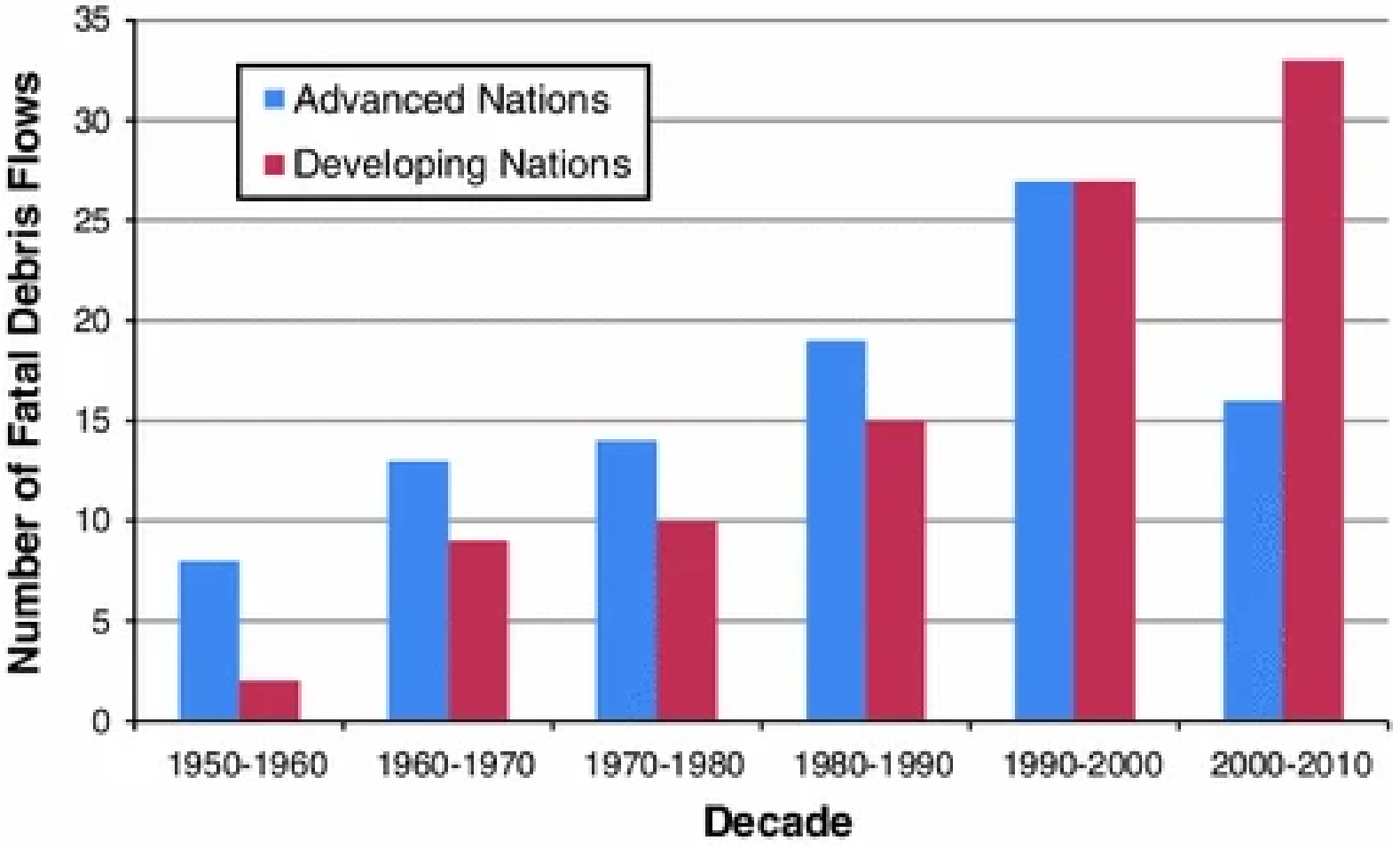
Debris flows

Debris flows

Debris flows

Between 1950 and 2011, 213 events caused 77 779 fatalities (365 event^{-1}, ~1300 yr^{-1})

Lacking comprehensive compilation

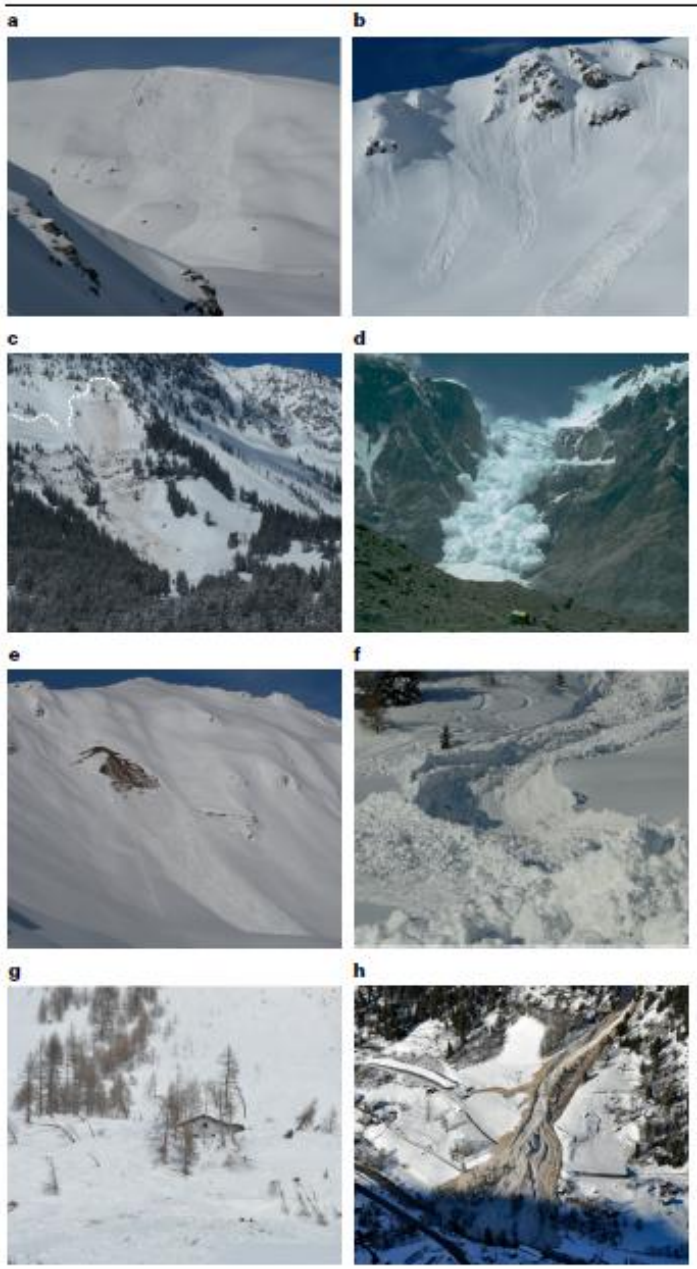


Dowling and Santi (2014)

Avalanches

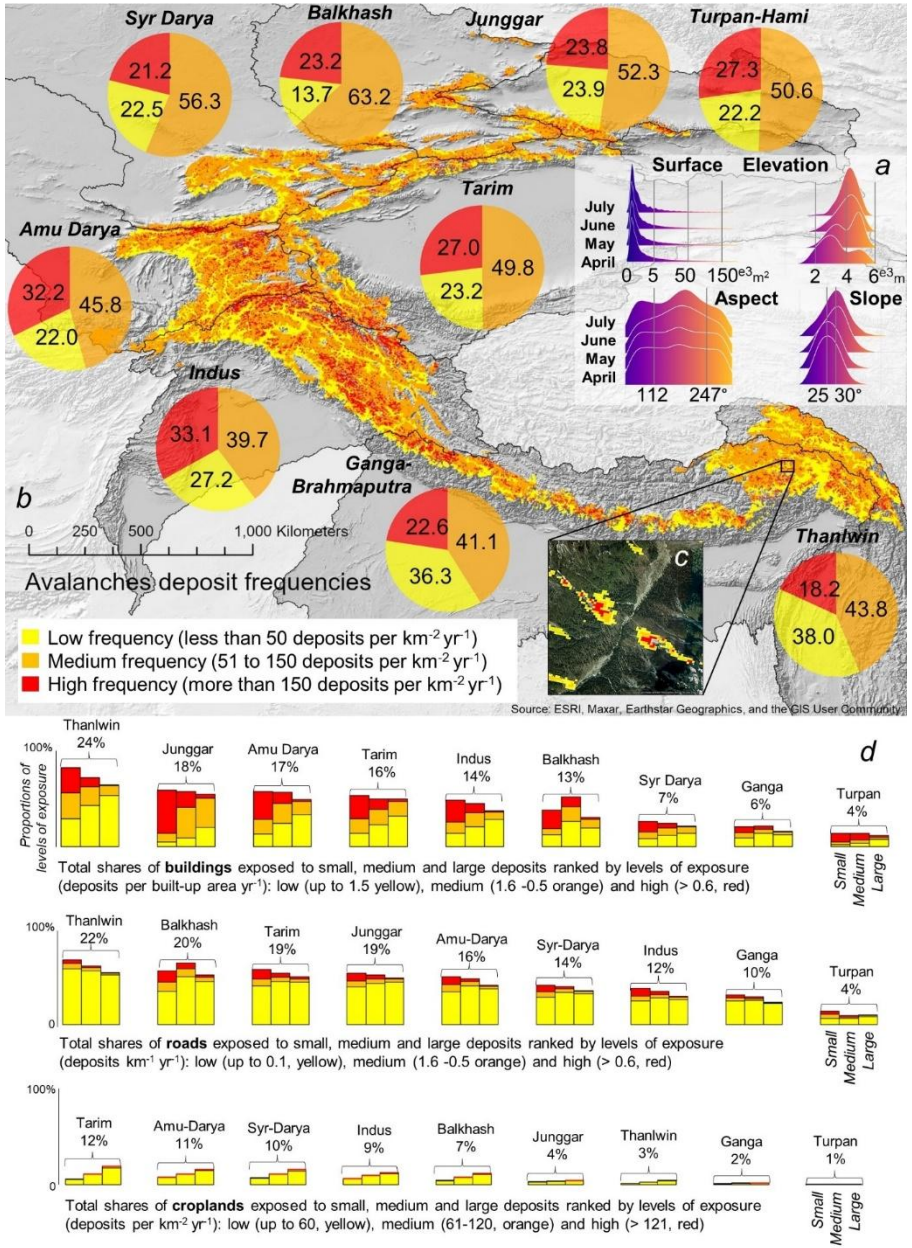
No global assessment but ca. 100 fatalities in Europe (mainly while doing sports) and 100 fatalities in High Mountain Asia (predominately non-sports related)

Climate change change signal but hard to trace



Eckert et al. (2024)

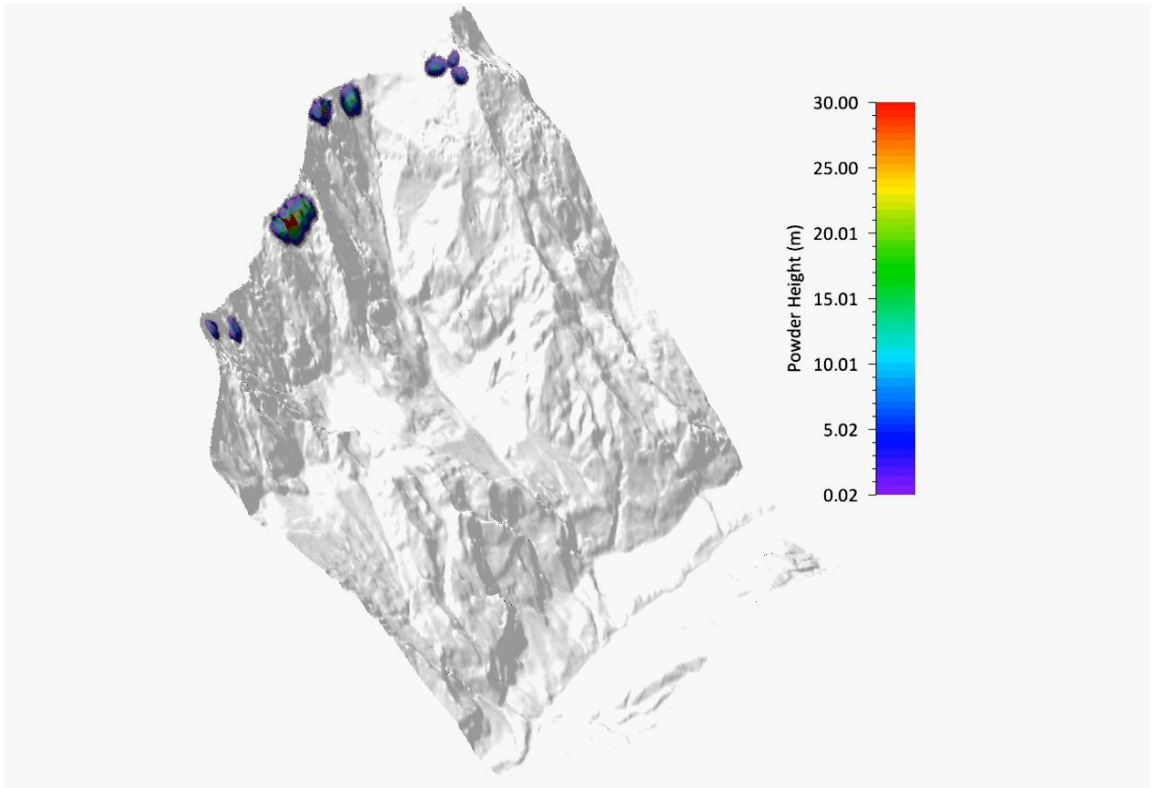
Acharya et al. (2023)



Caiserman et al. (in review)

Indigenous risk management

Acharya et al. (2023)

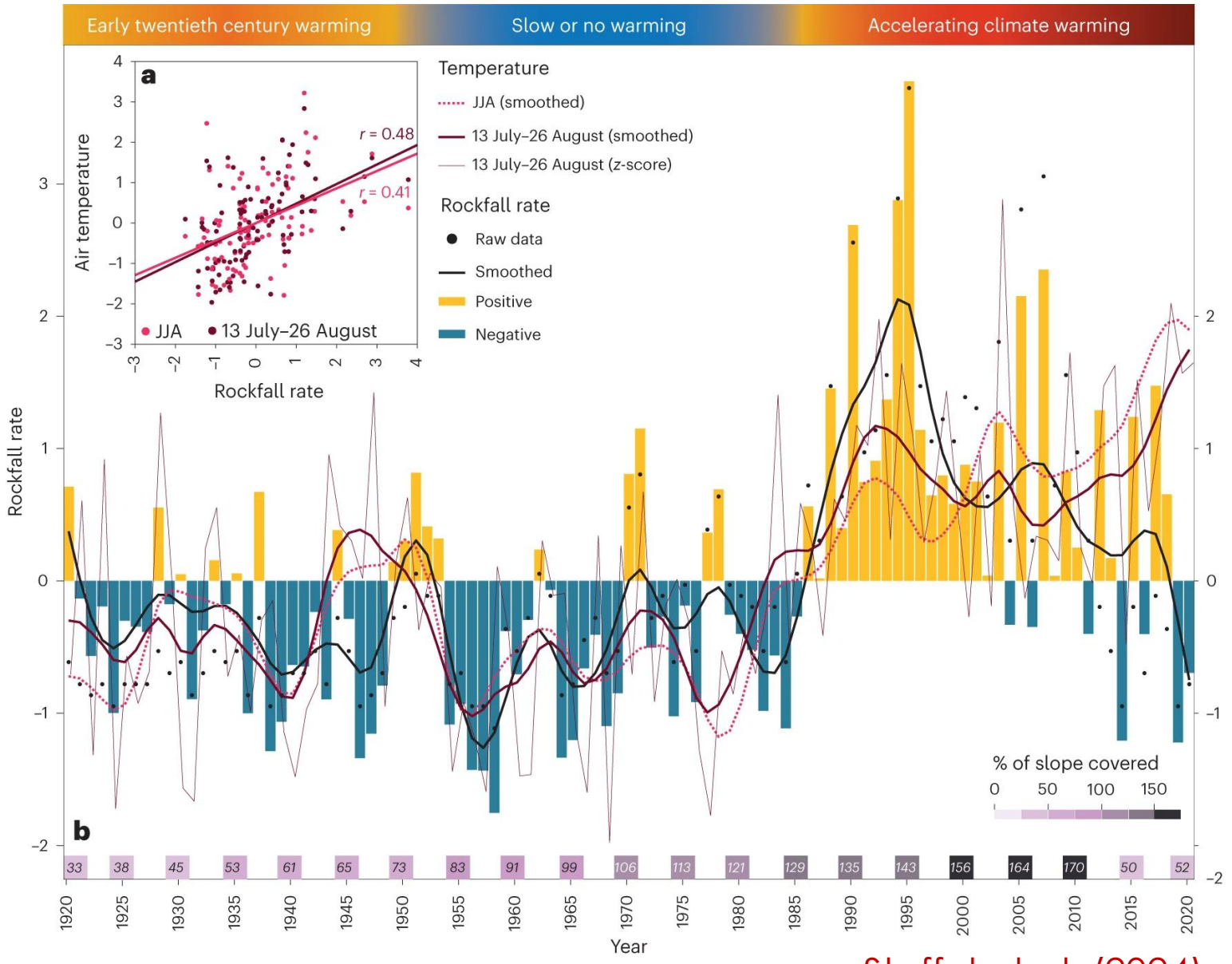


Zhuang et al. (2025)

Rockfalls

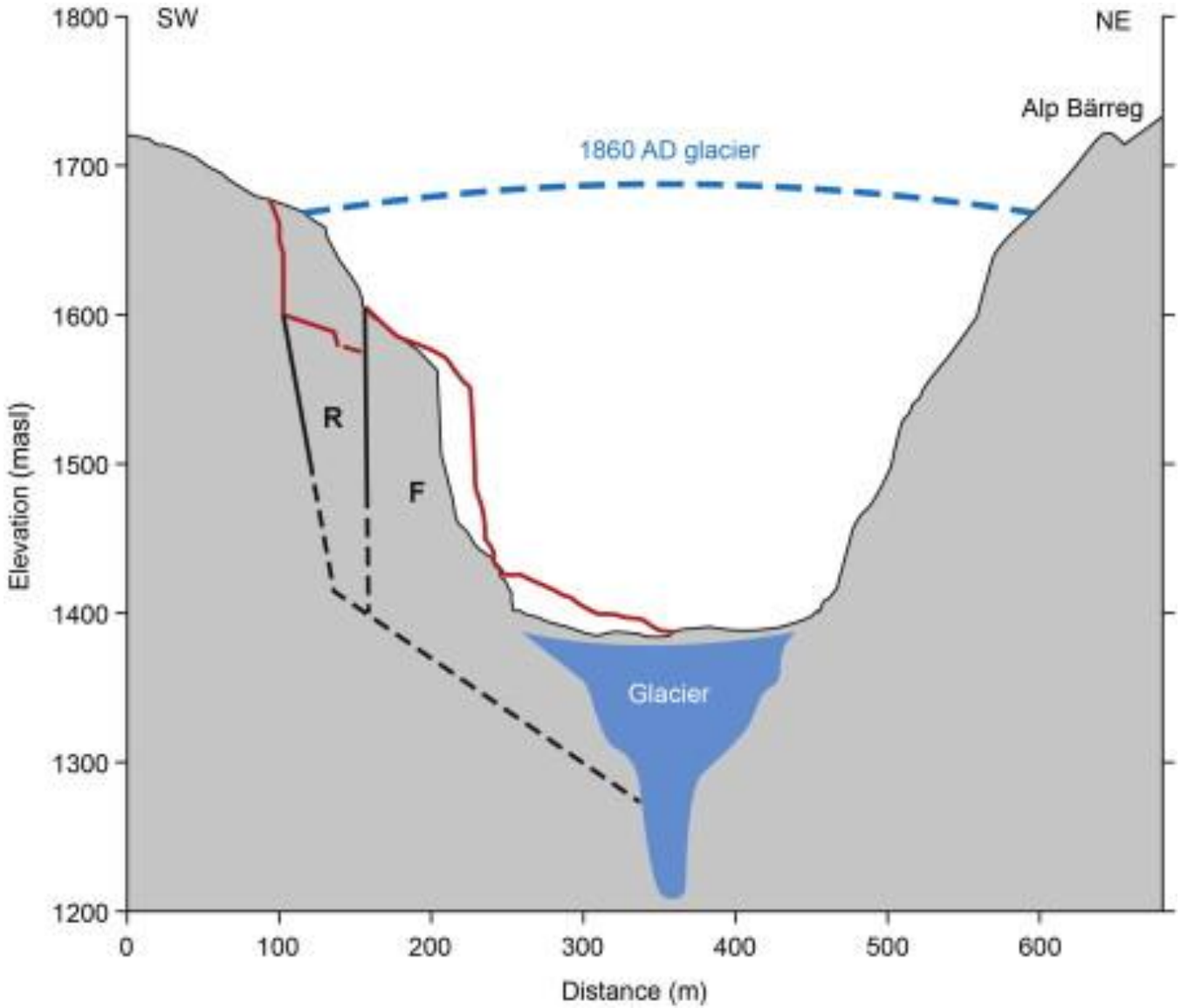
Research primarily in Switzerland and Germany

Clear indication that mountains are crumbling as temperatures rise



Stoffel et al. (2024)

Rockfalls



Deline et al. (2015)

Glacial lake outburst floods

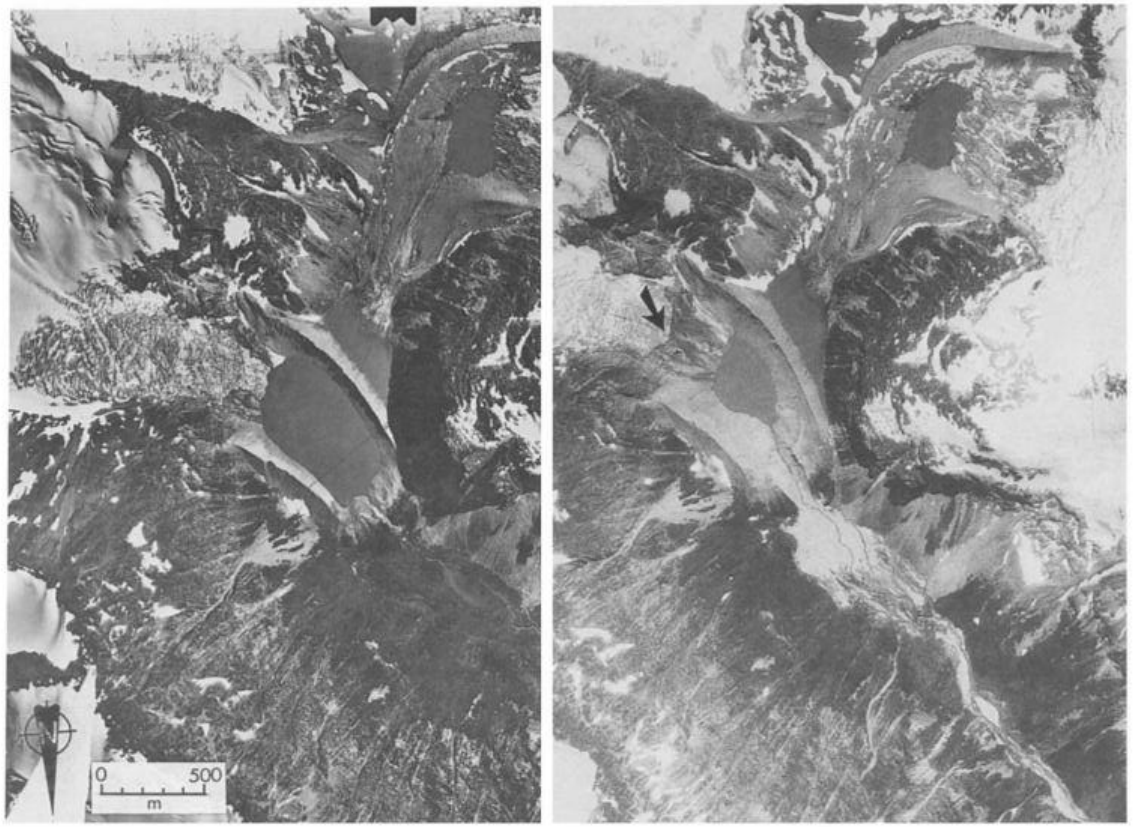
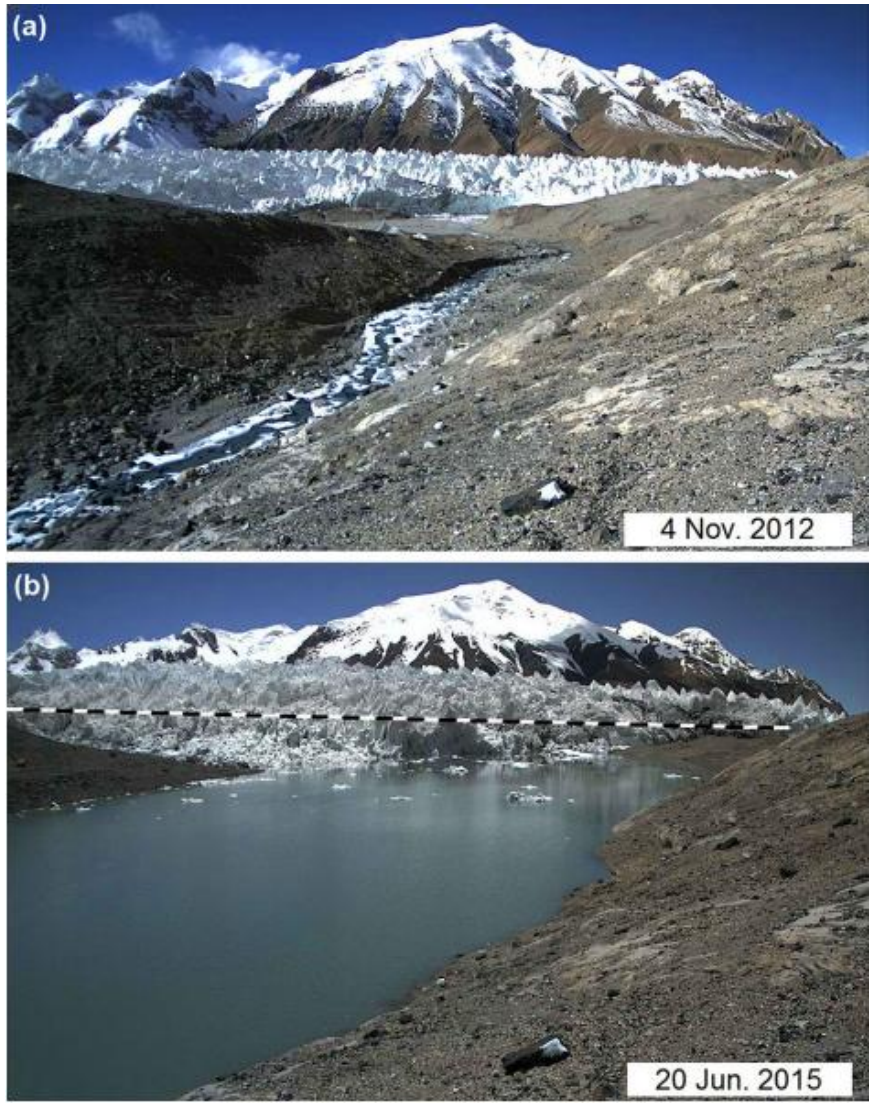
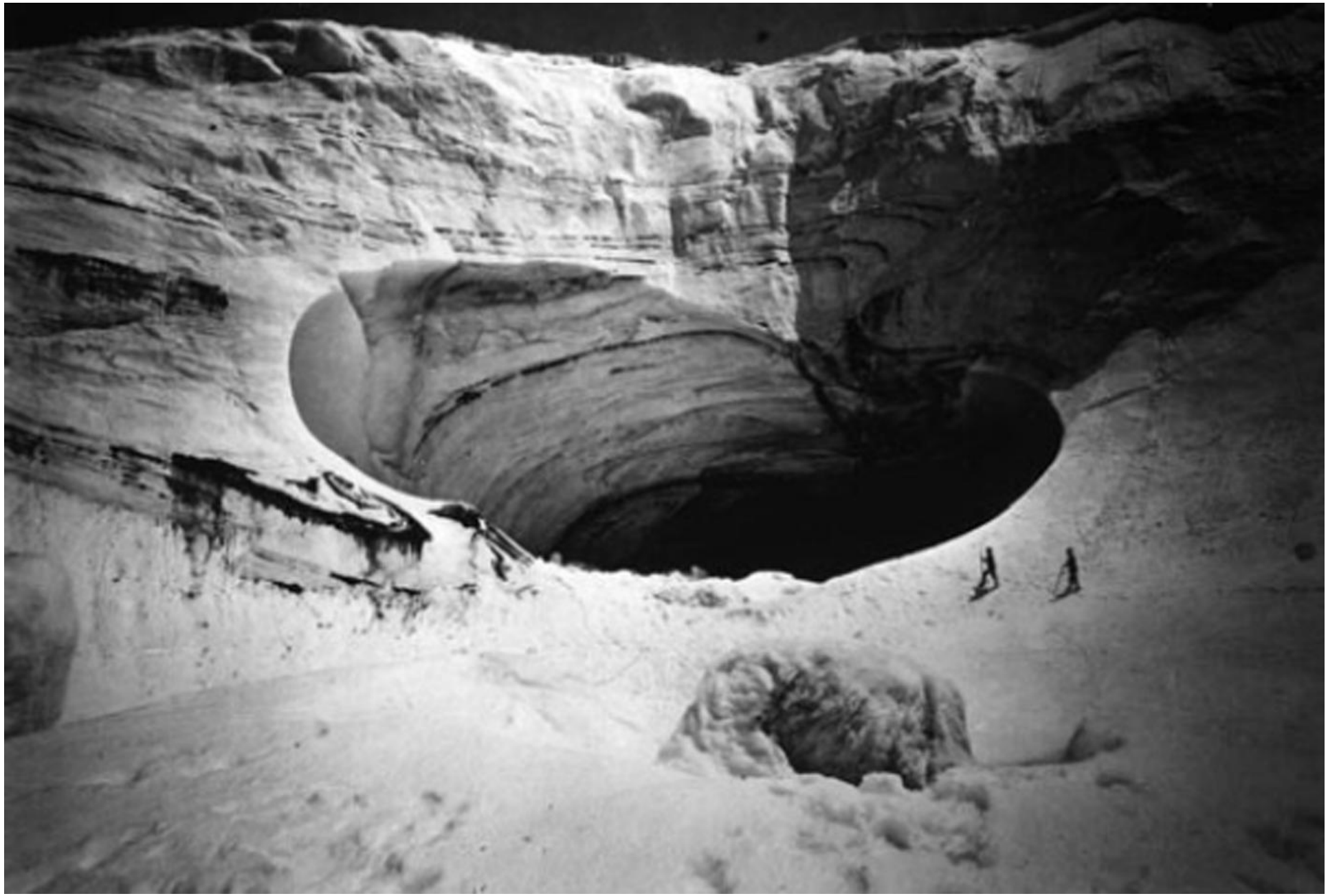


Fig. 12. Aerial photographs of Nostetuko Lake before (left, BC79069-190) and after (right, BC86048-147) the outburst flood of July 19, 1983. Failure occurred when toe of Cumberland Glacier (arrow) collapsed into the lake and generated waves that overtopped the moraine.

British Columbia
Canada



Kyagar Lake, Xinjiang, China, Geopraevent



Tête Rousse (Mont Blanc Massif) in 1892; Vincent et al., 2017

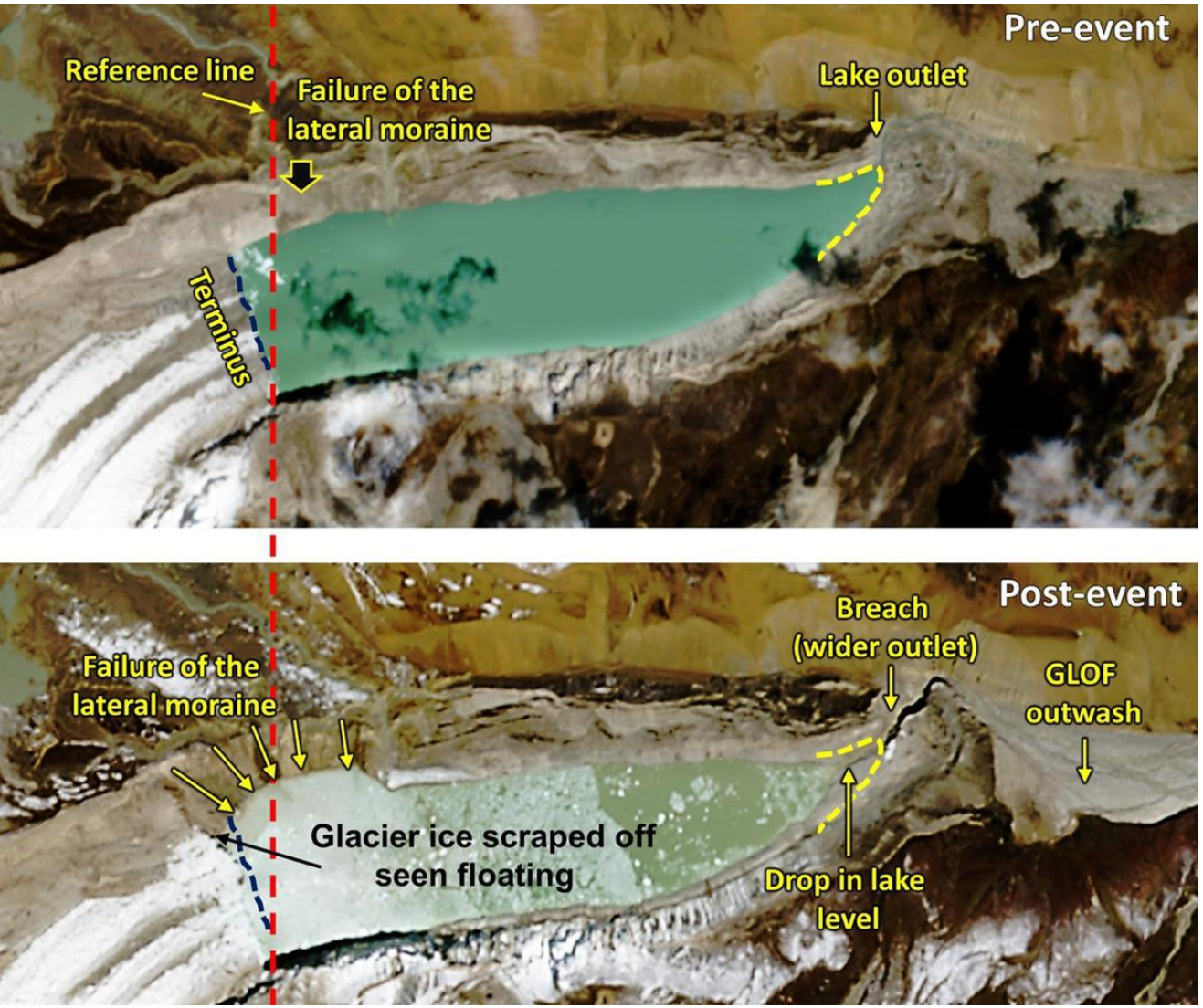


Figure 11: Schematic sketch showing a typical glacial lake outburst chain resulting from an initial mass movement (from, Worni et al., 2014). (1) A mass movement (ice, rock or debris) enters a lake, producing (2) a displacement wave that (3) overtops and (4) incises and erodes the dam area. (5) A flood then travels downstream where (6) populated areas and infrastructure are exposed. Note that displacement waves can be catastrophic with or without erosion of the dam area, and as such, can threaten also apparently stable bedrock dammed lakes.

Allen et al., 2017, GAPHAZ

Glacial lake outburst floods

~ 1 to 10 times yr^{-1} with overall not more than 20 000 people killed (ca 11000 in two events alone that were caused by many other processes)

No indication of an increase with climate change (for some even a decrease due to glaciers disappearing)

Historical a lot of attention due to the massive nature of these events ('biblical floods')

Many GLOFs misreported due to them getting more attention than other hazards → risk of mal-adaptation.

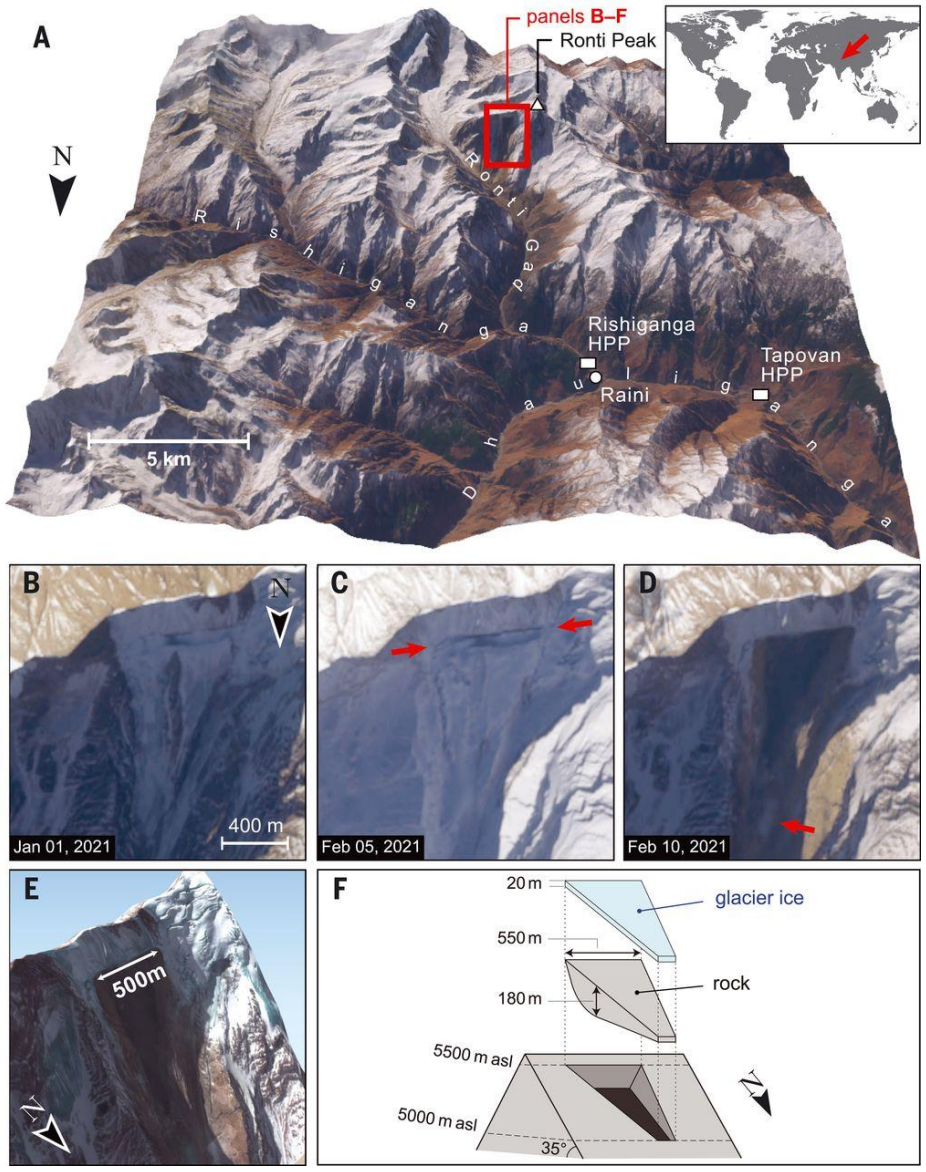
The new avalanches - multihazards

When ice, snow and rock mix, a deadly mix is created.

Increasing number of high impact events – not necessarily because more events happen, but also because more infrastructure at elevation is exposed!

Power dynamics depending on who the victims are

How to accurately report?



Shugar et al. (2022)

Uttarakhand
Chamoli
Glacier Burst
Flash Flood
& Devastation

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Mountain hazards – slow onset

(Snow) droughts

Snow to Rain transition

Permafrost creep

Subsidence

Urbanisation

Tourism

Mountain (snow) droughts

Drought is a ‘deficiency in water’, i.e. demands exceeds supply over a longer period of time

Meteorological drought: less than average precipitation over a prolonged period

Hydrological drought: water reserves fall short (lakes, rivers, behind dams)

Agricultural drought: stressed crops due to lack of moisture

Snow drought: well below average snow pack leading to lack of snow melt contribution to hydrology

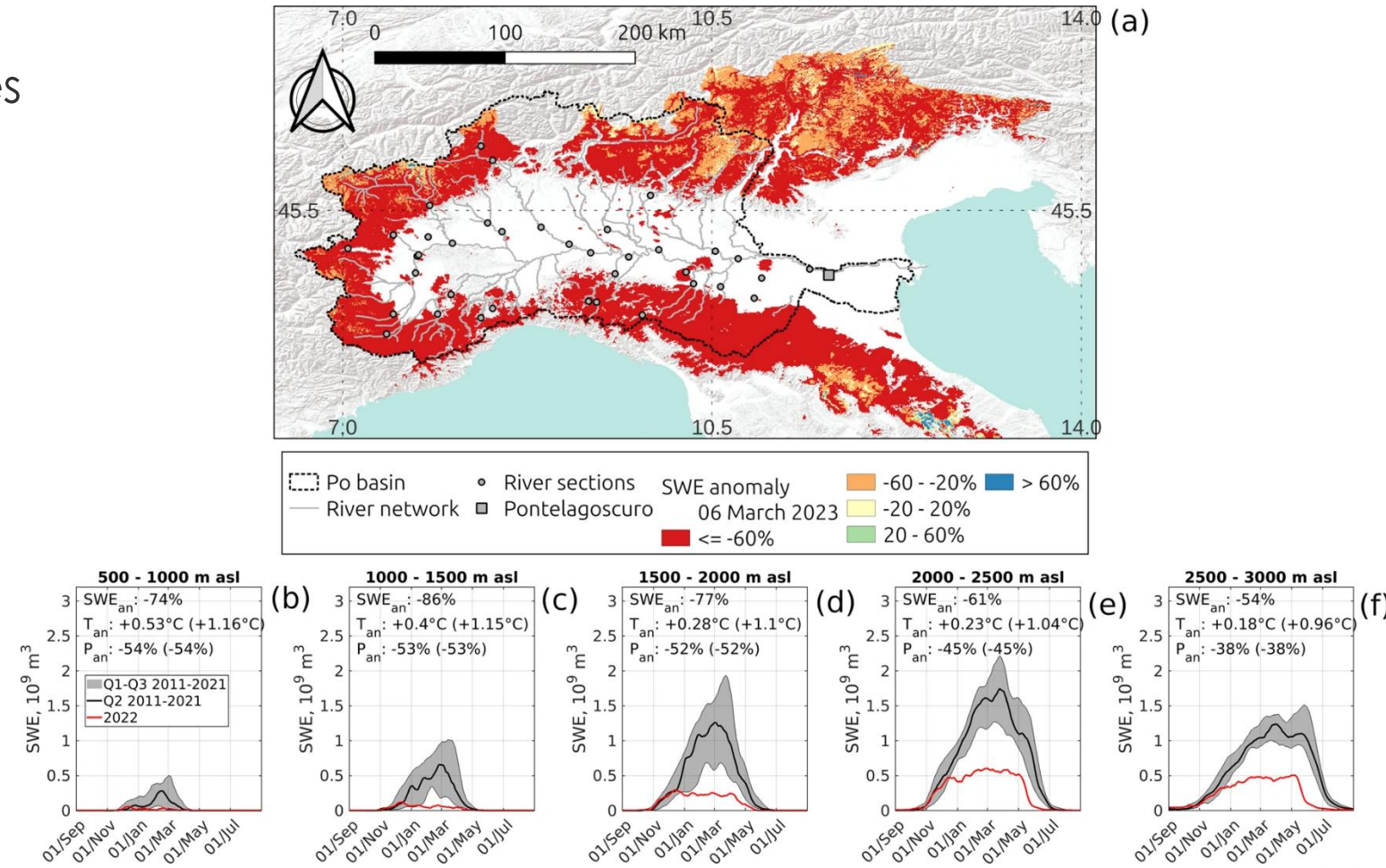
Groundwater drought: lack of groundwater supply

Mountain (snow) droughts

Snow drought happens when snow pack in winter is so small that spring/summer streamflow becomes inefficient

Example for a hazard that propagates downstream

Allows to anticipate months in advance



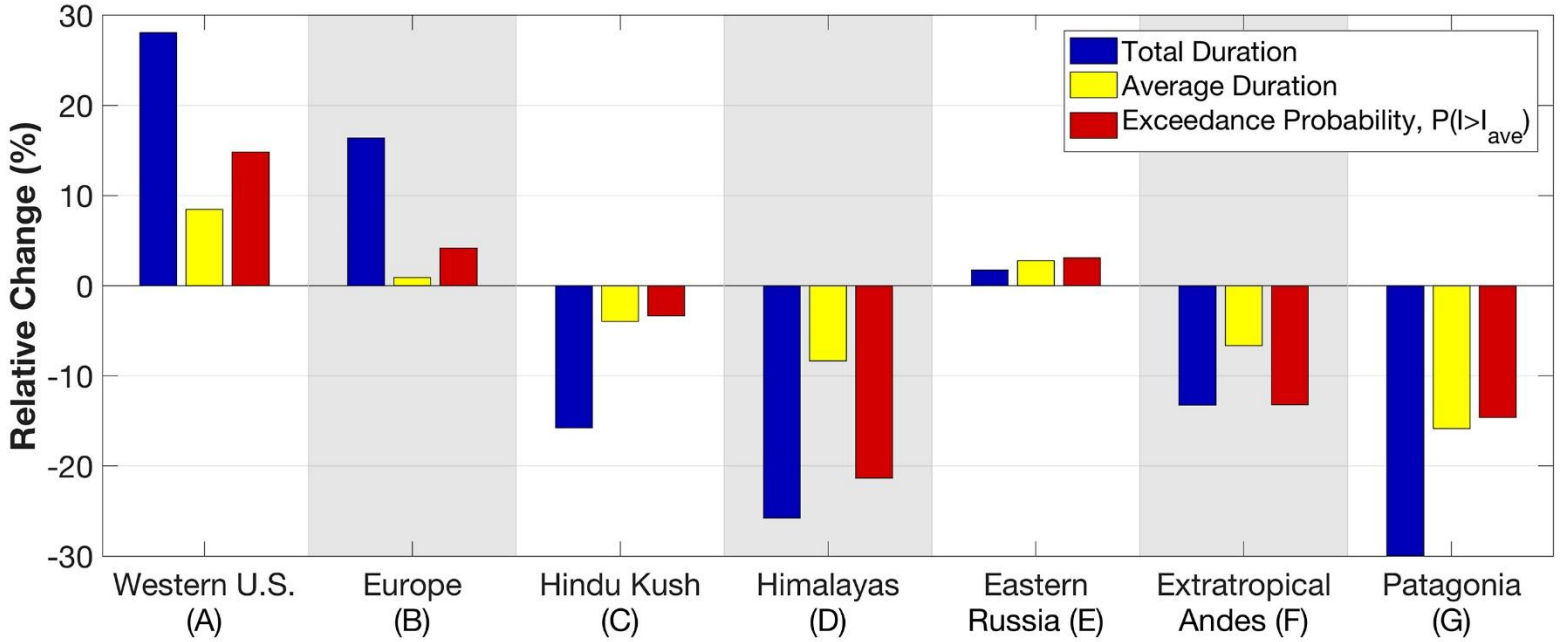
Avanzi et al. (2024)

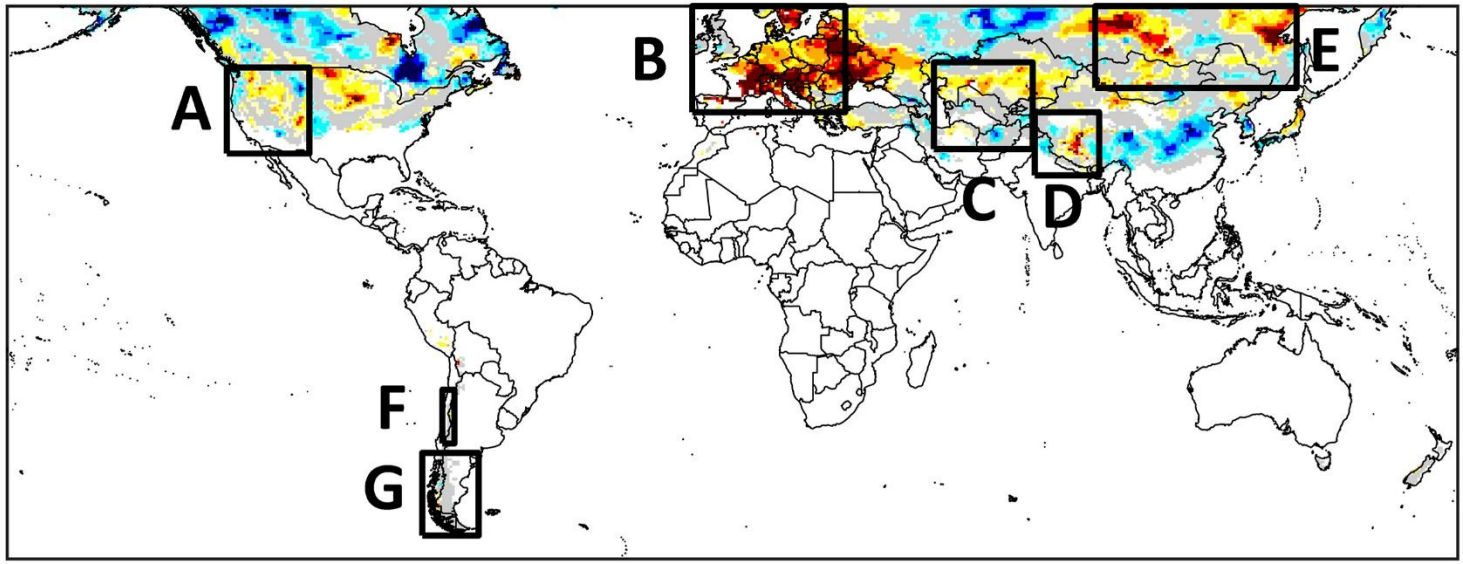
Snow droughts

Snow droughts have increased between 1980 and 2018 especially in the Western US and Europe

Afghanistan is especially vulnerable

"The 2024 snow drought will likely result in nearly an additional 110,000 children who will be severely acutely malnourished and at risk of dying. Along with recent increases due to worsening insecurity and COVID-19, there will be nearly 1 million children with severe acute malnutrition (SAM) in Afghanistan over the next year." (https://reliefweb.int/disaster/dr-2021-000022-afg)

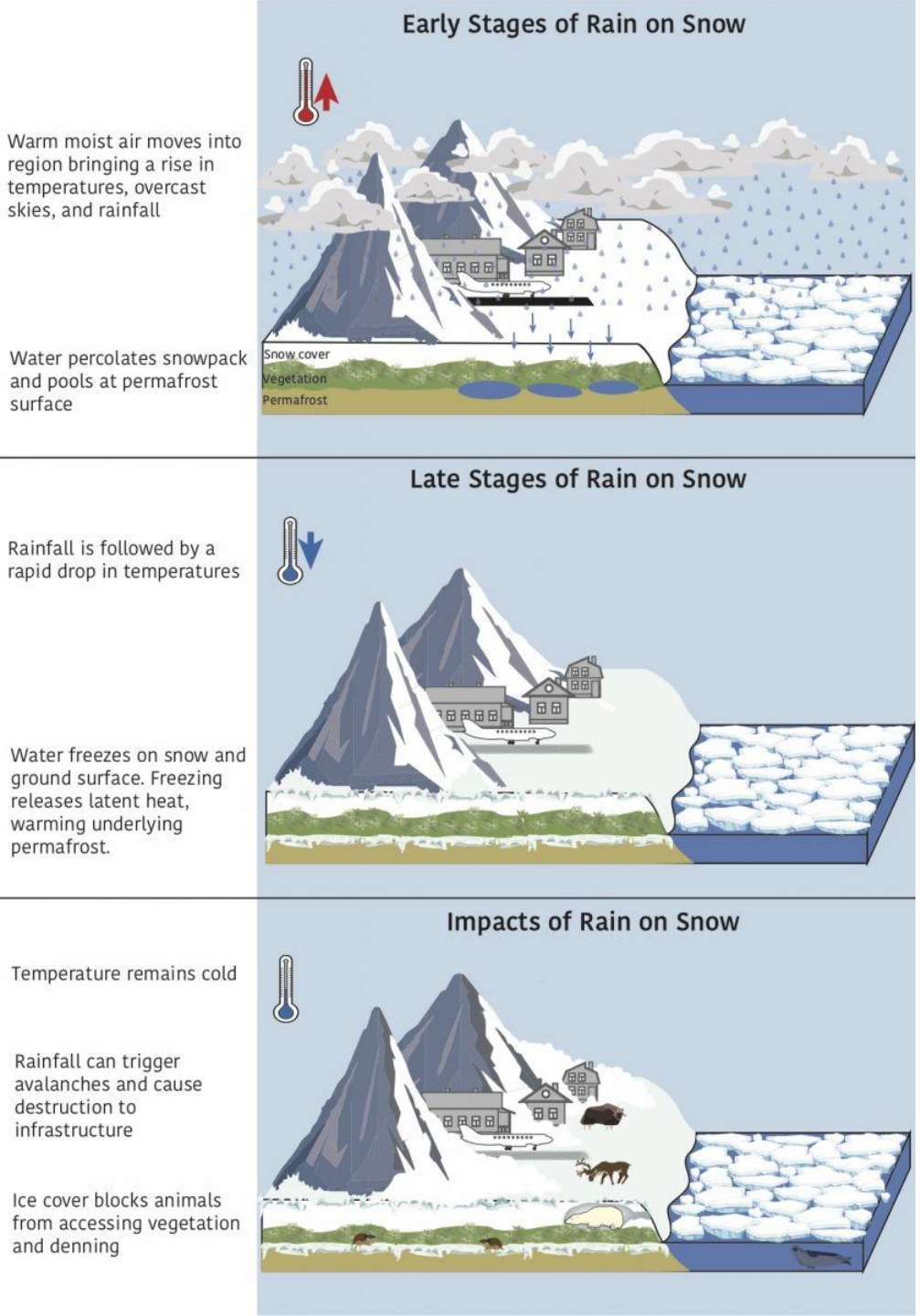


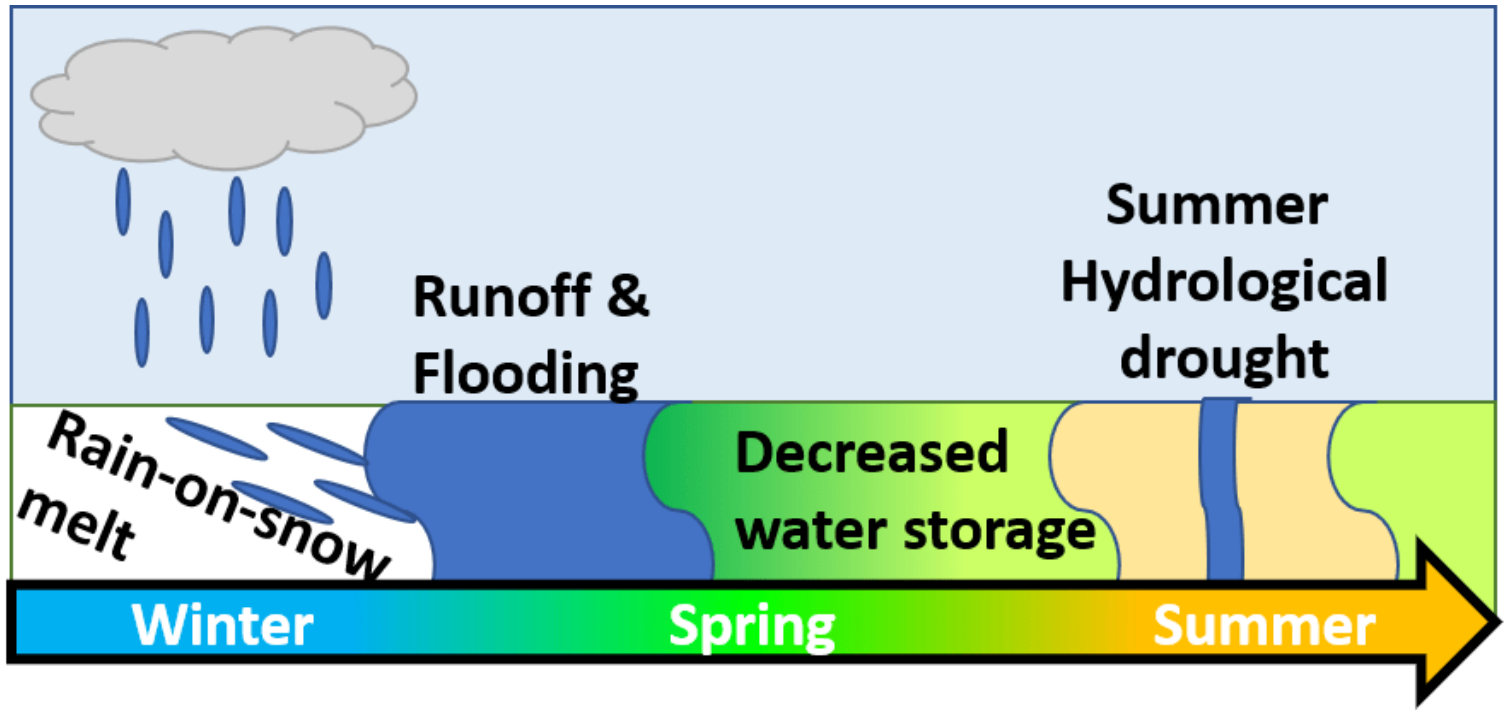


Huning et al. (2020)

Snow turning into rain

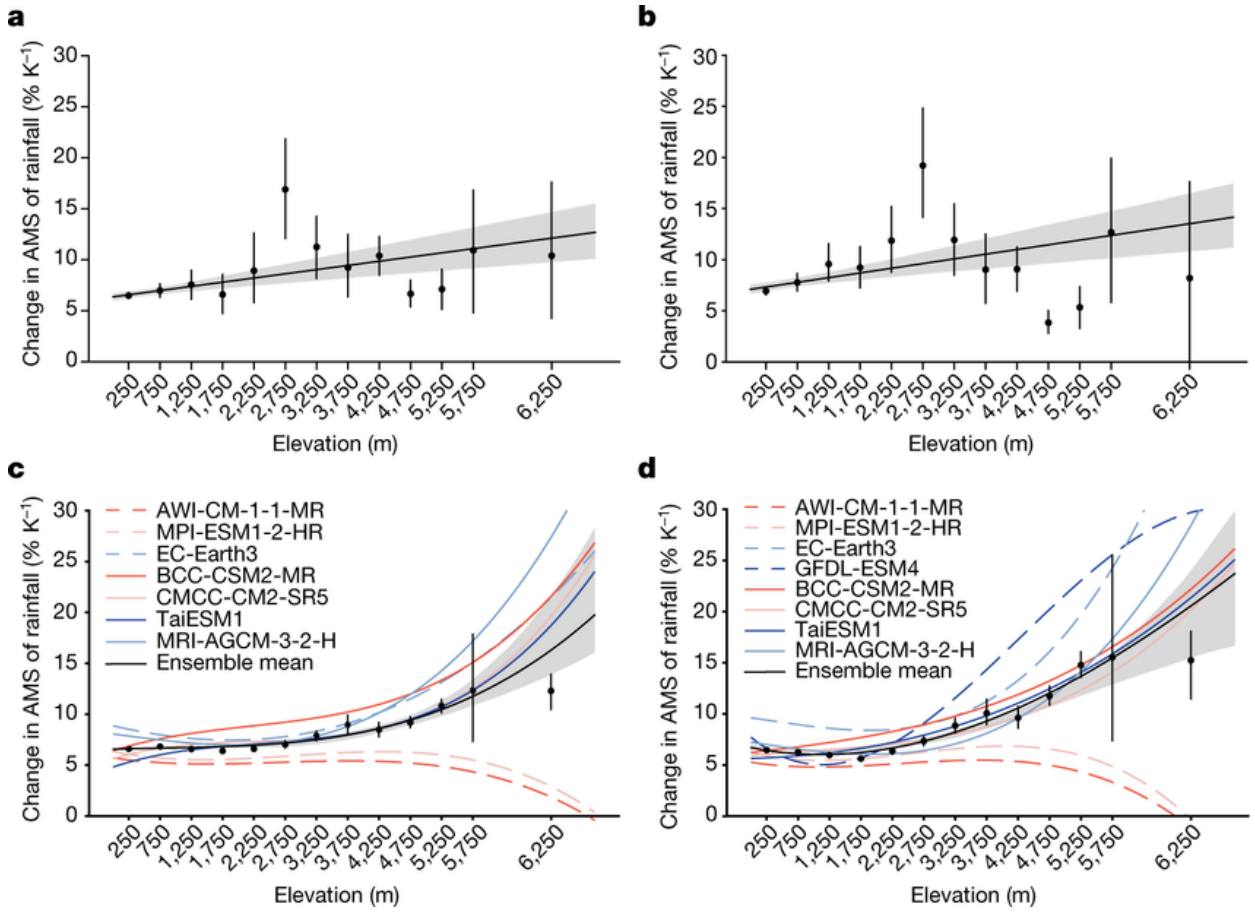
Why could that be a problem?

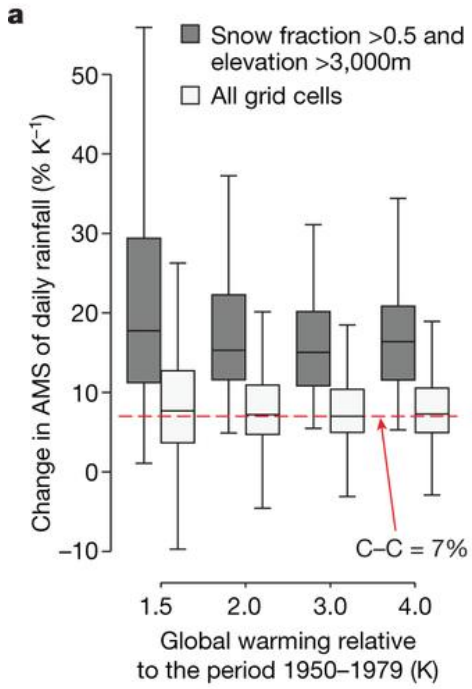


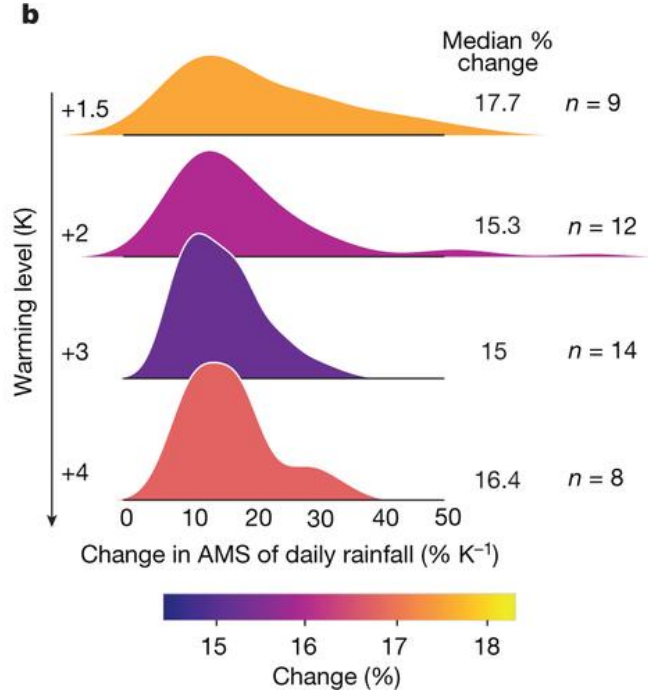


Potential lubricant for (rock-ice-snow-) avalanches resulting in much longer runouts

Snow to rain transition







Ombadi et al. (2023)

Permafrost thaw/creep

As previously constantly frozen ground warms, it looses stability

Threat to infrastructure, agriculture and downstream hydrology

Permafrost thaw/creep

As previously constantly frozen ground warms, it looses stability

Threat to infrastructure, agriculture and downstream hydrology

B Etzelmüller, UiO, Dawson, Alaska

Subsidence

Joshimath
(Uttrakhand, India)

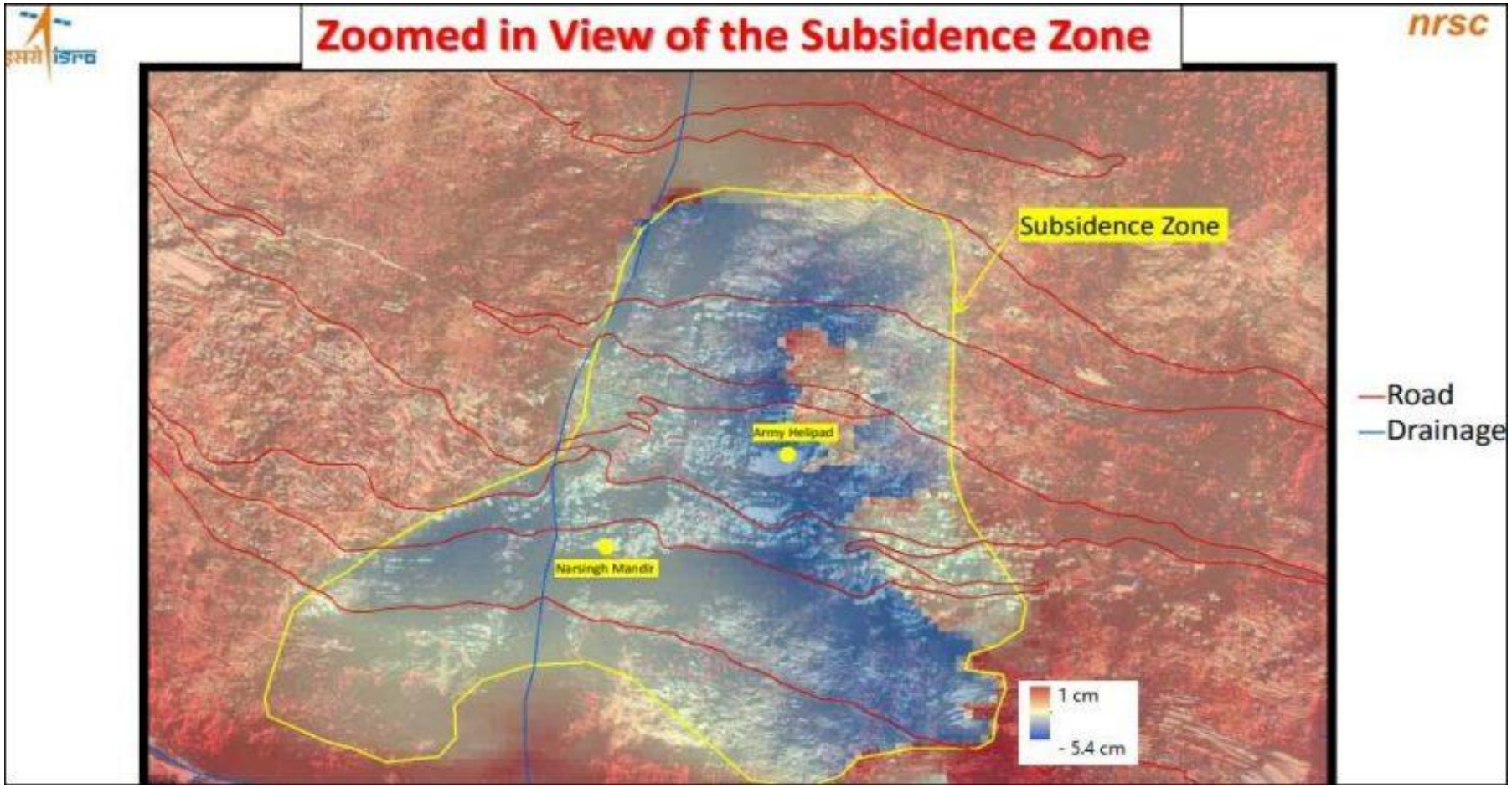
16 000 people

200 so far relocated

Intensive road construction

Chamoli disaster

Subsidence


Zoomed in View of the Subsidence Zone
nrsc
Subsidence Zone
Army Helipad
Narsingh Mandir
1 cm
- 5.4 cm
Road
Drainage

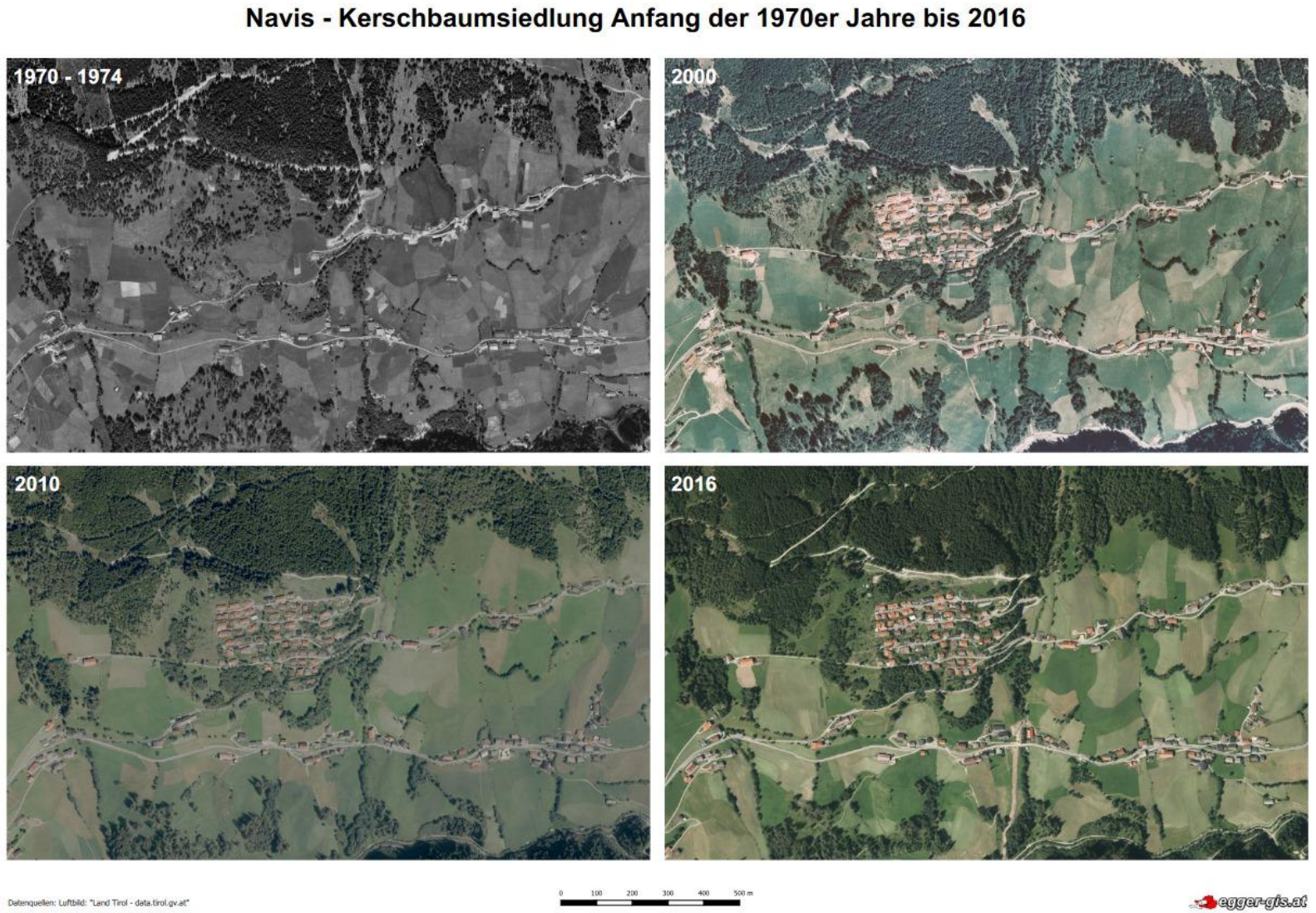
Subsidence and creep

Brenner Basis Tunnel

Subsidence and creep

Kerschbaumsiedlung, Navis, Tirol

84 houses (~250 people)



Drainage of slopes with 50 wells of 50 m depth

Total cost: >6 mio €

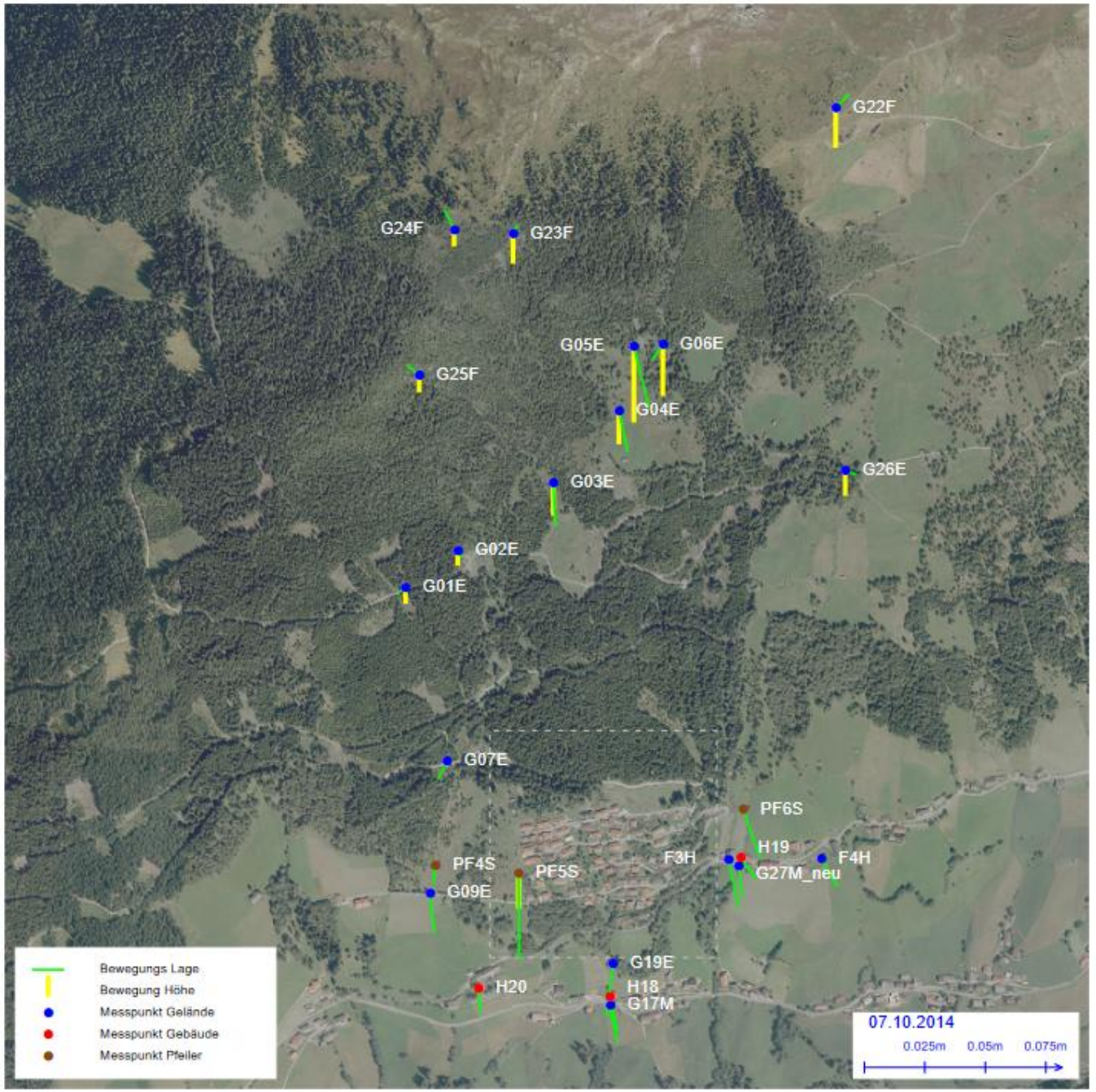


Abbildung 1: Absolutbewegung der Messpunkte seit der Nullmessung (Median aus 14 Tagen)

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Multihazards, compound events, cascading and transboundary risks

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Multihazards, compound events, cascading and transboundary risks

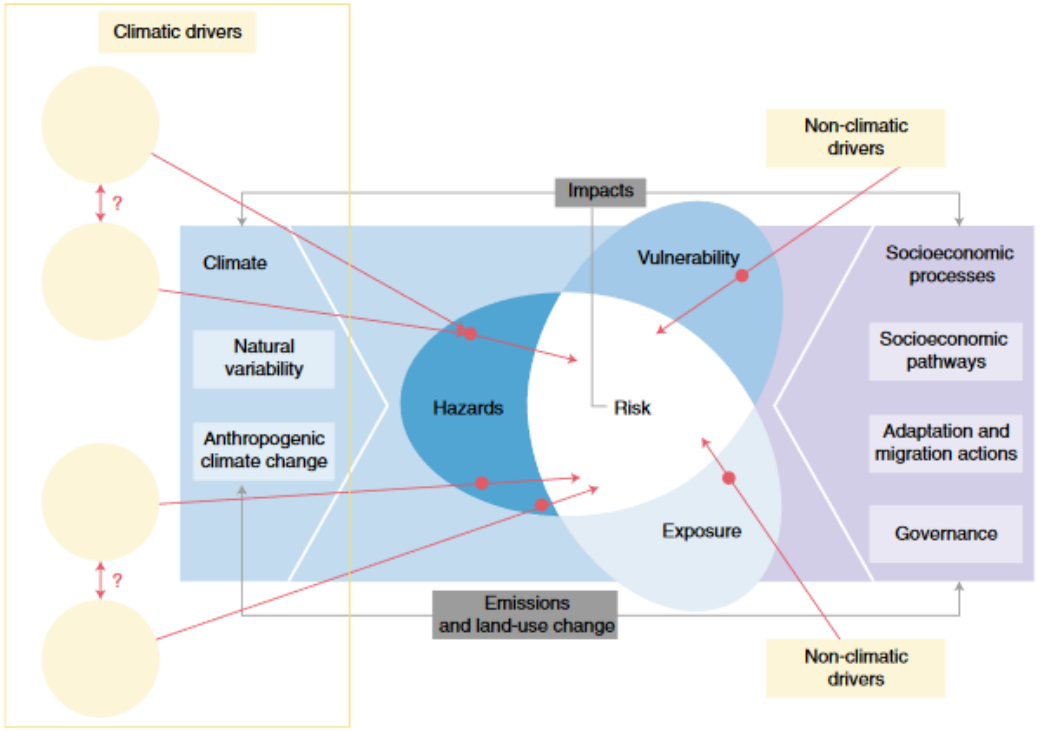
multi hazard: 'hazardous events may occur simultaneously, cascadingly or cumulatively over time, and taking into account the potential interrelated effects' (UNDRR)

multirisk: risk in a multihazard framework

cascading hazards: chain of hazards where an initial one sets off a number of others down the chain

Multihazards, compound events, cascading and transboundary risks

Compound events/drivers: The combination of processes (climate drivers and hazards) leading to a significant impact is referred to as a 'compound event'.



Zscheischler et al. (2018)

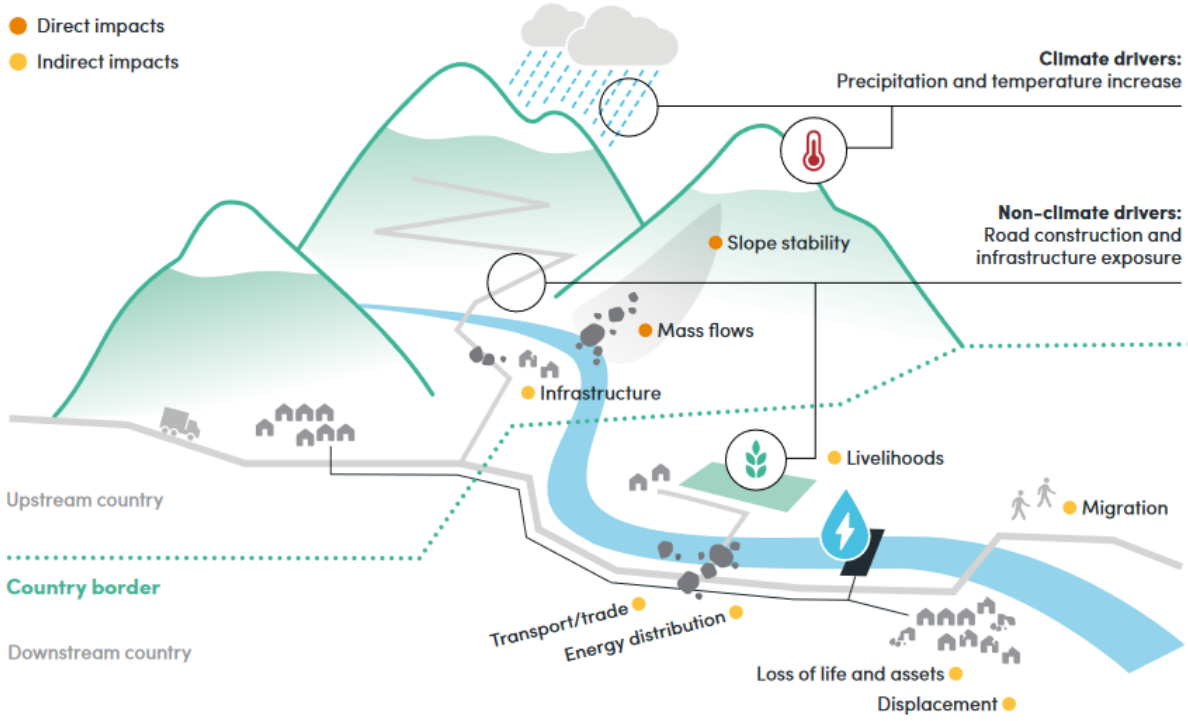
Fig. 1 | Extended risk framework. Multiple climatic drivers cause one or multiple hazards leading to societal and environmental risk. The climate drivers (which may vary from local-scale weather to large-scale climate modes, represented by yellow circles) and/or hazards may be mutually dependent. Non-climatic drivers related to vulnerability and exposure may also contribute to risk. Background risk figure adapted from ref. [96], IPCC.

Multihazards, compound events, cascading and transboundary risks

Compound events/drivers: The combination of processes (climate drivers and hazards) leading to a significant impact is referred to as a 'compound event'.

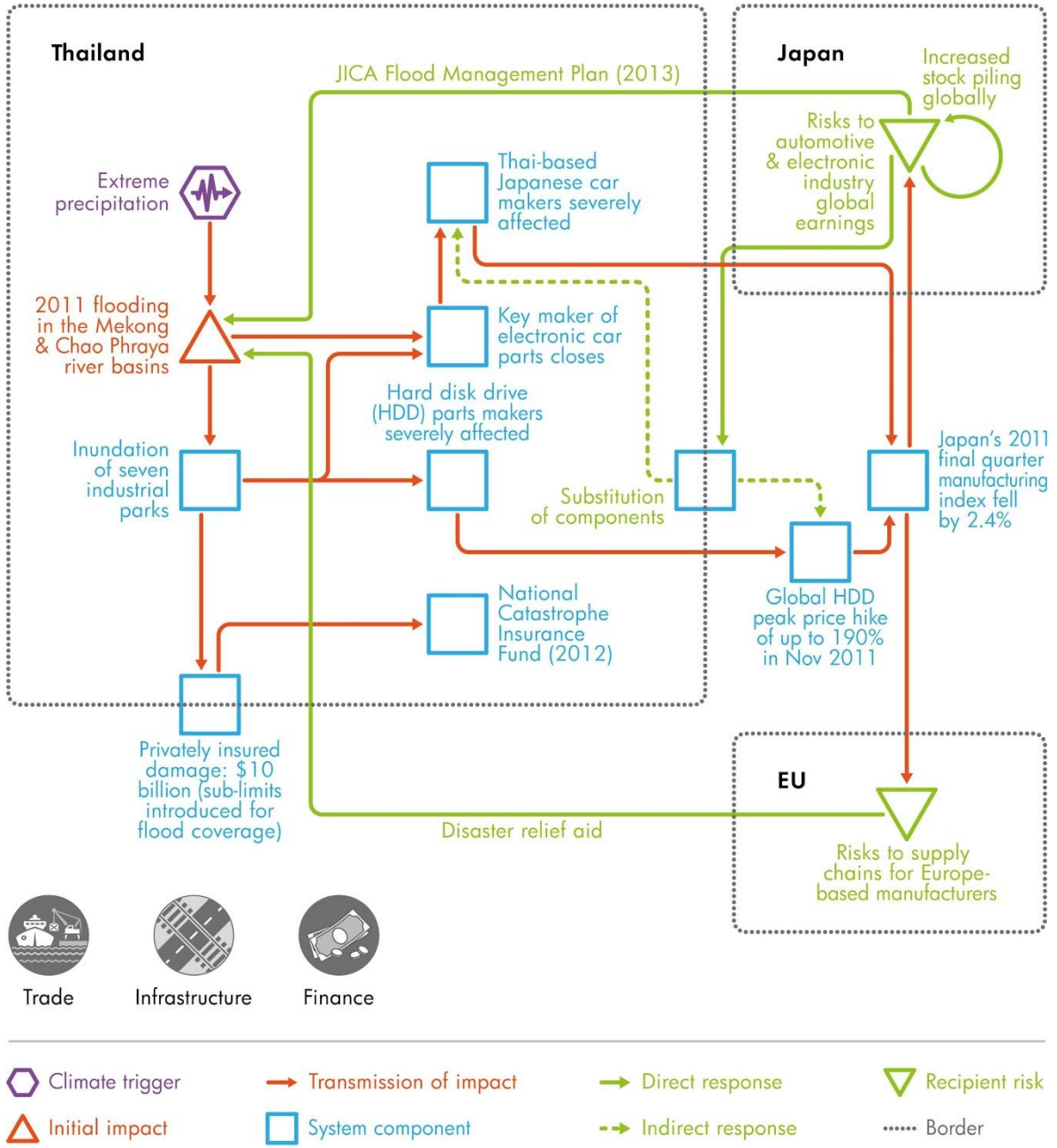
Cascading risks: risks propgating across domains (becoming transboundary when crossing borders)

Figure 4. Transboundary climate risks for shared terrestrial natural resources: the example of mass flows in cross-border river corridors



Notes: Conceptual overview on transboundary climate risks for terrestrial shared natural resources, using the example of mass flows in river corridors that cross multiple countries. The drivers of risk are related to both climate and non-climate variables, and potentially related directly or indirectly to anthropogenic drivers. The impacts can be direct (orange) and transboundary (e.g. flooding crossing borders threatening lives, damaging agricultural land and hydropower infrastructure), and indirect (yellow), the latter potentially transmitting risks across borders in the reverse direction (e.g. disruption of food or energy supply, effects on livelihoods). Different examples related to the concepts are discussed in the text.

Steiner et al. (2022)



Carter et al. (2021)

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28 June (1000 – 1800)

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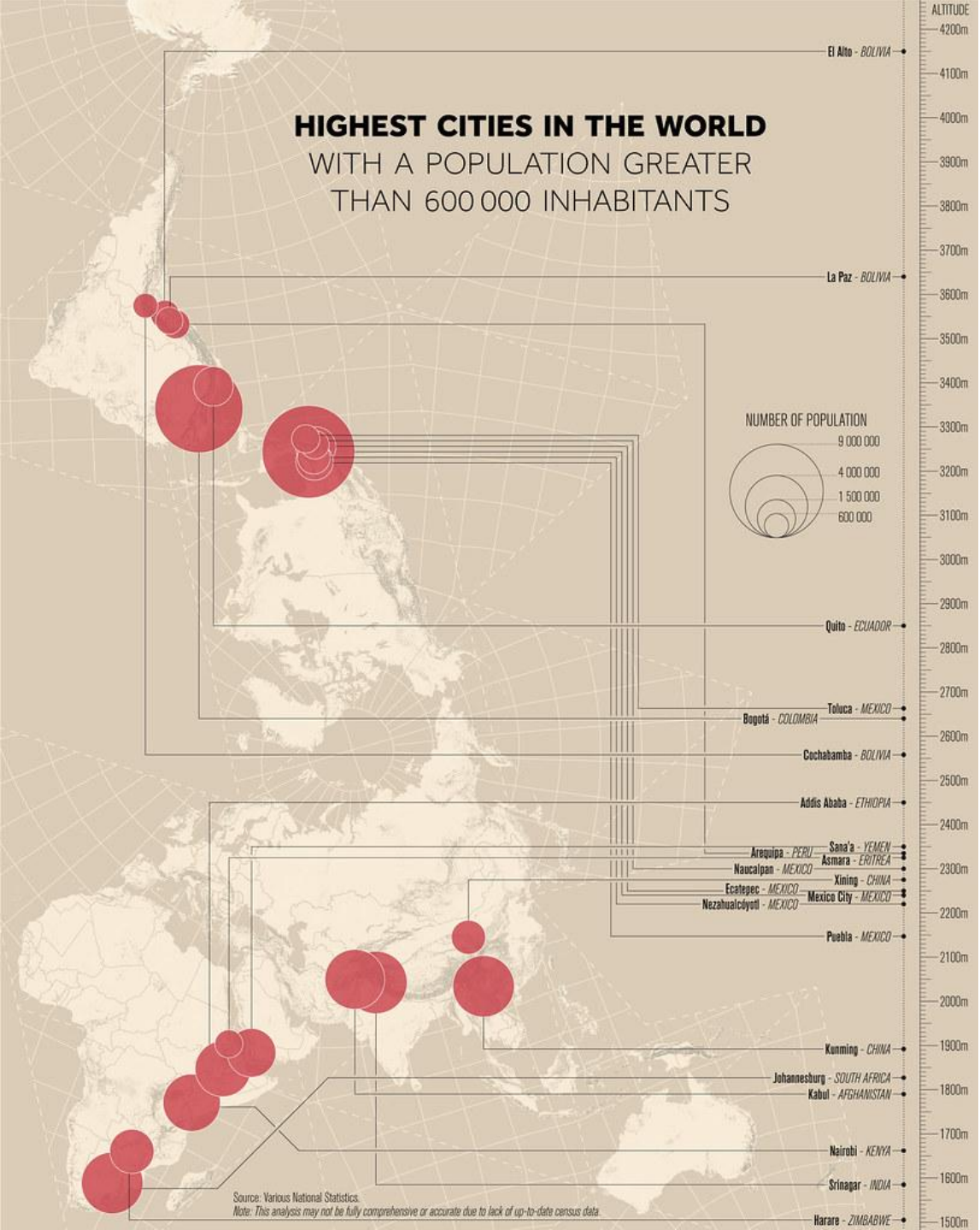
Urbanisation

Mountains are often associated with a *rural* context, resulting in risk assessments tailored for that direction.

How do we define cities/towns/urban?

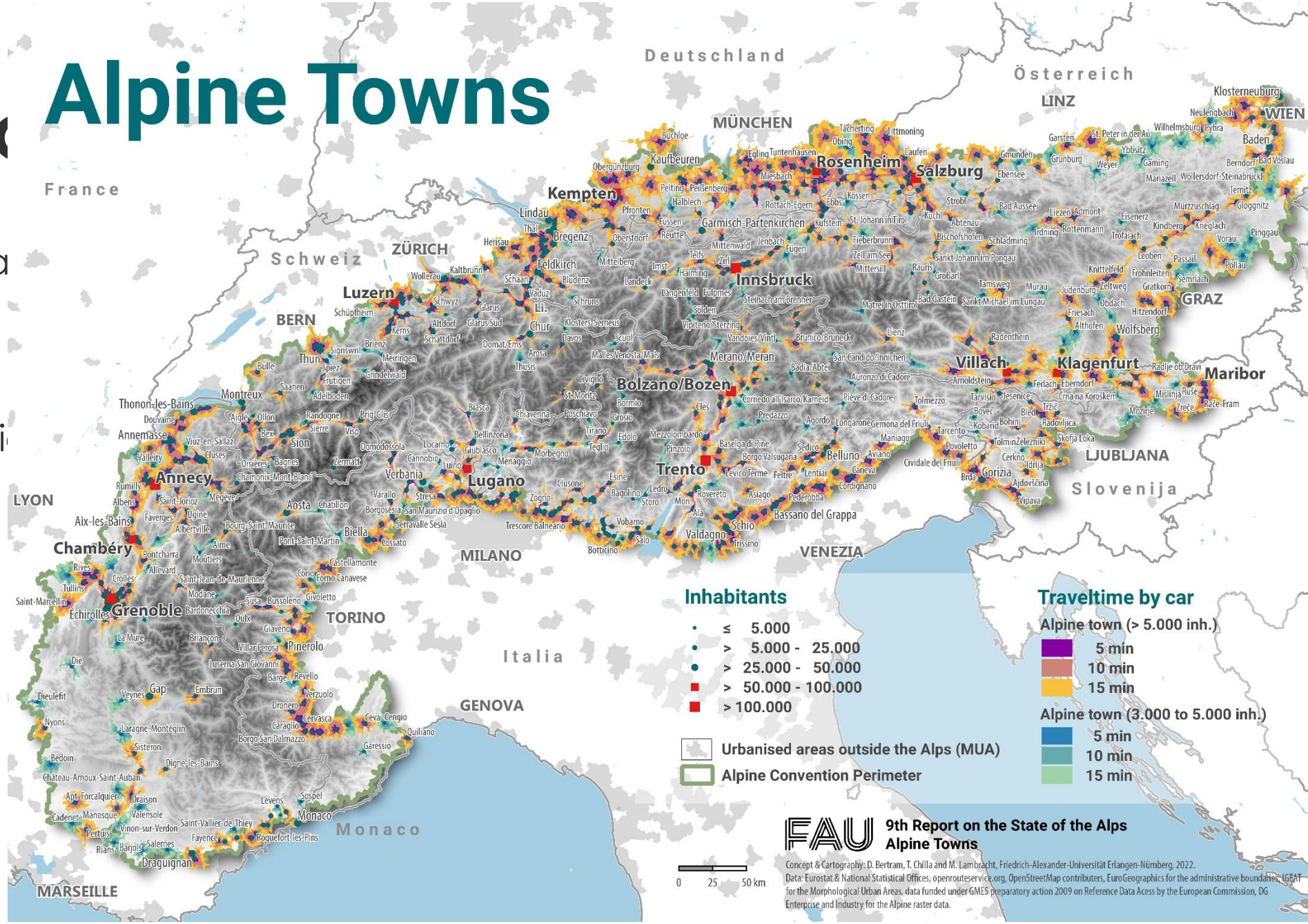


https://www.grida.no/resources/5029

Urbanisatio

Mountains are often a
that direction.

How do we define citi



https://alpinetowns.alpconv.org/

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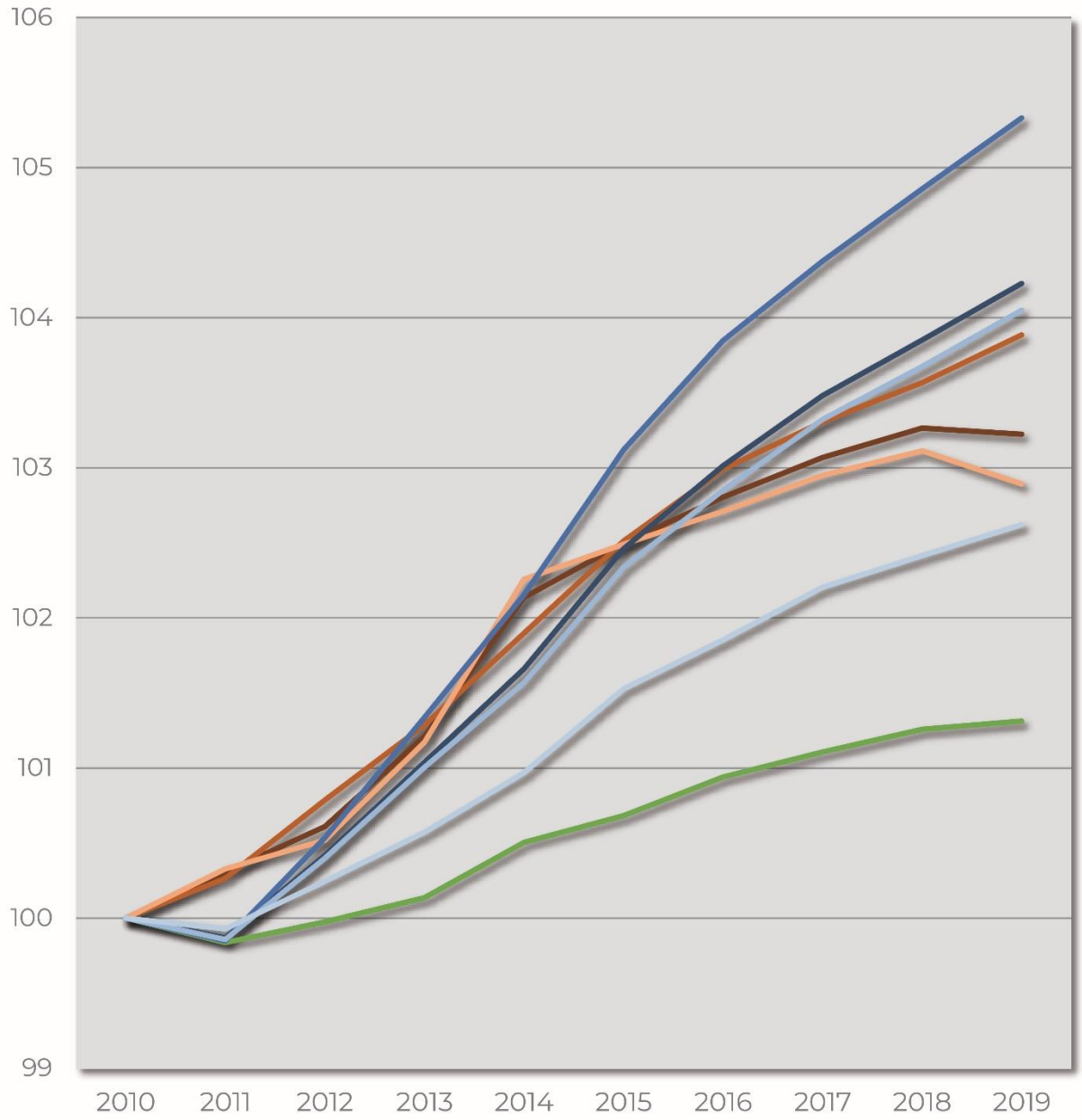


Population change
(2010 = 100)

Alpine area
Alpine Convention perimeter

- Total
- Alpine towns (≥ 3.000/5.000 Inh., RSA 9 definition)
- Rural areas

Peri-Alpine area
EUSALP perimeter without Alpine Convention area

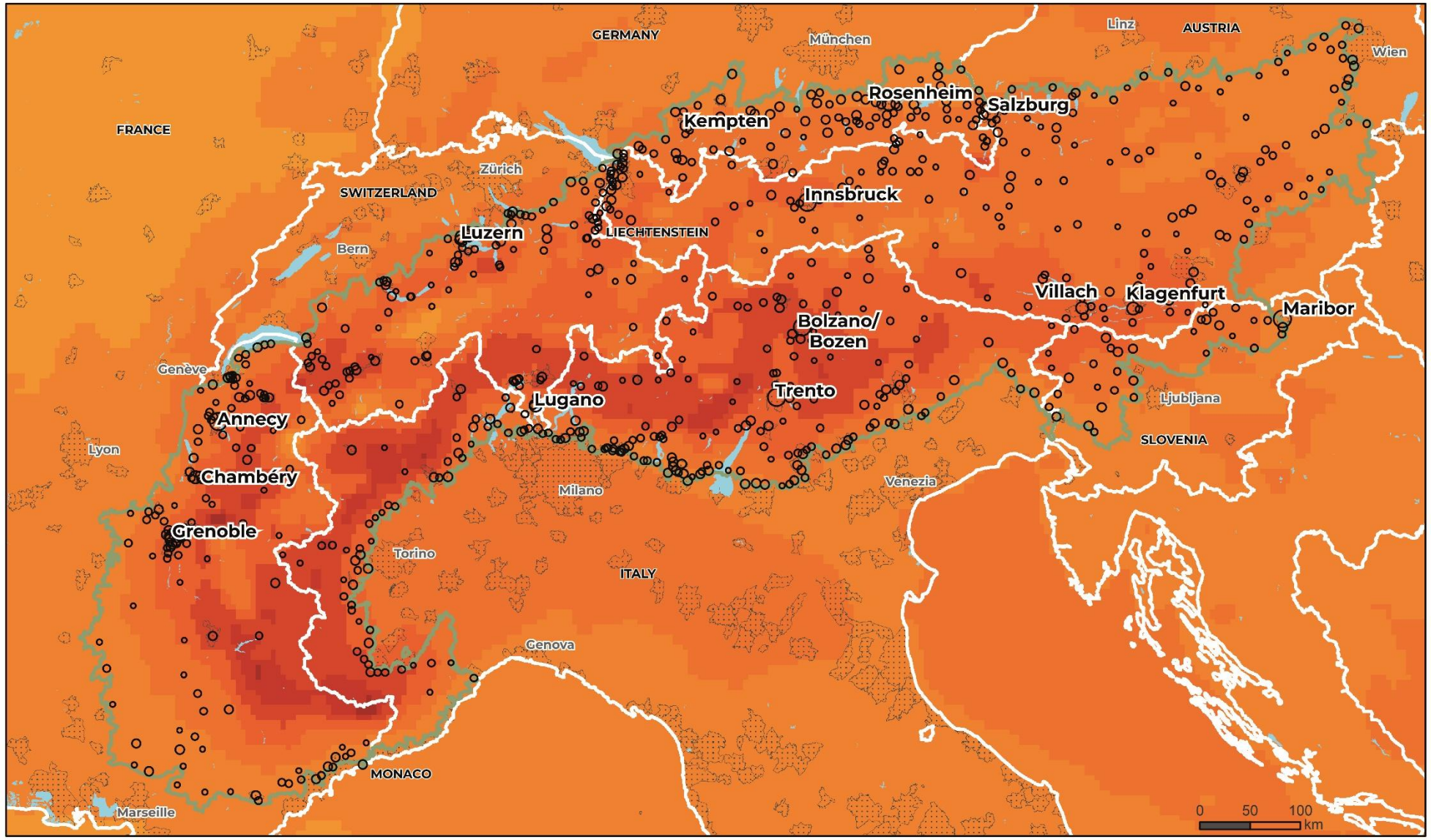
- Total
- Cities (DEGURBA 1)
- Towns and suburbs (DEGURBA 2)
- Rural areas (DEGURBA 3)

Europe

- Total EU 27

FAU **9th Report on the State of the Alps**
Alpine Towns

Data: Eurostat, 2021; National Statistical Offices, 2018 & 2021.
Concept & Visualisation: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, **2022.**



Temperature change

Population of Alpine towns (2015)

- 3.000 - 5.000
- > 5.000 - 10.000
- > 10.000 - 50.000
- > 50.000 - 100.000
- > 100.000

Changes in surface temperature between 2021-2050 (in Celsius)

0.81 - 0.9	1.41 - 1.5
0.91 - 1	1.51 - 1.6
1.01 - 1.1	1.61 - 1.7
1.11 - 1.2	1.71 - 1.8
1.21 - 1.3	1.81 - 1.9
1.31 - 1.4	1.91 - 2

FAU

9th Report on the State of the Alps
Alpine Towns

Data: Alpine Convention for projected changes in annual near surface temperature in Southern Europe 2021-2050 in Celsius (WMS).
Layout: © EuroGeographics for the administrative boundaries; IGEAT for the Morphological Urban Areas.
Concept & Cartography: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, **2022.**

Alpine Convention Perimeter

Urbanised areas outside the Alps (MUA)

Urbanisation

Mountains are often associated with a *rural* context, resulting in risk assessments tailored for that direction.

How do we define cities/towns/urban?

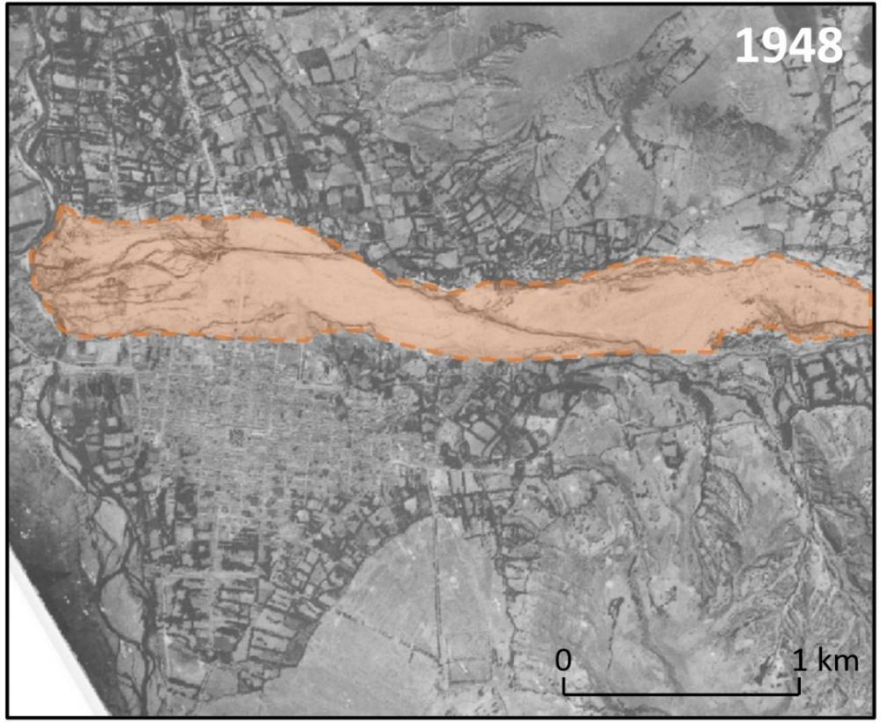
Why is urban space different in terms of risks?

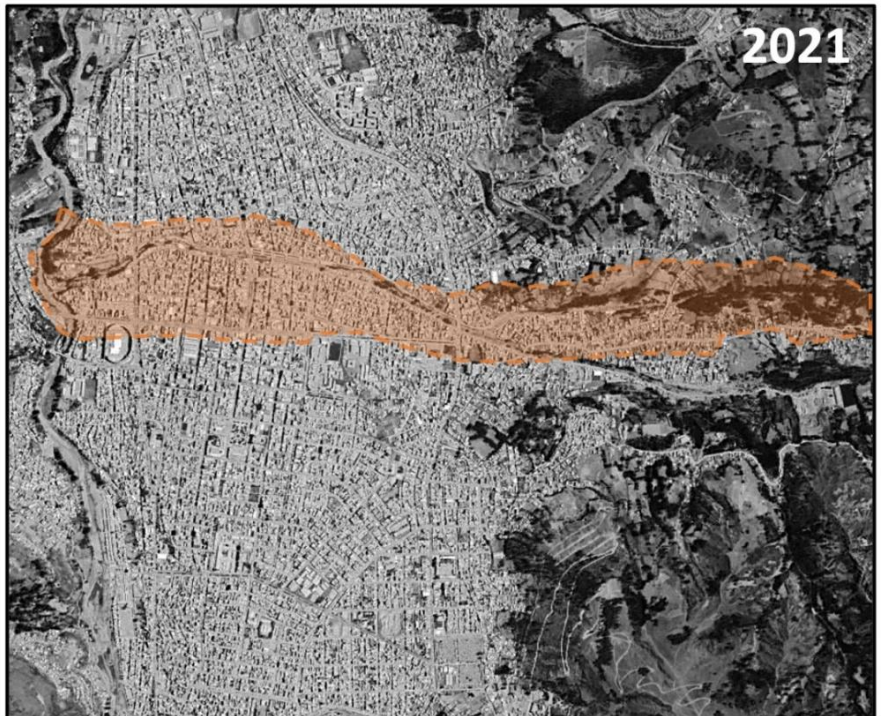
Urbanisation

Mountains are often associated with a *rural* context, resulting in that direction.

How do we define cities/towns/urban?

Why is urban space different in terms of risks?





Emmer, Steiner and Kalvoda (2024)

Hydropower

Germany: 3.5% of demand

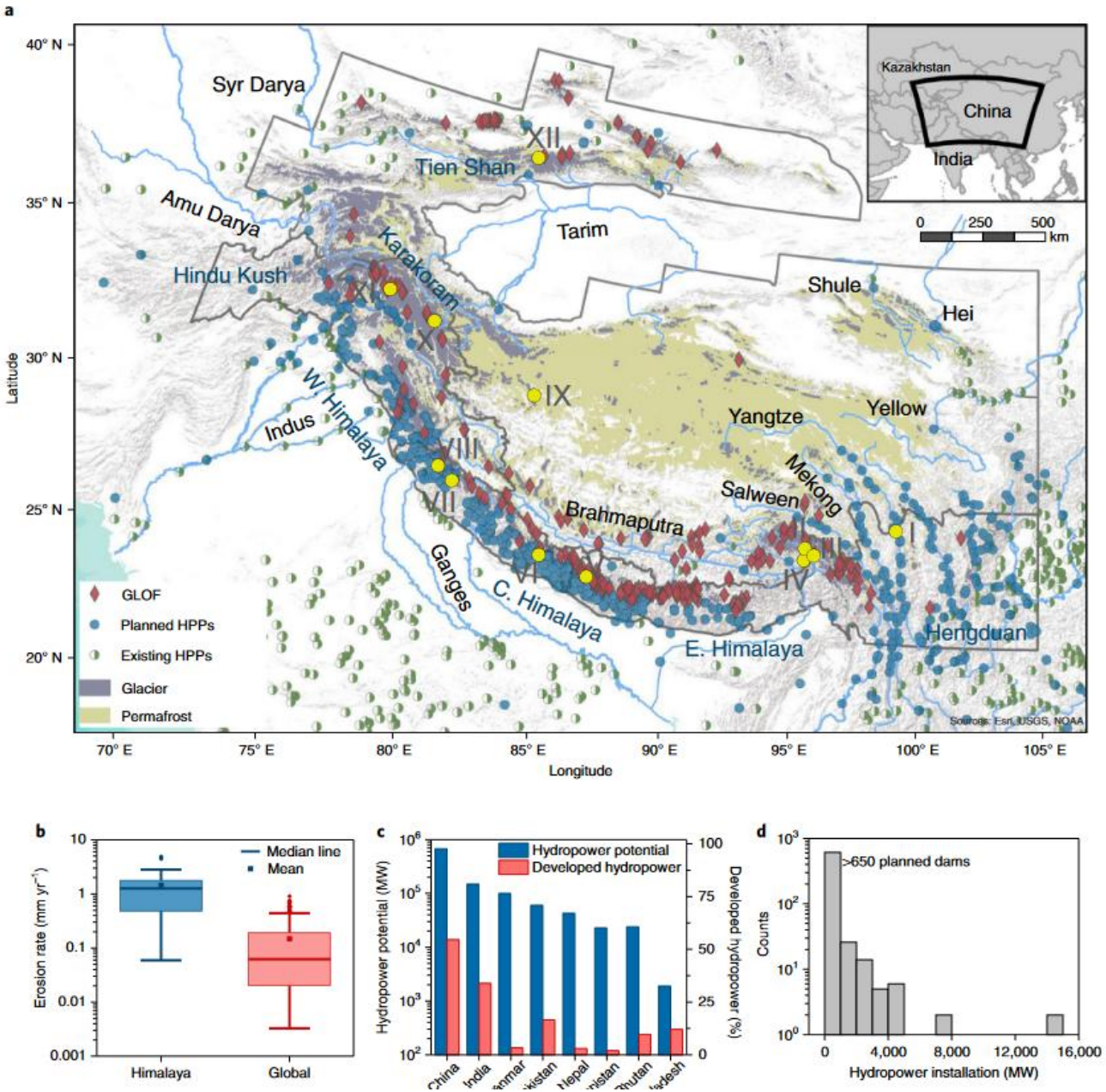
Austria: 58% of total generation

Switzerland: 62% of total generation

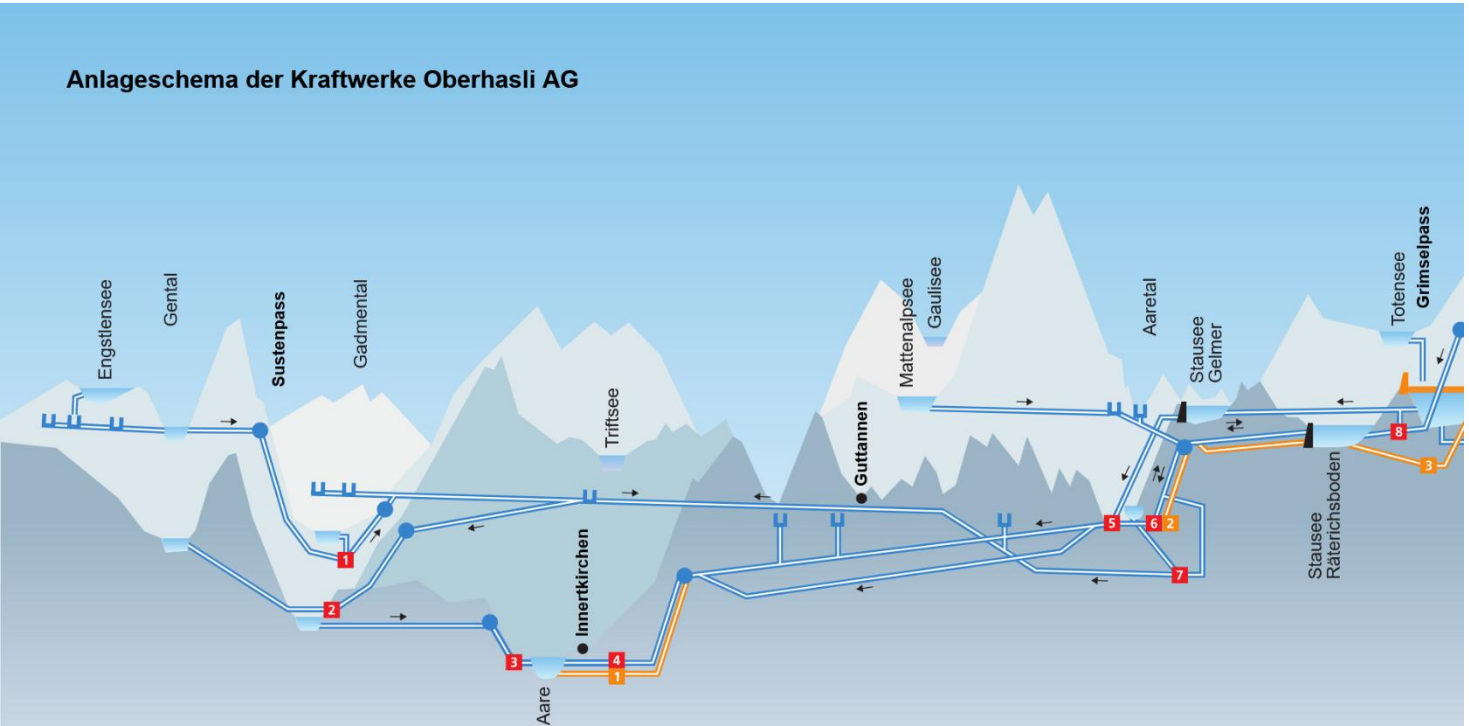
New Zealand: 55% of total generation

Bhutan: 99.5% of total generation

Nepal: 75% of total generation



Hydropower Infrastructure



- KRAFTWERKE
  - A Grimsel 1
  - B Grimsel 2
  - C Handeck 1
  - D Handeck 2
  - E Handeck 3
  - F Innertkirchen 1
  - G Innertkirchen 2
  - H Fuhren
  - I Hopflauenen
- SPEICHERSEEN
  1. Oberaarsee
  2. Grimselsee
  3. Räterichsbodensee
  4. Gelmersee
  5. Totensee
  6. Trübtensee
  7. Engstlensee
  8. Mattenalpsee
- BAHNEN
  1. Meiringen-Innertkirchen-Bahn (MIB)
  2. Gelmerbahn
  3. Triftbahn
  4. Tällibahn
  5. Luftseilbahn Handeck-Gerstenegg
  6. Oberaarbahn
  7. Sommerlochbahn
- GRIMSELHOTELS
  - Handeck · Kinder- und Familienhotel
  - Grimsel Hospiz · Historisches Alpinhotel
  - Oberaar · Restaurant und Berghaus
  - Bäregg · Ferien- und Alpinhütte
  - Tälli · Restaurant und Berghaus
- FIRMENSITZ

Konzessionsgebiet
Wasserführende Stollen
Zugangsstollen

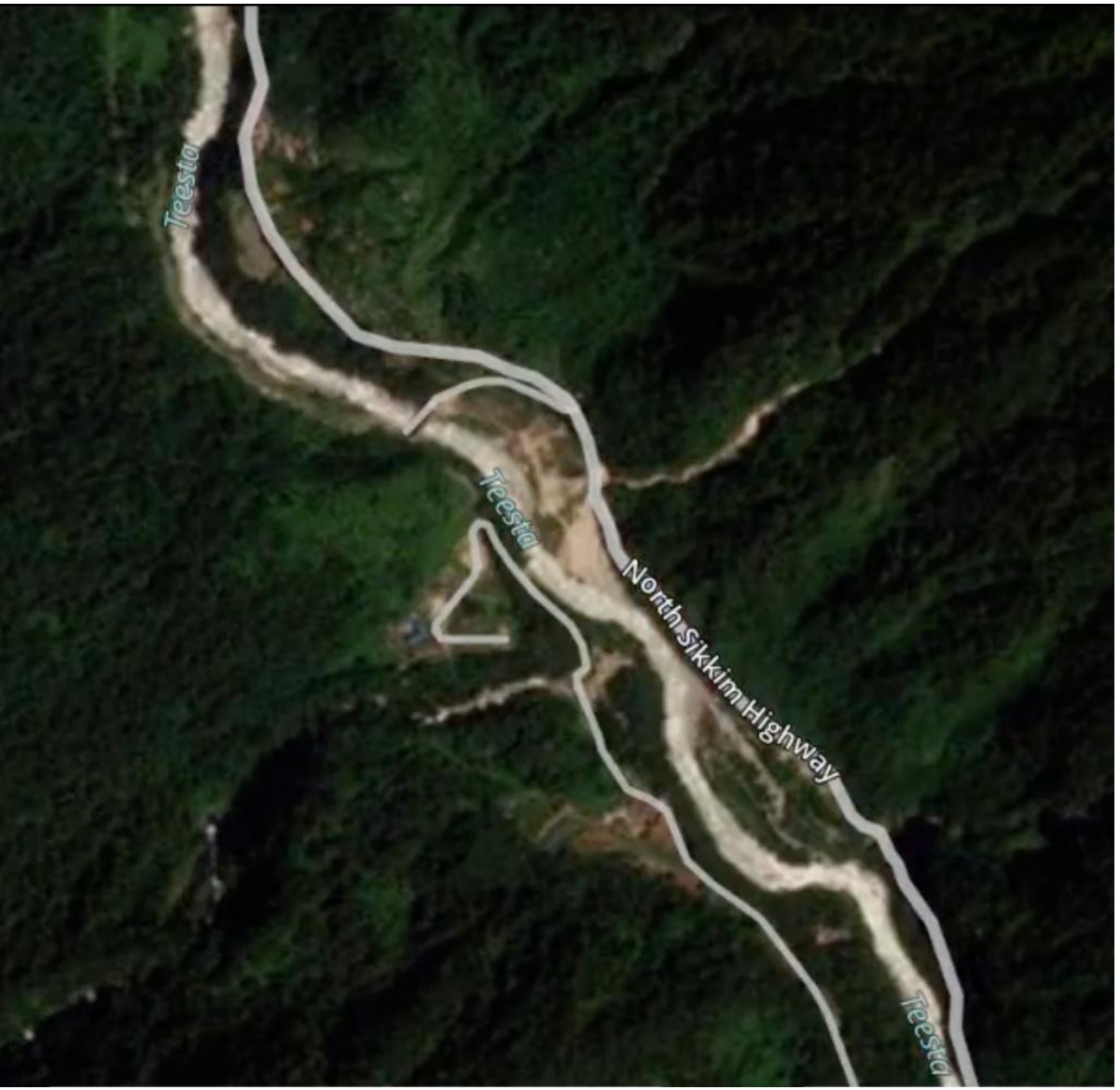
South Lonak GLOF – October 2023

- 100 people died
- Well known lake that was expected to cause problems

Hydropower infrastructure could help risk reduction if operated in accordance

In this case, it increased the impact, as the water was first retained and then breached with even higher force

04-10-2023
Earlier
TEESTA 3 DAM
BEFORE & AFTER


Teesta
Teesta
North Sikkim Highway
Teesta

Teesta
Teesta
North Sikkim Highway
Teesta

ZDF
Gesamtsendung
Unwetter in der Schweiz:
Suche nach Verschütteten
heute JOURNAL

Framing of risks in media

Examples of recent mountain floods in Switzerland

Parallels to Germany

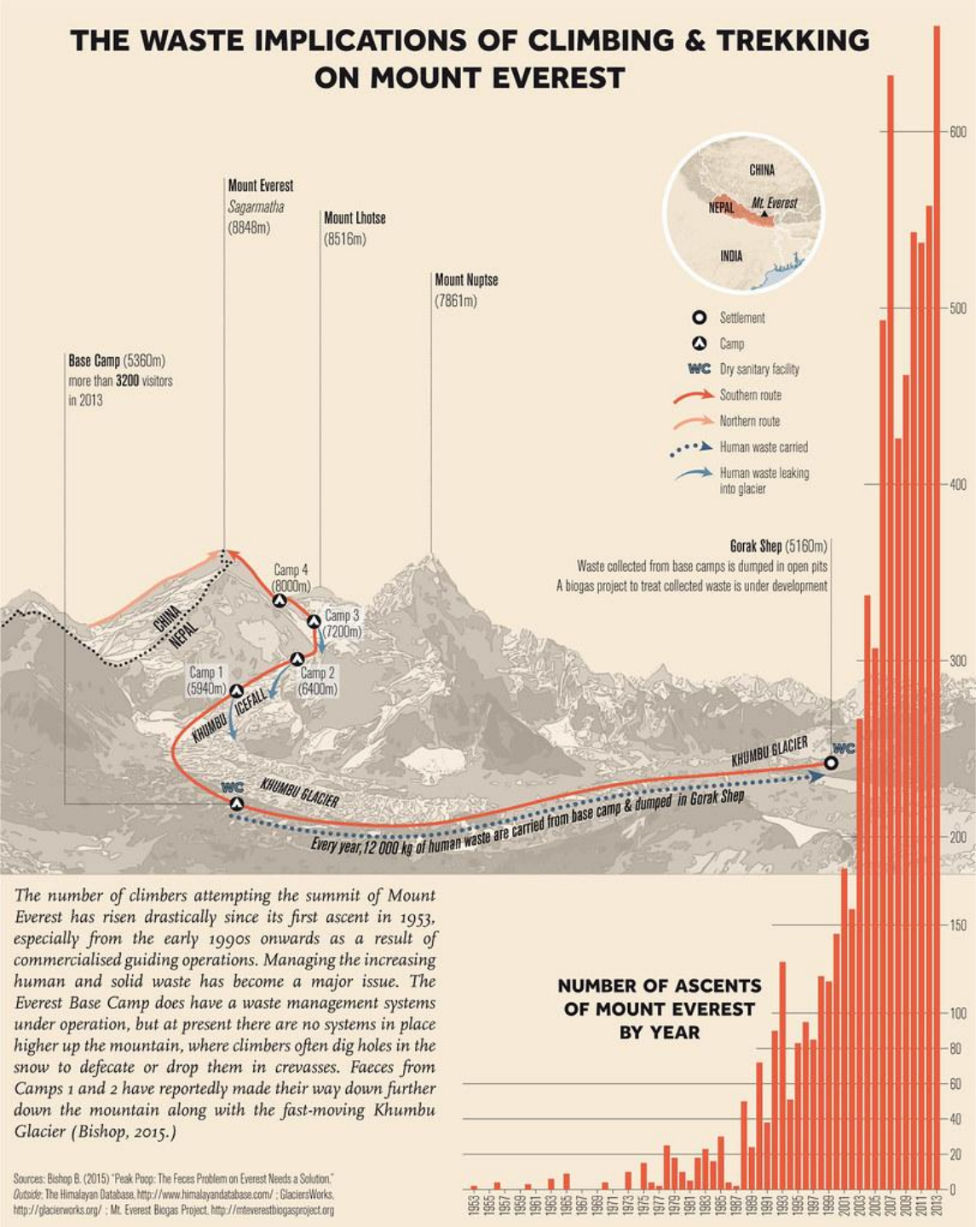
Mediterranean that is too warm, resulting in extreme precipitation

Tourism

Tourism

Risk for tourism

Tourism as risk



Focus has been long on *risk for foreign climbers*

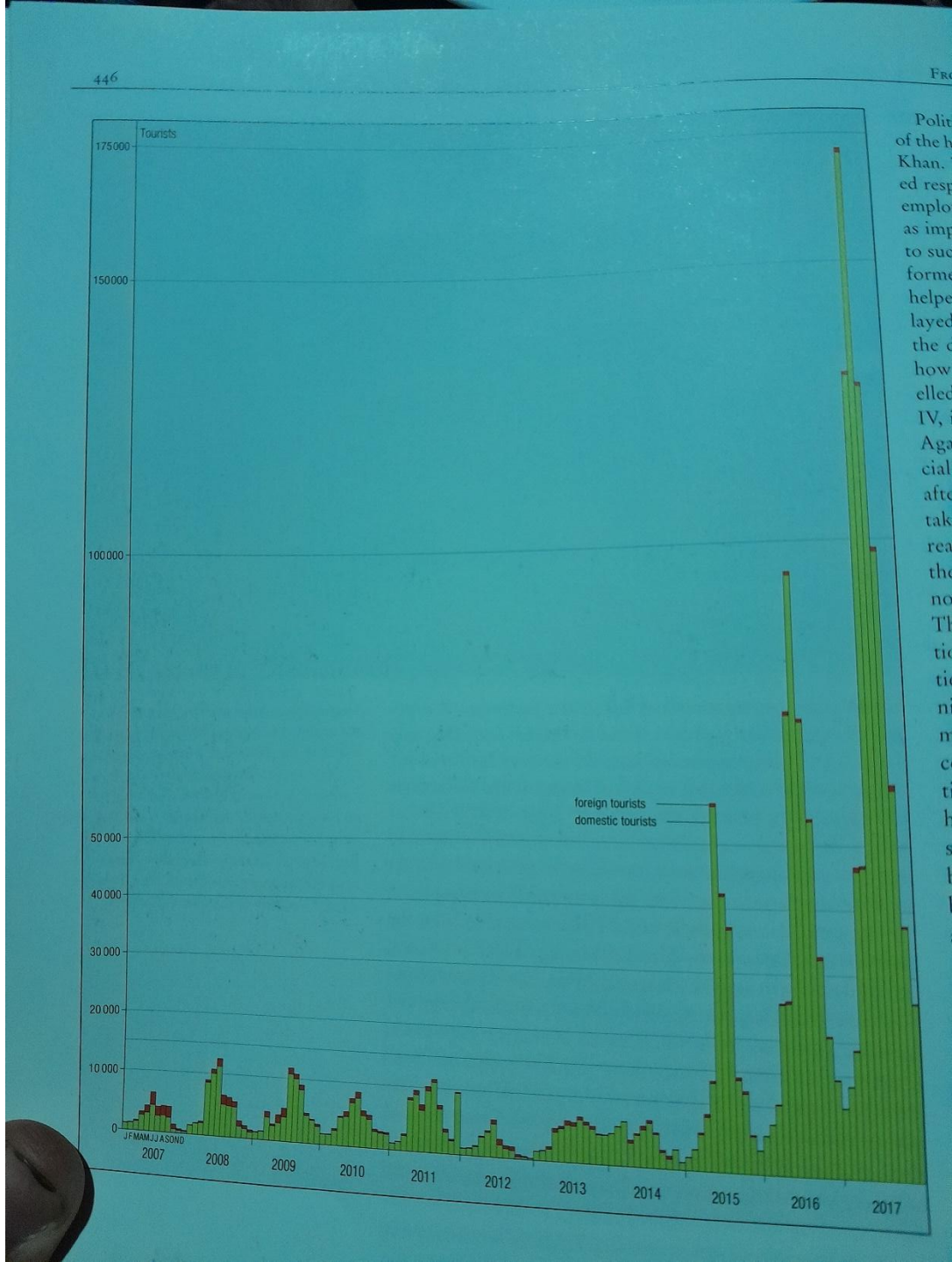
Slowly shifting to *risks posed by tourists* (overtourism)

https://www.grida.no/resources/5028

Myths of mountain tourism's relevance

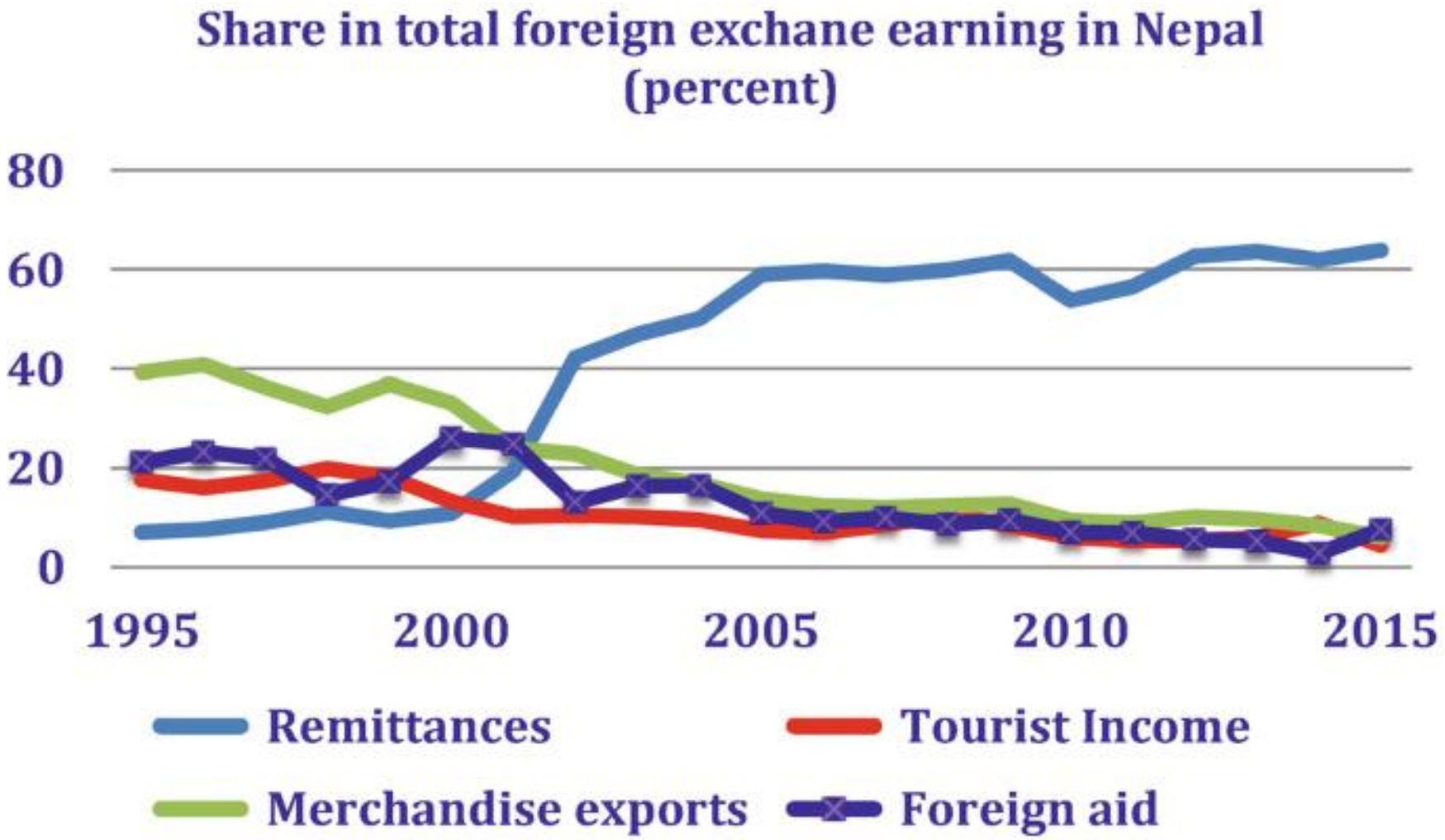
How relevant is it for the local economy?

How relevant are foreign tourists?



Total tourists Gilgit Baltistan, Pakistan, from Kreutzmann, 2020

Tourism earnings vs. remittances

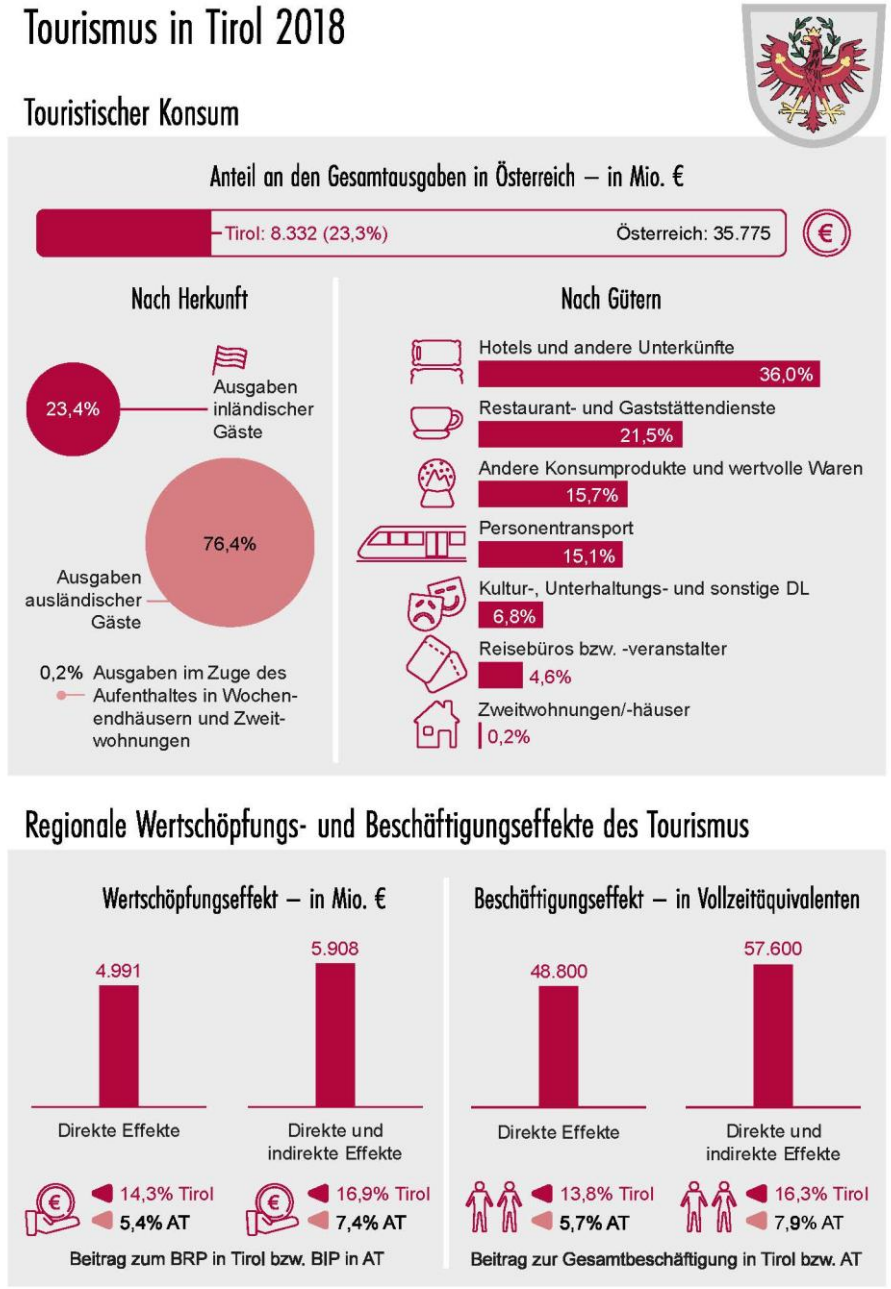


What is the role of remittances/migrant labour with respect to climate risks?

Sharma, 2019

Tourism relevance in the Alps

Similar challenges of adapting to a changing climate (no more skiing in future?, more hazards?)



- Lack of studies for Africa and South America

- Very different challenges between Alps/Rocky Mountains and e.g. High Mountain Asia (differences in vulnerability)

Natural hazard	Description	Potential impact level	Sources
Desertification	Higher aridity of agricultural, forestry and pastoral areas with consequent rise in erosion and loss of organic matter in forest areas as a result of increased fire risk in connection with drought-related events.	Low	Corrado et al., eds., 2014 Matasci and Altamirano - Cabrera, 2010 Probst et al., 2013
Ecosystem changes (land and aquifer)	Changes in the phenological cycle and inland water and ecosystem transition (i.e. shift) due to habitat and soil mutations.	High	Beniston, 2012 Cannone et al. 2008 Cantonati et al., 2006 Ianni et al., 2015 Mourier et al., 2010 Revernmann et al., 2012 Wieser et al., 2008
Forest fires	Expected increase in the danger of forest fires throughout the year, mainly in the spring season.	Medium	Dupire et al., 2019 Moriondo et al., 2006 Moser et al., 2010 Schumacher and Bugmann, 2006 Wastl et al., 2012
Hydro-geological and hydraulic instability	Variation in seasonality and magnitude of phenomena associated with snow dynamics, instability of rock complexes, debris flows and surface landslides.	High	Ellena et al., 2020 Palladino et al., 2018 Probst et al., 2013 Prudent-Richard et al., 2008 Winkler and Reichl, 2014
Water scarcity	Decrease in the availability and quality of water resources related to the reduction of precipitation in winter and summer seasons.	Medium-high	Brunner et al., 2019 Hohenwallner et al., 2011 Klug, 2011 Mastrotheodoros et al., 2020 Zampieri et al., 2016

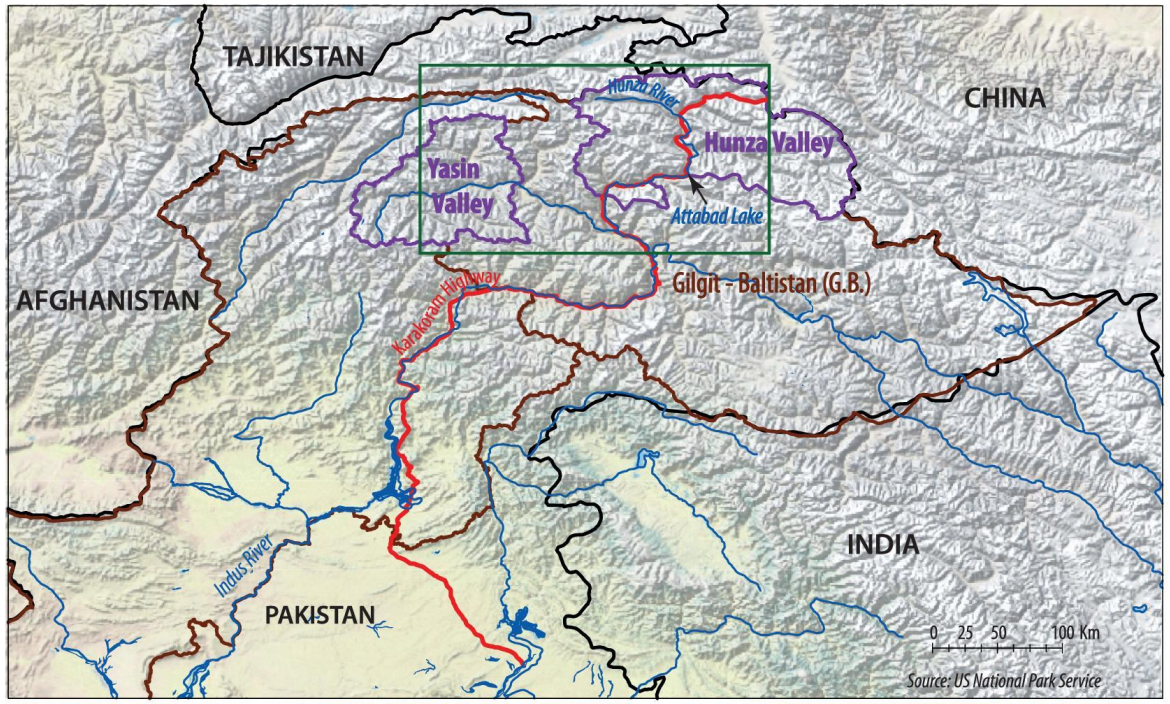
Tab.1 Predicted natural hazards due to climate change and their potential impact levels on tourism

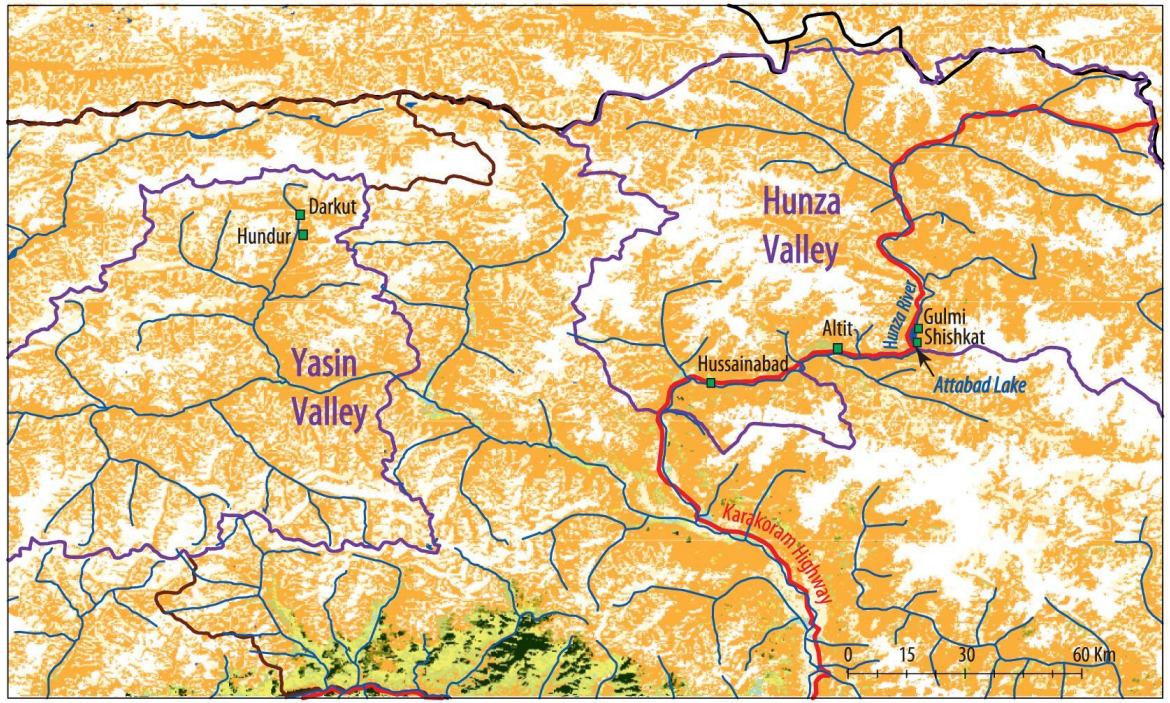
Pede et al. (2022)

Migration and climate risks

Do people move away because of (climate) risks in mountains?

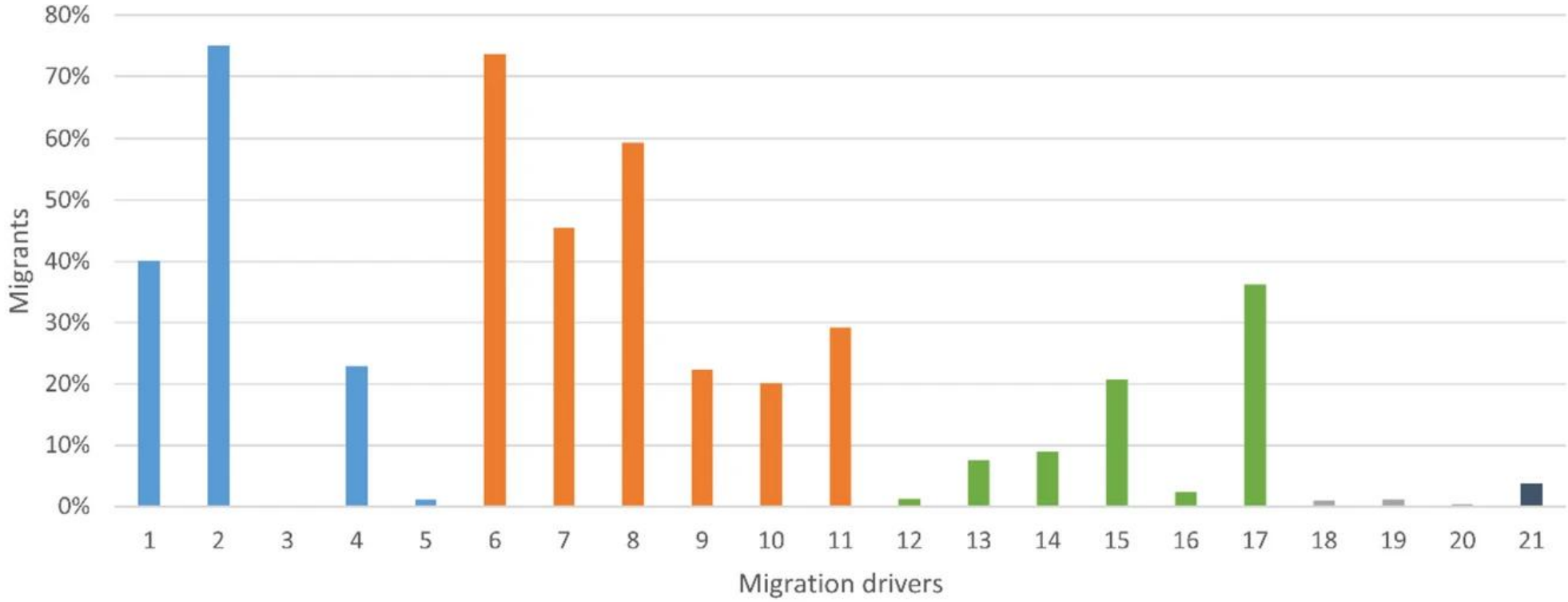
Why do people move? Why do people stay?





Gioli et al., 2014

Reasons to migrate



Social
1. better education at destination
2. better healthcare at destination
3. marriage
4. family connections at destination
5. discrimination at the place of origin

Economic
6. employment opportunity
7. unemployment at place of origin
8. higher income at destination
9. lack of agricultural land
10. crop failure
11. not enough income

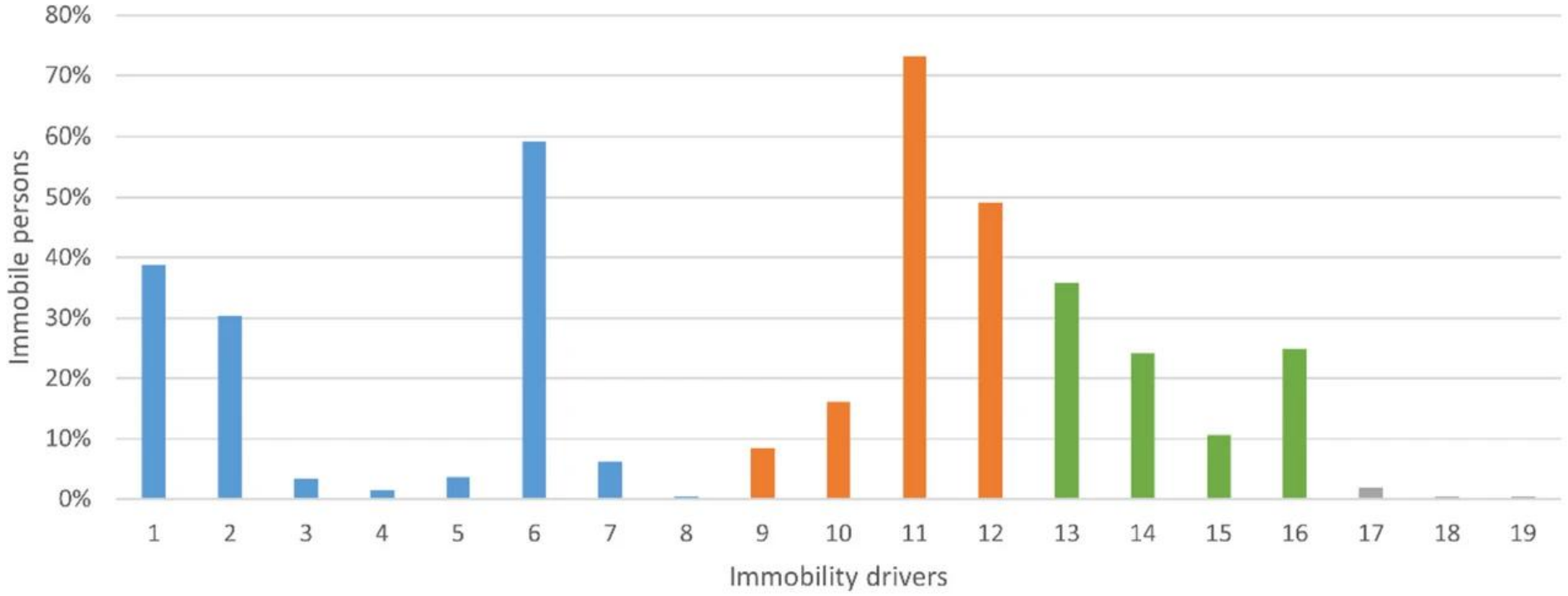
Environmental
12. water shortage or drought like conditions
13. land degradation
14. floods
15. landslides/ avalanche
16. cyclone/ storm
17. heavy snowfall/ rainfall

Political
18. conflict with family/ community
19. lack of political freedom
20. government incentives

other reasons

Khan et al., 2024

Reasons to (involuntarily) stay



Social
1. lack of education
2. health reasons
3. disability
4. gender
5. Age
6. household care
7. lack of family connections
8. discrimination at destination

Economic
9. lack of transportation costs
10. no employment opportunity at destination
11. no financial resources
12. resource for basic needs

Environmental
13. impact of extreme events on income
14. loss of economic assets
15. collapse of transportation network
16. labor for land rehabilitation

Political
17. legal & administrative barrier
18. conflict at destination
19. lack of political freedom

Khan et al., 2024

Ex ante adaptation strategy
(insurance against shocks)
39%

Time of first migration:
before 2010

Average income:
USD 1930

Remittances:
Invested in better education for both genders, health, and increased food security

Ability to absorb shocks:
Fair

Ex post adaptation strategy
(coping mechanism after a shock)
25%

Time of first migration:
After 2010

Average income:
USD 710

Remittances:
Spent on basic survival needs (food and firewood); little if any left over for investment

Ability to absorb shocks:
Low

Not feasible (due to lack of financial, human, and social assets)
36%

Average income:
USD 560

Remittances:
None

Ability to absorb shocks:
Extremely low; vulnerable households at risk of becoming trapped population

Resilience **Vulnerability**

Gioli et al., 2014

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In class discussion

We want to cover:

- Communicating mountain risks
- How do existing initiatives address mountain risks?

In class discussion

Watch any of the clips provided (max. 10 minutes) and then discuss in the breakout rooms:

a) Is the **language** used in these clips **comprehensible** to you? Would it be to anyone, for example people living in mountain environments?

b) To what degree do these initiatives or assessments address **experiences from people** living with risks in mountains and to what degree do they simply feel academic? Is there a reasonable balance?

c) Who among the people speaking addresses the issues of **long term risks**, i.e. not just the immediate impacts of a single event? Do you feel like 'risk' in its entirety is captured or are there crucial aspects missing?

d) What is **crucially missing** in the clips for you to be convinced of what they are trying to get across?

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Disaster Risk Reduction

Disaster risk reduction (DRR) is defined by United Nations Office for Disaster Risk Reduction (UNDRR) as those actions which aim to "prevent new and reducing existing disaster risk and managing residual risk, all of which contribute to strengthening resilience and therefore to the achievement of sustainable development".

UNDRR
UN Office for Disaster Risk Reduction

GFDRR
Global Facility for Disaster Reduction and Recovery

Disaster Risk Reduction – Sendai Framework

Chart of the Sendai Framework for Disaster Risk Reduction

2015-2030

Scope and purpose

The present framework will apply to the risk of small-scale and large-scale, frequent and infrequent, sudden and slow-onset disasters, caused by natural or manmade hazards as well as related environmental, technological and biological hazards and risks.
It aims to guide the multi-hazard management of disaster risk in development at all levels as well as within and across all sectors

Expected outcome

The substantial reduction of disaster risk and losses in lives, livelihoods and health and in the economic, physical, social, cultural and environmental assets of persons, businesses, communities and countries

Goal

Prevent new and reduce existing disaster risk through the implementation of integrated and inclusive economic, structural, legal, social, health, cultural, educational, environmental, technological, political and institutional measures that prevent and reduce hazard exposure and vulnerability to disaster, increase preparedness for response and recovery, and thus strengthen resilience

Targets

Substantially reduce global disaster mortality by 2030, aiming to lower average per 100,000 global mortality between 2020-2030 compared to 2005-2015	Substantially reduce the number of affected people globally by 2030, aiming to lower the average global figure per 100,000 between 2020-2030 compared to 2005-2015	Reduce direct disaster economic loss in relation to global gross domestic product (GDP) by 2030	Substantially reduce disaster damage to critical infrastructure and disruption of basic services, among them health and educational facilities, including through developing their resilience by 2030	Substantially increase the number of countries with national and local disaster risk reduction strategies by 2020	Substantially enhance international cooperation to developing countries through adequate and sustainable support to complement their national actions for implementation of this framework by 2030	Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to people by 2030

Priorities for Action

There is a need for focused action within and across sectors by States at local, national, regional and global levels in the following four priority areas.

Priority 1
Understanding disaster risk

Disaster risk management needs to be based on an understanding of disaster risk in all its dimensions of vulnerability, capacity, exposure of persons and assets, hazard characteristics and the environment

Priority 2
Strengthening disaster risk governance to manage disaster risk

Disaster risk governance at the national, regional and global levels is vital to the management of disaster risk reduction in all sectors and ensuring the coherence of national and local frameworks of laws, regulations and public policies that, by defining roles and responsibilities, guide, encourage and incentivize the public and private sectors to take action and address disaster risk

Priority 3
Investing in disaster risk reduction for resilience

Public and private investment in disaster risk prevention and reduction through structural and non-structural measures are essential to enhance the economic, social, health and cultural resilience of persons, communities, countries and their assets, as well as the environment. These can be drivers of innovation, growth and job creation. Such measures are cost-effective and instrumental to save lives, prevent and reduce losses and ensure effective recovery and rehabilitation

Priority 4
Enhancing disaster preparedness for effective response, and to «Build Back Better» in recovery, rehabilitation and reconstruction

Experience indicates that disaster preparedness needs to be strengthened for more effective response and ensure capacities are in place for effective recovery. Disasters have also demonstrated that the recovery, rehabilitation and reconstruction phase, which needs to be prepared ahead of the disaster, is an opportunity to «Build Back Better» through integrating disaster risk reduction measures. Women and persons with disabilities should publicly lead and promote gender-equitable and universally accessible approaches during the response and reconstruction phases

Guiding Principles

- Primary responsibility of States to prevent and reduce disaster risk, including through cooperation
- Shared responsibility between central Government and national authorities, sectors and stakeholders as appropriate to national circumstances
- Protection of persons and their assets while promoting and protecting all human rights including the right to development
- Engagement from all of society
- Full engagement of all State institutions of an executive and legislative nature at national and local levels
- Empowerment of local authorities and communities through resources, incentives and decision-making responsibilities as appropriate
- Decision-making to be inclusive and risk-informed while using a multi-hazard approach

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- Coherence of disaster risk reduction and sustainable development policies, plans, practices and mechanisms, across different sectors
- Accounting of local and specific characteristics of disaster risks when determining measures to reduce risk
- Addressing underlying risk factors cost-effectively through investment versus relying primarly on post-disaster response and recovery
- «Build Back Better» for preventing the creation of, and reducing existing, disaster risk
- The quality of global partnership and international cooperation to be effective, meaningful and strong
- Support from developed countries and partners to developing countries to be tailored according to needs and priorities as identified by them

Target A: Disaster-related deaths and missing persons (excluding COVID-19)

- Average disaster-related mortality per 100,000 population declined from 1.60 in 2005-2015 to 1.15 in 2013-2022 (a decline of 28%).
- During 2015-2022, the average annual disaster-related mortality stood at 41,789 persons in absolute terms.

Target B: Disaster-affected population

- Number of disaster-affected people Increased from 1,092 per 100,000 population during 2005-2015 to 2,034 during 2013-2022 (an increase of 86%).
- On an average, more than 130 million people were affected by disasters each year globally during 2015-2022.

Target C: Disaster-related economic losses

- Direct economic loss averages 0.37% of global GDP of reporting countries during 2015-2022.
- Provisional data shows that in 2022, direct economic loss accounts for 0.83% of global GDP.

Target D: Disaster damage to critical infrastructure and disruption of basic services

- From 2015 to 2022, over 800,000 critical infrastructure units and facilities, including schools and hospitals, were damaged or destroyed due to disasters.
- In the same period, more than 12 million educational, health, and other basic services were disrupted by disasters.

Target E: National and local disaster risk reduction (DRR) strategies

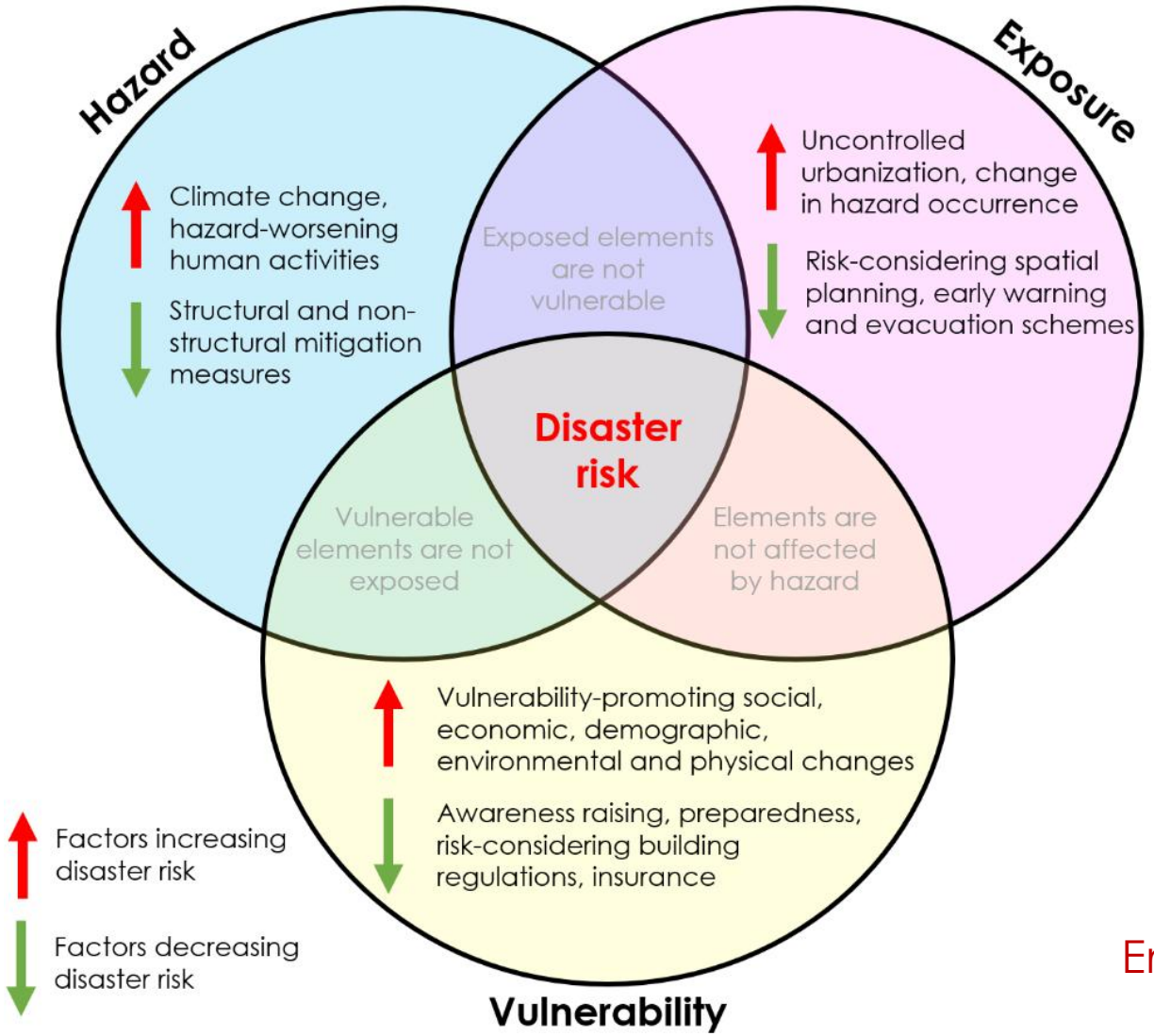
- 126 countries have reported having national disaster risk reduction strategies, compared to 55 in 2015 (two-third of countries in the world and an increase of 130% since the adoption of Sendai Framework).
- 102 countries have reported having local disaster risk reduction strategies. Average proportion of local governments in such countries is 72%.

Target F: International cooperation for disaster risk reduction

- International cooperation for developing countries remains limited, as developing countries reported only US$ 304 million of official development assistance (ODA) plus other official flows for supporting disaster risk reduction actions in 2021.
- From 2005 to 2022, 1,742 programmes and initiatives on science, technology and innovation transfer in disaster risk reduction were reported by 39 countries.
- In the same period, 33,441 instances of capacity development on disaster risk reduction were reported by 42 countries.

Target G: Multi-hazard early warning systems (MHEWS)

- 101 countries reported having MHEWS, compared to 47countries in 2015 (an increase of 114%).
- Two third (66%) of the countries reported having MHEWS have self-assessed their MHEWS with “limited to moderate achievement.”



Emmer, Steiner and Kalvoda (2024)

Risk management – disaster risk reduction

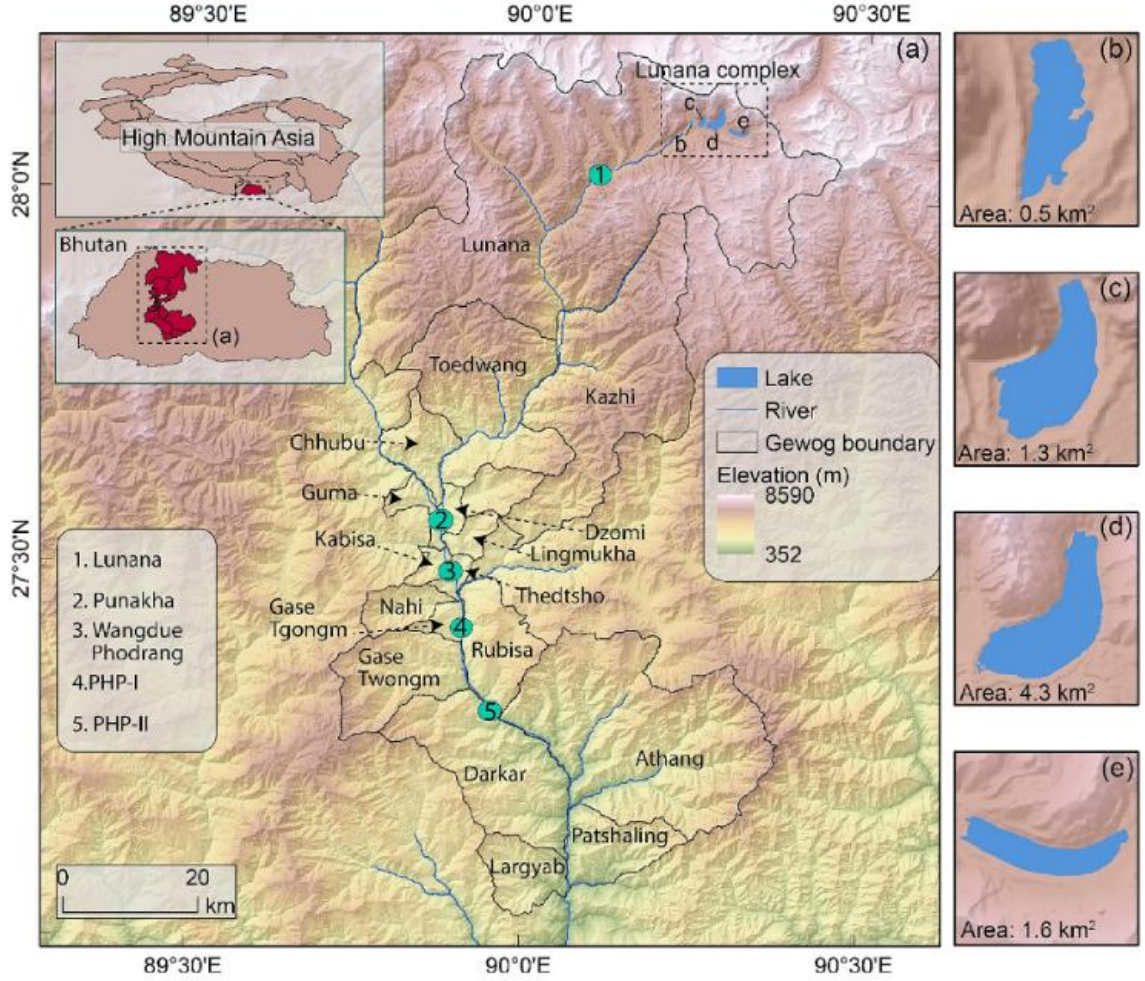


Fig. 1. Lunana glacier complex and downstream settlements. Map (a) shows settlements downstream of the Lunana glacier complex along the Phochu and Punatsangchu basins. Downstream settlements are presented following the subdistrict block administrative boundary. Panels (b–e) show four glacial lakes, which we studied here: Bechung Tsho, Raphstreng Tsho, Lugge Tsho, and Thorthormi Tsho, respectively. Background image: ALOS-PALSAR DEM.

Rinzin et al., 2023

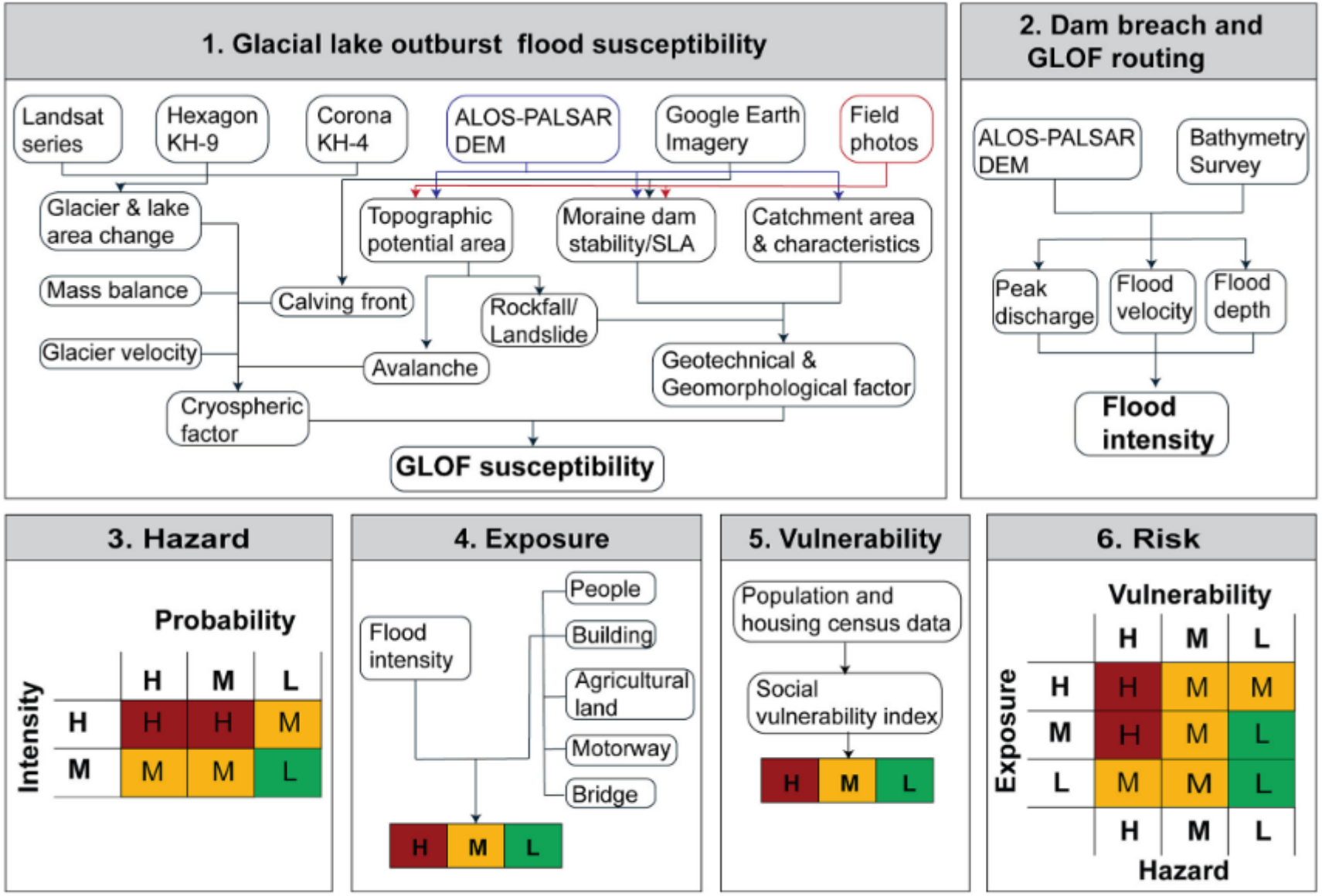


Fig. 2. Methodological flow chart adopted for GLOF susceptibility, downstream hazard, exposure, vulnerability, and risk assessments for the potentially dangerous glacial lakes in the Lunana glacier complex of the Bhutan Himalaya: downstream GLOF hazard, vulnerability, exposure, and risk are categorized into three classes; high (H), medium (M) and low (L). For the hazard computation, the GLOF intensity used here is translated from the modelled flood depth: high (more than 1 m) and medium (less than 1 m) (after GAPHAZ, 2017).

Rinzin et al., 2023

Hazards

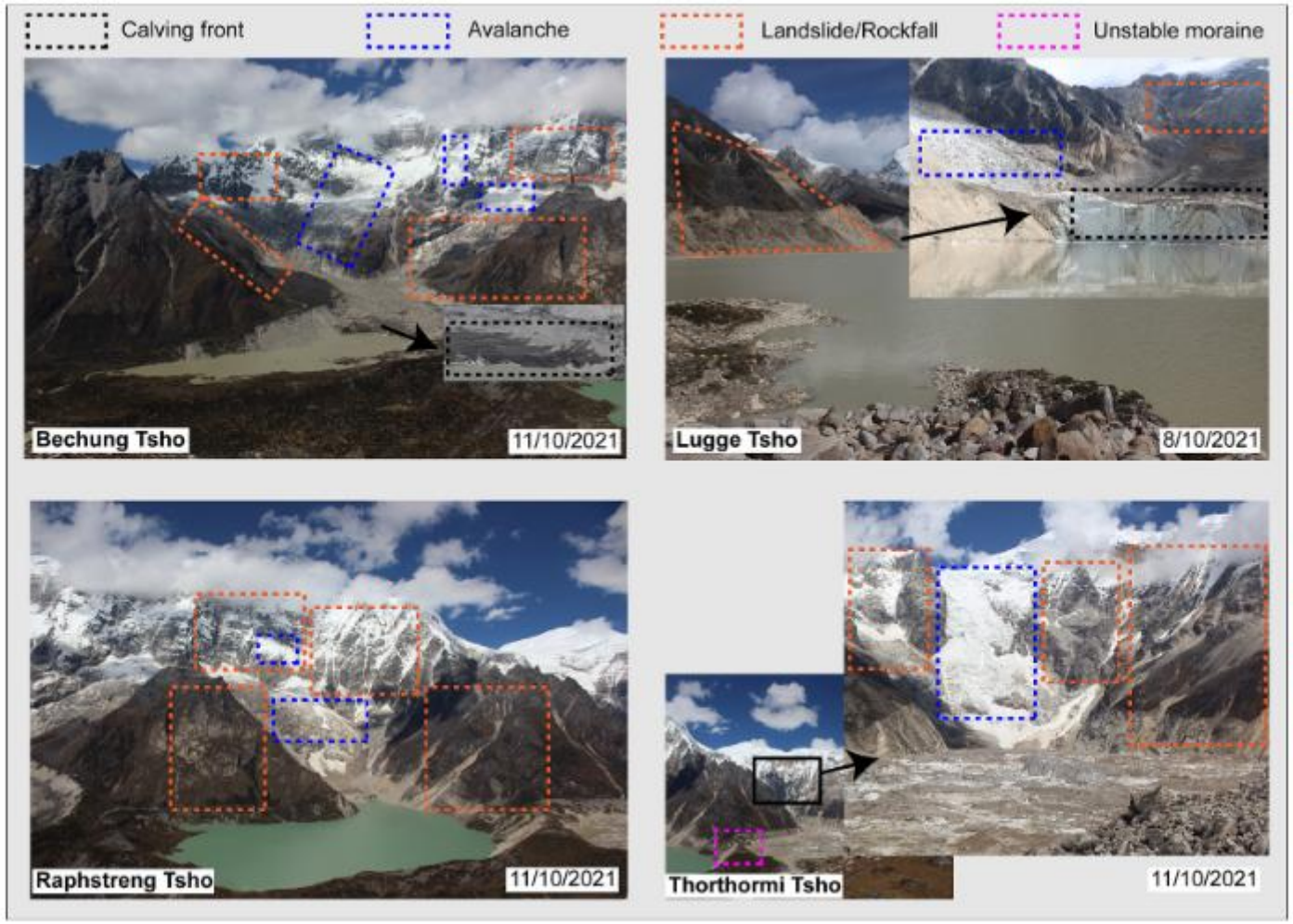
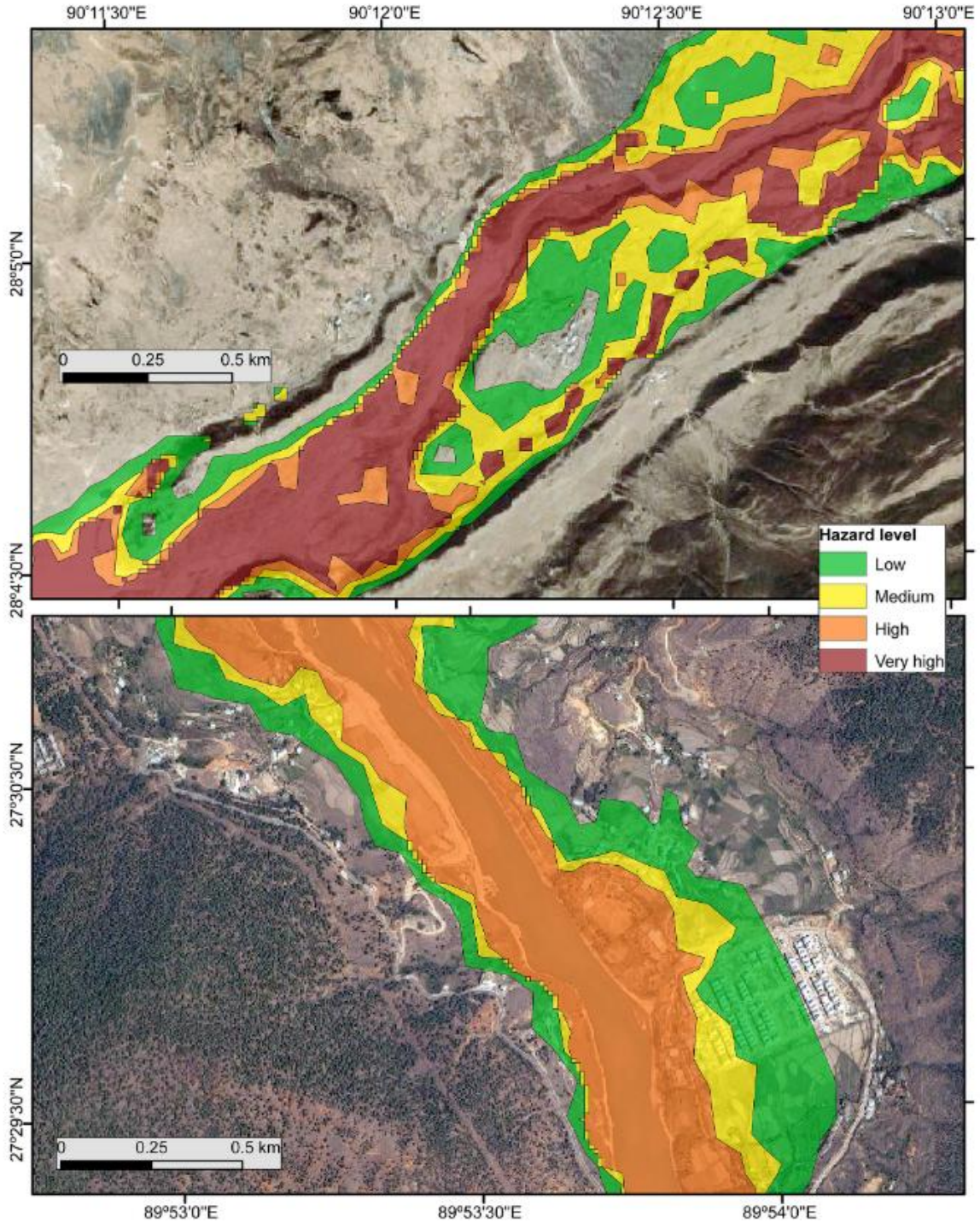


Fig. 3. Field photographs show the surrounding conditions that make lakes susceptible to GLOF, including calving front, avalanche, landslide/rockfall, and unstable moraine dam. Photo courtesy: NCHM, Bhutan.

Exposure


90°11'30"E
90°12'0"E
90°12'30"E
90°13'0"E
28°5'0"N
28°4'30"N
27°30'30"N
27°29'30"N
89°53'0"E
89°53'30"E
89°54'0"E
0 0.25 0.5 km
0 0.25 0.5 km
Hazard level
Low
Medium
High
Very high

Vulnerability

Table 2
List of indicators and corresponding represented components considered for calculating the social vulnerability index for the downstream settlements. The indicators are extracted from the Bhutan Population and Housing Census 2017 data (NSB, 2018), following the convention used by Allen et al. (2016) and Cutter et al. (2003). The absolute and normalized values for each indicator are also provided in supplementary Table S2.

No.	Social vulnerability indicator	Component represented
1	Old age dependence ratio	Sensitivity, capacity to prepare, respond and recover
2	Child dependence ratio	Sensitivity, capacity to prepare, respond and recover
3	Disability prevalence rate	Sensitivity, capacity to prepare, respond and recover
4	Female population	Sensitivity, capacity to prepare, respond and recover
5	Household without high school education attainment (adequate literacy)	Capacity to prepare, respond and recover
6	Household with insufficiency in food	Sensitivity, capacity to prepare, respond and recover
7	Rented household	Capacity to recover
8	Household without a reliable source of water	Capacity to prepare, respond and recover
9	Household without an improved sanitation	Capacity to recover
10	Household with more than a 30-minute walk to the road point	Capacity to prepare, respond and recover
11	Household without a smartphone	Capacity to prepare and respond
12	Household living in the house without housing concrete wall	Capacity to recover
13	Household without a car	Capacity to prepare, respond and recover
14	Household without a TV	Capacity to prepare and respond
15	Household without electricity as a main source of energy	Capacity to prepare and respond

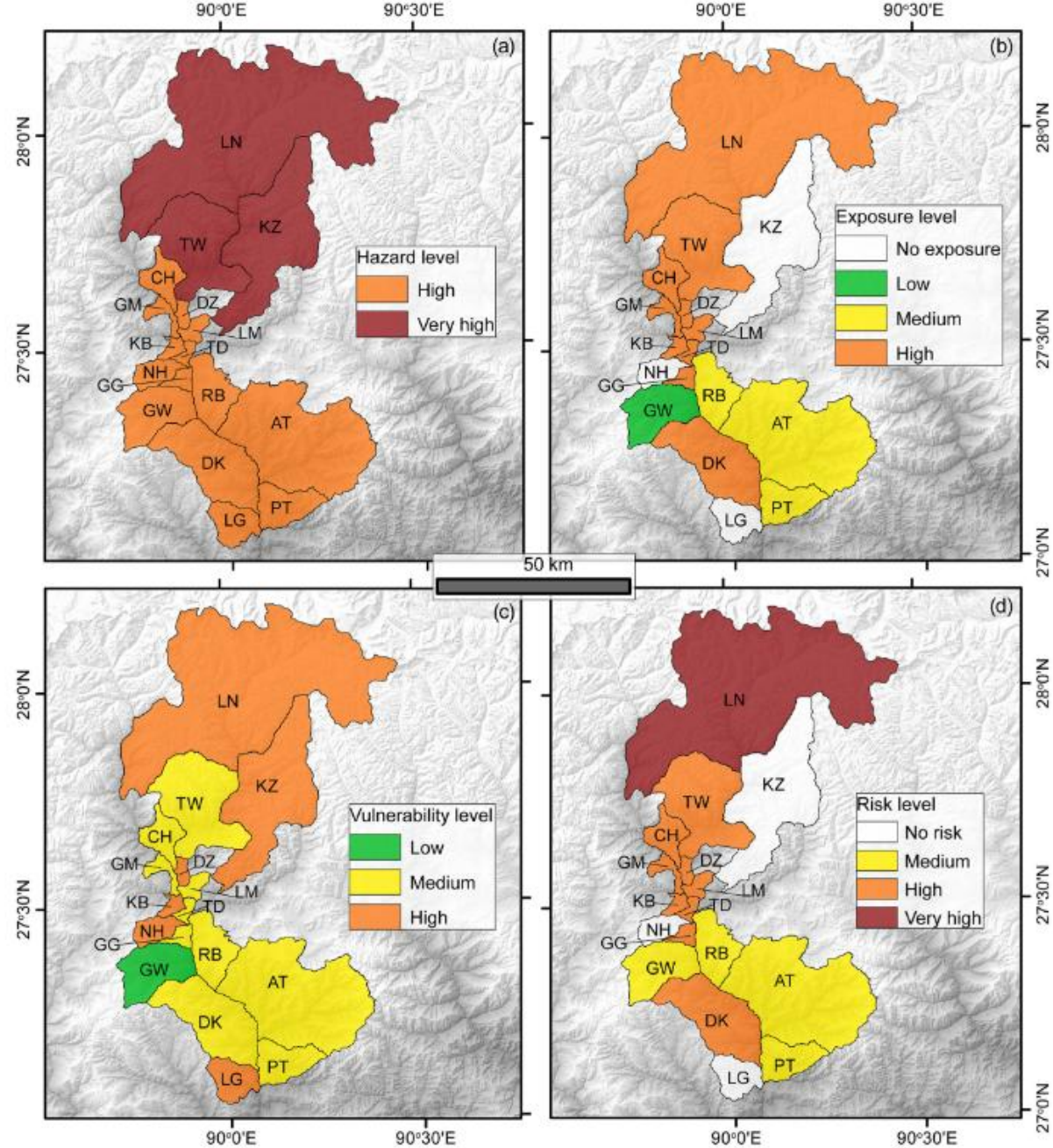


Fig. 9. The hazard (a), exposure (b), vulnerability (c), and risk (d) levels in 2020 of 17 subdistrict blocks downstream of the Lunana glacier complex under three different scenarios of GLOF magnitude (large, medium, and small) from each lake. The risk is calculated as a function of hazard, exposure, and vulnerability. The 17 subdistrict blocks are: Lunana (LN), Kazhi (KZ), Toedwang (TD), Chhubu (CH), Guma (GM), Dzomi (DZ), Lingmukha (LM), Kabisa (KB), Thedtsho (TD), Nahi (NH), Gase Tshogongm (GG), Gase Tshowongm (GW), Rubisa (RB), Darkar (DK), Athang (AT), Patshaling (PT), Largyab (LG). Background image: ALOS-PALSAR DEM.

RISK ASSESSMENT AND RISK MANAGEMENT PLAN

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
Construction and completion	Medium	The completion risk of any hydropower projects falls mainly on geological conditions and their relevant time and cost overruns. Many geological investigations and drilling studies undertaken for more than 3.5 years during project preparation found 80% of the rock tested to be in good order. The detailed project report prepared by a qualified engineering firm and the PPTA conducted due diligence studies including extensive field and laboratory investigations, such as hydrological, geological, geotechnical and geophysical, sedimentation, mechanical, and electrical requirements. Uncertainties of geological conditions remain as residual risks of construction delays and cost overruns, which could be affected by price increases in labor and materials. To address these risks, contingencies total $32 million or 22% of the base cost. Contractual mitigations are included in engineering, procurement, and construction contracts with liquidity damage clauses on a milestone basis. DGPC has adequate experience in procuring and managing contractors for construction of hydropower projects, demonstrated with Dagachhu hydropower plant implementation. Experienced staff of Dagachhu will support project construction. Project site accessibility is very good, i.e., along the main east–west national highway close to access roads and transmission connections.
Operations	Low	The plant's run-of-river hydropower operations are straightforward. DGPC has operated five run-of-river hydropower plants (HPPs) with capacity of 60 MW, 64 MW, 126 MW, 336 MW, and 1,020 MW. THyE will be staffed with experienced staff from DGPC and its subsidiaries—management and staff with adequate experience in operating HPPs. DGPC plans to select a reputed private joint venture partner with extensive experience in similar project development. The project's powerhouse will release water discharge to the Mangdechhu HPP reservoir (720 MW, run-of-river), expected to be available in 2017. The reservoir, under construction, is large enough to absorb the project's water release (confirmed by the Mangdechhu HPP's dam safety assessment under the PPTA due diligence studies). Project operation can begin regardless of completion of Mangdechhu HPP construction, as the existing Mangdechhu's diversion tunnel could easily absorb the project's water discharge.
Offtake	Low	A long-term 25-year PPA was agreed with PTC, the leading power trading company in India. DGPC has already exported almost 75% of its generation through other PPAs with PTC. The project's PPA is a take-or-pay contract in fixed volume terms for 80% of salable power; the volume will be less than 1% of the entire PTC trading power, which PTC can easily absorb. Given the prolonged power shortage in India, all of the project's power will be readily absorbed. If PTC fails to pay as per the PPA, the right for third-party sale will be triggered, but PTC's financial
Transmission	Low	The 132 kV transmission line works are contracted with BPC for project's full annual operations in 2020. The nearest substation u construction is on track to be available in 2017 as part of the ong Mangdechhu HPP. Once the project's transmission line (18.6 km connected to the substation, the project's power can alternatively exported through existing grid connections to India. The project's po will be mostly evacuated to India through the Mangdechhu–Gol Jigmeling–Alipurdwar transmission line corridor (400 kV) u construction for Mangdechhu operations scheduled in 2017.
Hydrological	Low	The hydrological risk primarily comprises drought risks, which significantly affect hydropower generation. The project's design energ based on 22-year hydrological data; the risk of any deviatio considered minimal. The financial analysis is based on the design en of 90% dependability (P-90 generation and plant availability of 9 which is quite conservative. This means that more water is likely t available for 90% of the operating time than designed on the hydrolo data. The assessment of potential future hydrological impacts, u various climate models found the project to be hydrologically soun projects that annual water flow in the Nikachhu watershed would incre by an average by 1%–3% for non-monsoon months and 5%–8% monsoon months to 2041–2060.
Other climate change (disaster)	Low	The climate change modeling analysis was extended to risks other potential hydrological issues, including risks of possible flo earthquakes, landslides, and glacial lake outbursts. The enginee design and environmental management plan are adaptive to extrem severe conditions created by such events. The analysis concludes overall climate change is unlikely to impose negative effects on project and its operation. Details of the assessment are in the clim change impact assessment. Effective adaptation measures will adopted for the project's structures, facilities, and adjoining area address such potential disasters as floods, landslides, and earthqua The dam safety design was reviewed and continues to be monitore DGPC's dam safety committee and a panel of experts. The project include an early warning system and awareness programs for pu safety in spite of no human settlement along the river downstream to confluence with the main river of the Mangdechhu.
Funding and debt	Low	The cofinancing loan from Indian commercial banks has been appro in principal. The cofinancing framework was agreed and its signing

Currency	Medium	Since the project is earned in Indian rupees from power exports, foreign debt from the OCR loan will have a currency exchange rate fluctuation

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
		risk if the Indian rupee depreciates. The Indian commercial bank cofinancing in local terms will reduce the overall debt service risk. The PPA stipulates that tariff adjustments in dollar terms may be introduced sometime in the future if both parties agree. Given the currency risk assessment, THyE will fully meet the debt service requirements even under an extreme scenario of continued annual 7% depreciation of Indian rupees over the debt repayment period. To mitigate the residual risks arising from the foreign debt services, THyE will adopt a foreign exchange fluctuation reserve fund. THyE and DGPC will also explore periodic currency hedging contracts once this becomes a feasible option in the financial market as covenanted.
Safeguards	Low	All safeguard documents are in accordance with ADB's Safeguard Policy Statement (2009). The compensation relating to resettlement will be fully paid in a satisfactory manner to the affected people. The current mitigation measures in the environmental management plan are appropriate and adequate to contain environmental impacts during project construction and operation, and environmental monitoring will be carried out during the project. To ensure minimum ecological flows, particularly in the lean seasons, the recommended level of 10% will continue to be monitored and finalized during implementation to ensure the habitat requirements combined with integrated water resources management, including relevant plant operations in the river basin. The ADB financing and loan agreements will ensure compliance with safeguard measures and implementation requirements, and an attached TA will support the safeguard activities. Extensive consultations and
		monitor and audit implementation of safe
Financial management	Low	DGPC has adopted comprehensive c covering accounting and financial procurement management, internal a safeguards, along with enterprise res external independent auditor reported t have adequate internal control system adequate competitive bidding, and s control to carry out their operations in a THyE will follow the same policies and pr
Procurement	Low	Increased transparency, efficiency, and ADB requirements for major constructior will conduct regular site inspections a timeliness of progress on contracts where
Corruption	Low	Increased transparency and accountab requirements and regular stakeholc agreements ensure that ADB's Antico measures are conformed with. ADB will related information on the website, and expenditure by ADB.
Overall	Low	

ADB = Asian Development Bank, BPC = Bhutan Power Corporation, DGPC = D kilovolt, MW = megawatt, OCR = ordinary capital resources, PPA = power pu preparatory technical assistance, PTC = Power Trading Corporation of India, Tangsibji Hydro Energy.
Source: Asian Development Bank.

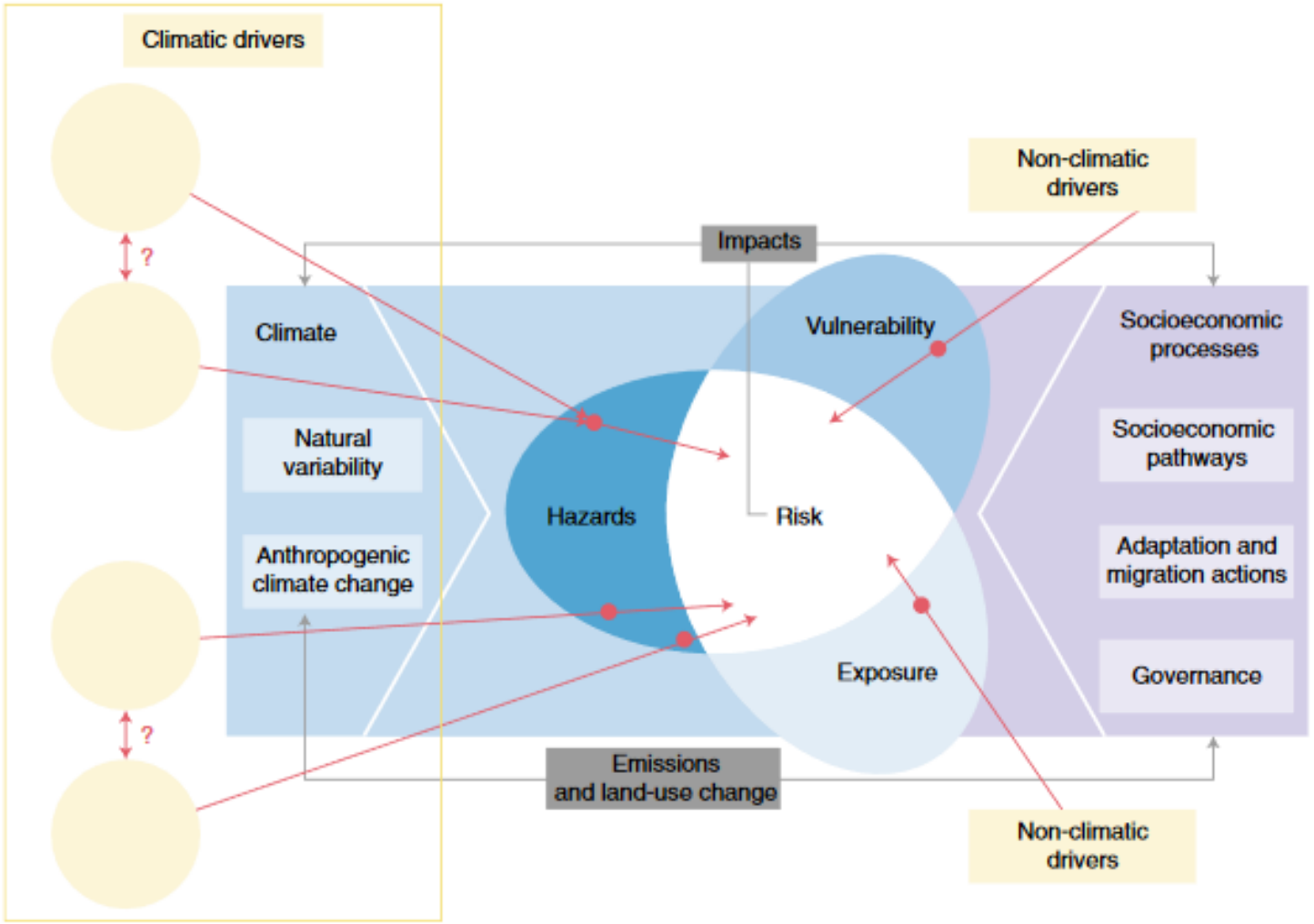
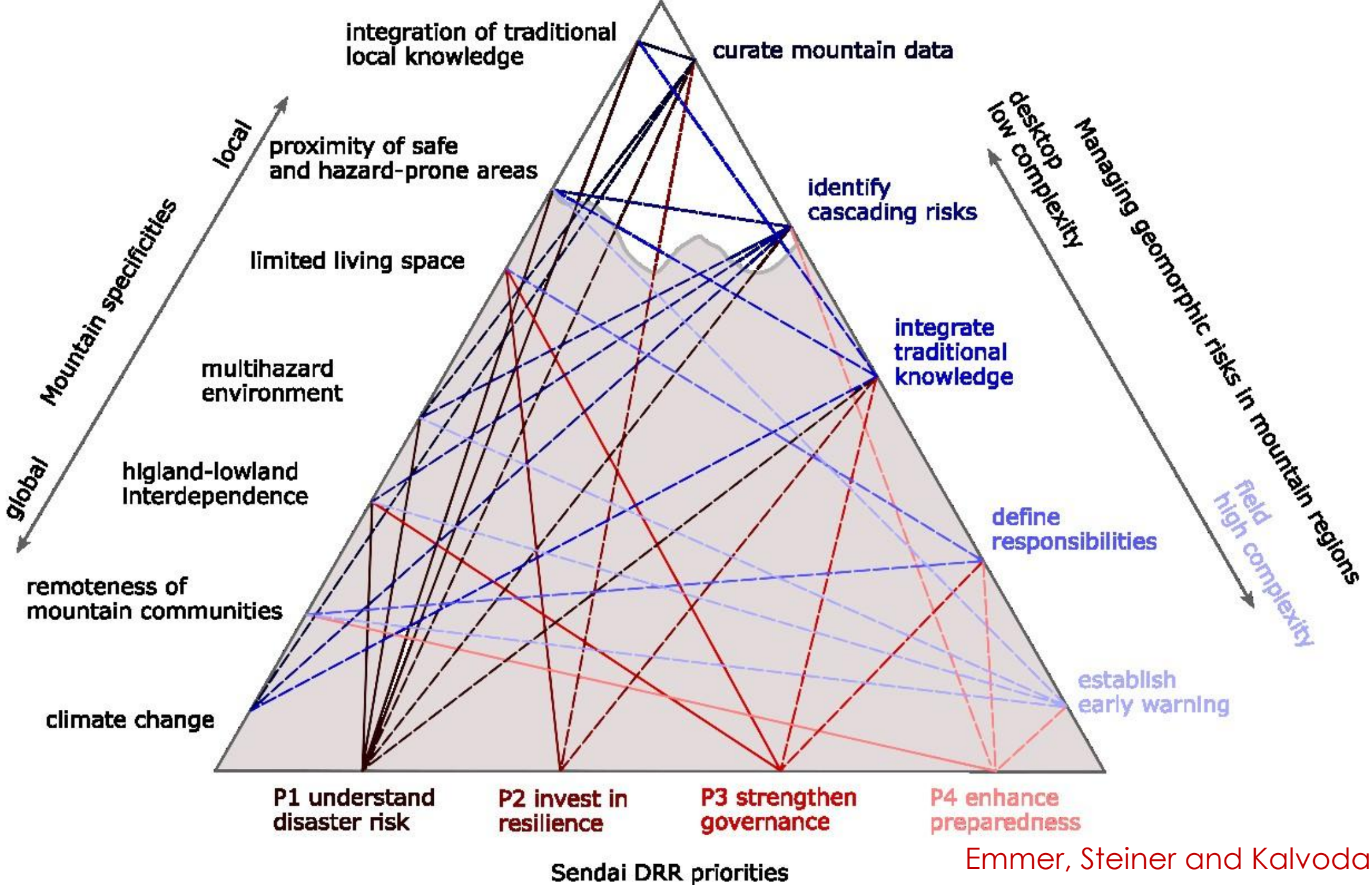


Fig. 1 | Extended risk framework. Multiple climatic drivers cause one or multiple hazards leading to societal and environmental risk. The climate drivers (which may vary from local-scale weather to large-scale climate modes, represented by yellow circles) and/or hazards may be mutually dependent. Non-climatic drivers related to vulnerability and exposure may also contribute to risk. Background risk figure adapted from ref. [96], IPCC.

Zscheischler et al. (2018)



Emmer, Steiner and Kalvoda (2024)

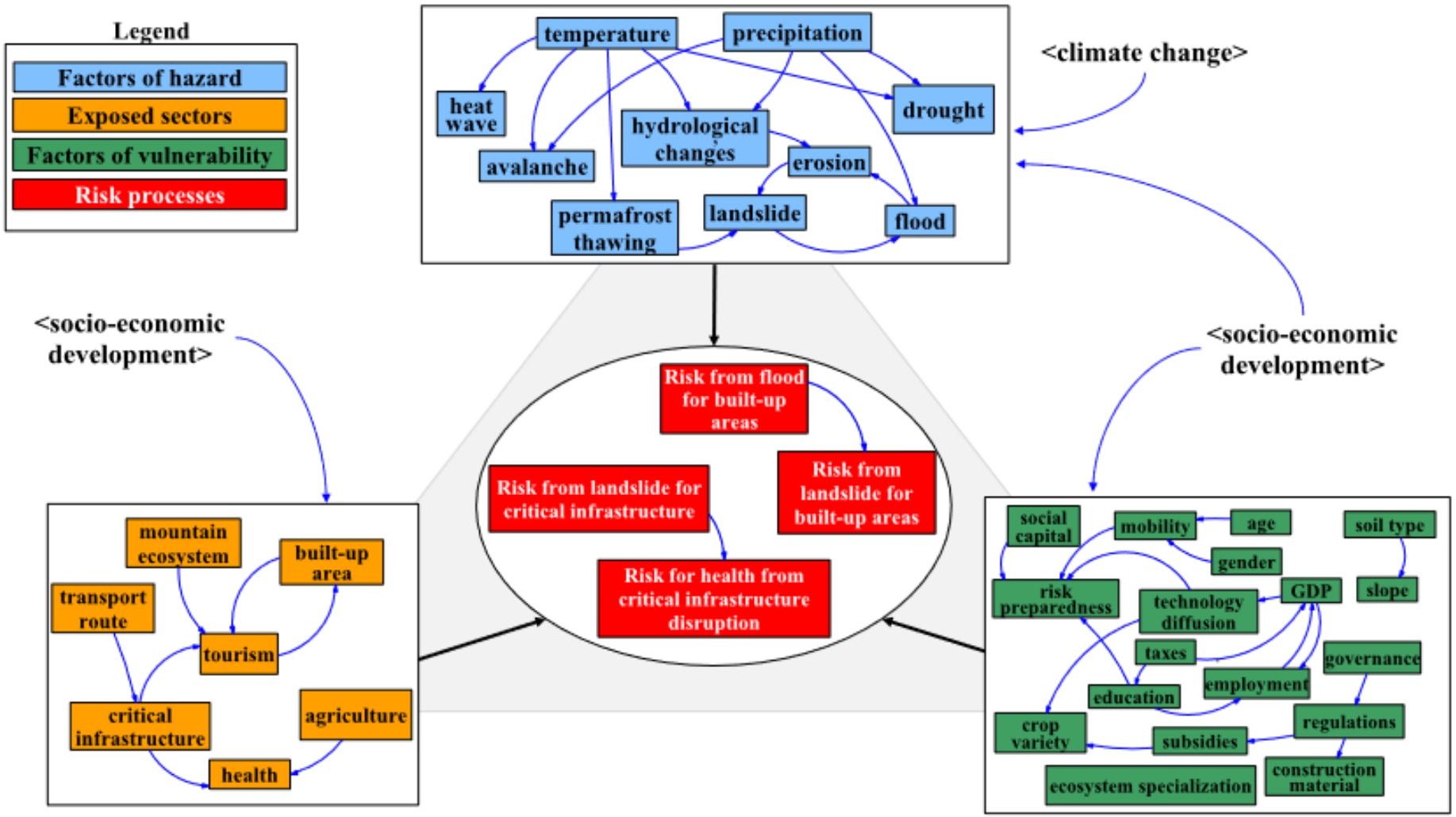


Fig. 2. Conceptual framework for mountain regions expanding the risk components into multi-hazard, exposure, multi-vulnerability and multi-risk.

Terzi et al. (2019)

Soneri Bank
بیرون ملک سے بھیجی جانے والی تمام رقوم سونیری بینک
کی تمام برانچز سے وصول کی جا سکتی ہیں
IME
International Money Express
Fast, reliable and secure remittance
MoneyGram.
bringing you closer
TRANSFAST
Worldwide Money Transfer
ALPINE EXCHANGE INC
sw small world
Money Transfer
GOLDEN
MONEY TRANSFER
WORLDWIDE
CASH EXPRESS
MONEY TRANSFER
الأنصاري للصرافة
AL ANSARI EXCHANGE
WALL STREET FINANCE WSF
ria Money Transfer

PIZZA PAMIR
TASTE OF MOUNTAIN HERBS
DINE IN / TAKEAWAY / DELIVERY
Pizza in the Mountains

LONG LIVE
PAK-CHINA FRIENDSHIP
中巴友谊万岁
PAK FC P52 OKZ
Pak-FC

Course content

28 June (1000 – 1800)

10:00 - 11:00	Fast onset hazards
11:00 - 11:15	Break
11:15 - 12:00	Slow onset hazards
12:00 - 13:00	Multihazards, compound events and cascading risks
13:00 - 14:00	Lunch Break
14:00 - 14:45	Urbanisation, Hydropower and Tourism
14:45 - 15:00	Breakout Groups - Discuss
15:00 - 15:30	In class discussion (see Moodle)
15:30 - 15:45	Recap on Group Discussion – dimensions of risks and local knowledge
15:45 – 16:30	Disaster Risk Reduction
16:30 – 17:00	Break
17:00 – 18:00	International Initiatives on Mountain Risks

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International initiatives addressing mountain risks

Loss and Damage (https://www.lossanddamagecollaboration.org)

Mountain Research Initiative (MRI) (https://mountainresearchinitiative.org/)

Adaptation at Altitude (https://adaptationataltitude.org/)

Mountain Partnership (https://www.fao.org/mountain-partnership/home/en/)

UNFCCC and Loss and Damage

United Nations Framework Convention on Climate Change (UNFCCC)

- An international treaty to limit dangerous climate change
- Signed by 154 states in 192 (Rio de Janeiro)
- Secretariat in Bonn, Germany
- Hosts the Green Climate Fund (GCF) – goal of $100 billion annually by 2020 – the Adaptation Fund (AF), as well as smaller LDCF, SCCF and GEF
- Invites the IPCC (Intergovernmental Panel on Climate Change) to provide assessments
- Conference of Parties (COPs) are arranged under the framework (many of them!)

Home

UNCCD 16th session of the Conference of the Parties

RIYADH, SAUDI ARABIA | 2-13 DECEMBER 2024

COP 16 Overview | Registration | High-level segment | Official documents | Daily journal | Special and side events | Media

Loss and Damage Fund

[…] Parties recognised by the UNFCCC as the most responsible for the climate crisis would provide predictable, and new and additional finance to address loss and damage.

[…] Loss and Damage Fund will tackle the gaps that current climate finance institutions such as the Green Climate Fund (GCF) and humanitarian assistance […] do not fill on Loss and Damage. The current international climate finance system is already stretched to breaking point due to insufficient funds, lack of political will and poor quality climate finance that indebts countries.

Loss and Damage Fund

[…] Parties recognised by
provide predictable, and

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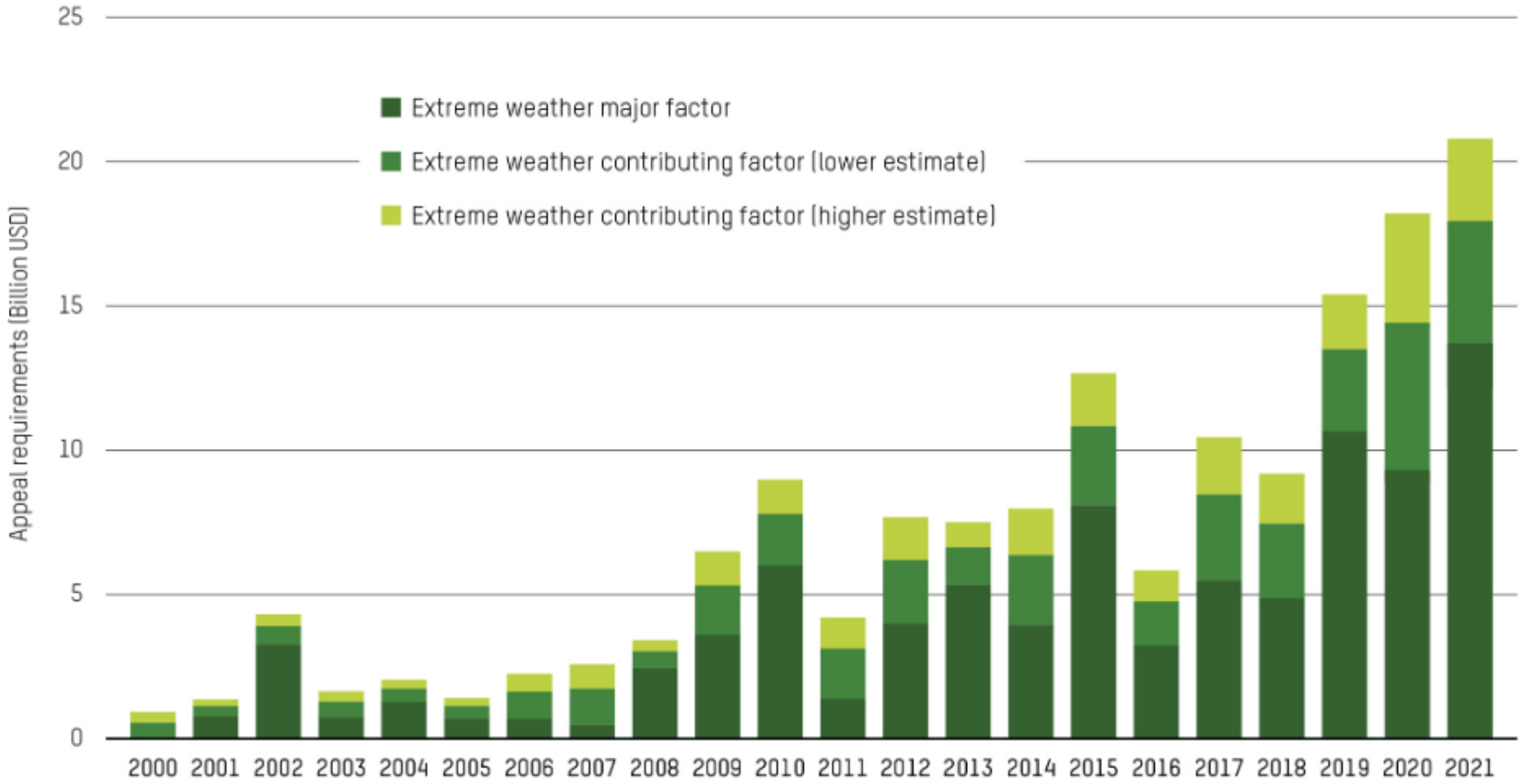
indebts countries.

Figure 3: Funding requirements for UN humanitarian appeals linked to extreme weather, 2000–2021



USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal requirement is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of these appeals.

https://oxfamilibrary.openrepository.com/bitstream/handle/10546/621382/bp-fair-finance-loss-and-damage-070622-en.pdf

Loss and Damage Fund

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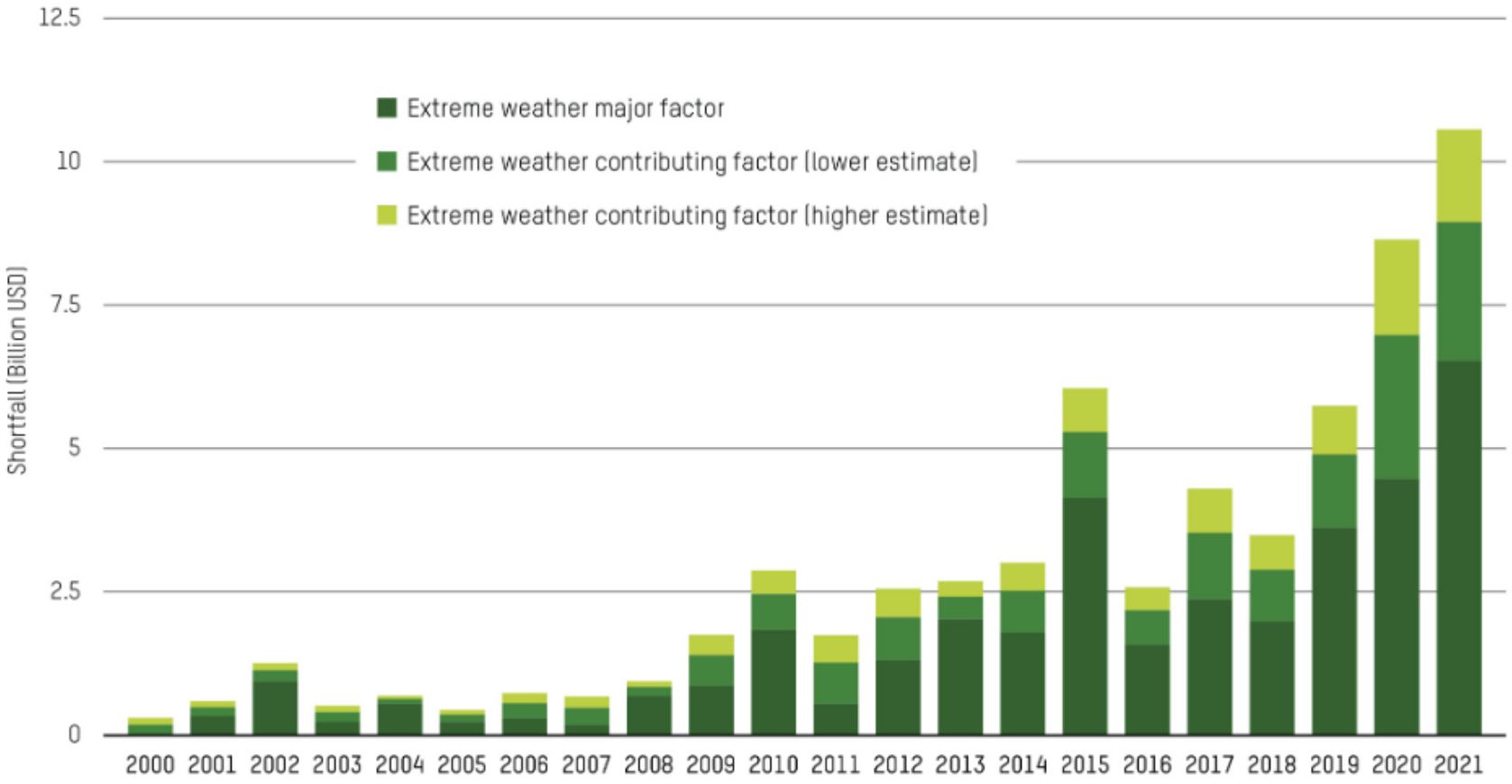
Figure 4: Funding shortfalls for extreme-weather-related UN appeals, 2000–2021



Notes: USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal shortfall is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of the appeal shortfall.

Loss and Damage Fund

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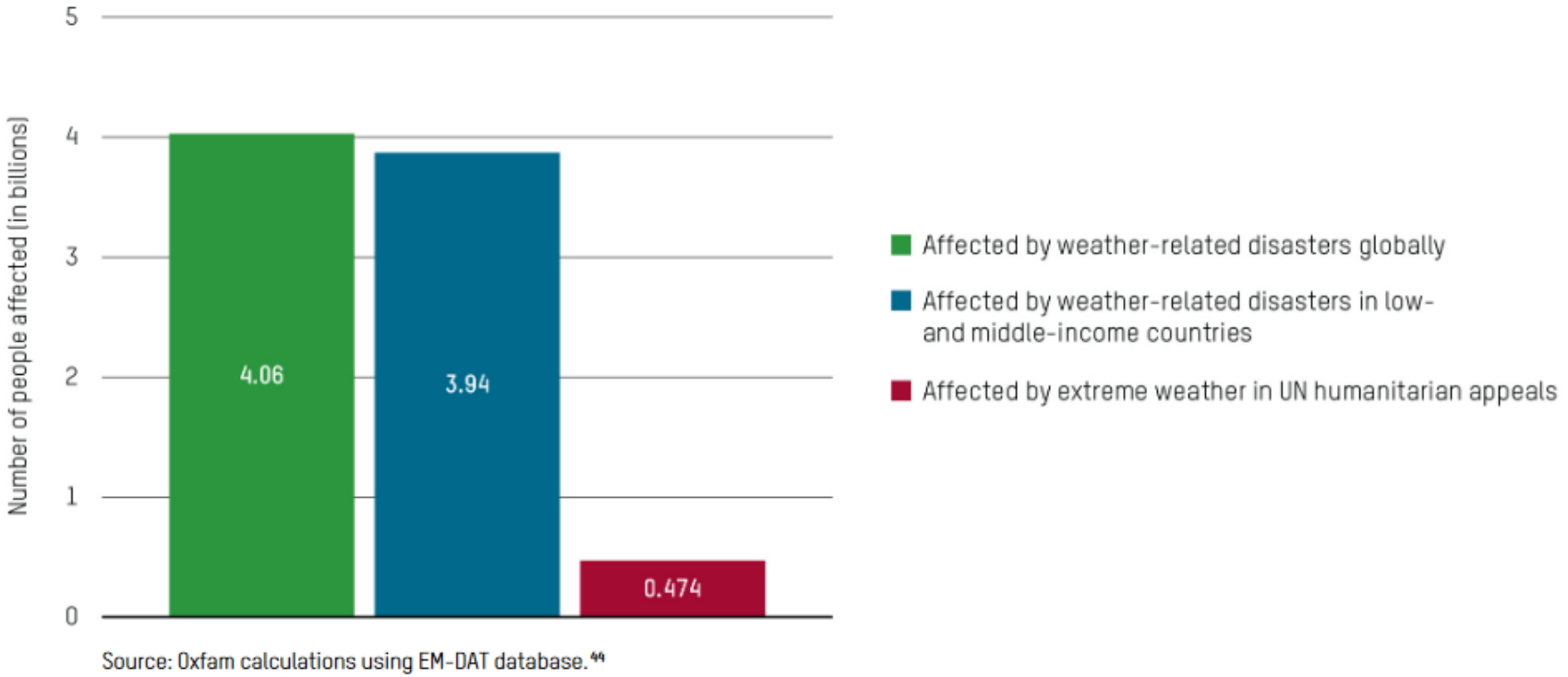
inadequacy of data
for mountain regions
hirisk.org

Figure 6: Number of people affected by weather-related disasters, 2000–2021



Source: Oxfam calculations using EM-DAT database.[44]

Loss and Da

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for mountain regions
hirisk.org

Table 3: Estimated future costs of loss and damage

Source, year	Estimated costs
IFRC 'The cost of doing nothing' (2019)[59]	2030: humanitarian funding requirements for climate-related disasters could grow to $20bn per year.
Economist Intelligence Unit (2019)[60]	Measuring each country's direct exposure to loss from climate change and extreme weather events found that by 2050, GDP at risk could be: Africa (most vulnerable) 4.7% Latin America 3.8% Middle East 3.7% Eastern Europe 3% Asia Pacific 2.6% Western Europe 1.7% North America 1.1%
Markandya and González-Eguino (2018)[61]	Total annual residual damages for developing countries: 2030: $290–$580bn 2040: $551bn–$1tn 2050: $1.1–$1.7tn
Climate Analytics (2015)[62]	Projected macro-economic damage of climate change for developing countries: 2030: $400–$431bn 2050: $1–$1.8tn
Deloitte (2022)[63]	3°C trajectory could cost the U.S. economy $14.5 trillion by 2070.

Where do mountains come in?

Continuous attempts to bring *mountains* on the agenda since 1992

Less congruence in approach than SIDS (Small Island Developing States)

In some regions (like High Mountain Asia) complicated by political conflict

Often central governments themselves not located in mountains (perceived *marginal issue*) – leading to an increased emphasis of '*upstream-downstream linkages*' and '*cascading risks*'

Mountain Partnership (FAO)

Formal and government led

MOUNTAIN PARTNERSHIP STEERING COMMITTEE

The Steering Committee of the Mountain Partnership provides programmatic orientation to the Mountain Partnership, oversees the preparation of a four-year strategy and a two-year workplan. The 18-member body monitors the work of the Mountain Partnership Secretariat, the implementation of the 'Mountain Agenda' and the achievements and impacts of their respective electoral groups.

Steering Committee members are elected to four-year terms by the various membership categories and regions. They meet annually and select a Chair and Vice Chairs.

The following governments and organizations are serving on the 2022-2025 Steering Committee:

Donor Organization to the MPS
Italy (Chair)

Governments
Asia and the Pacific: Nepal
Europe: Andorra, Romania (Alternate)
Central Asia: Kyrgyzstan
Near East and North Africa: Yemen
North and Central America and the Caribbean: **Dominican Republic (Vice Chair)**, Guatemala (Alternate)
South America: Peru, Argentina (Alternate)
Sub-Saharan Africa: Malawi, Uganda (Alternate)

Host Institution of the MPS
Food and Agriculture Organization of the United Nations

Intergovernmental Organization
United Nations Environment Programme, United Nations University (Alternate)

- Our Vision and Mission
- Governance and Strategy
- **Mountain Partnership Steering Committee**
- Mountain Partnership Secretariat
- Logos and PowerPoint presentations
- FAQ
- Definitions

2024
- MP Steering Committee meeting report (22 February 2024)

2023
- MP Steering Committee meeting report (26 October 2023)
- MP Steering Committee meeting report (20 April 2023)

2022
- MP Steering Committee meeting report (1 December 2022)
- MP Steering Committee meeting report (29 September 2022)
- MP Steering Committee meeting report (30 August 2022)
- MP Steering Committee meeting report (21 June 2022)
- MP Steering Committee meeting report (30 March 2022)
- MP Steering Committee Documents (13 January 2022)

2021
- MP Steering Committee

Mountain Range Organisations

Alps: Alpine Convention - https://www.alpconv.org/en/

- Convention that includes a lawyer and proposes legal texts that are not binding but are used by arbitration courts in case of inter-governmental as well as country-internal disagreements

Carpathians: Carpathian Convention http://www.carpathianconvention.org/

- Similar to above, less developed/younger

Andes: CONDESAN https://condesan.org/

- Network of organisations working in the Andes (research heavy)

Hindukush-Himalaya: ICIMOD https://www.icimod.org/

- Intergovernmental organization (8 countries) that supports governments with expertise and provides platform for exchange

Mountain Research Initiative

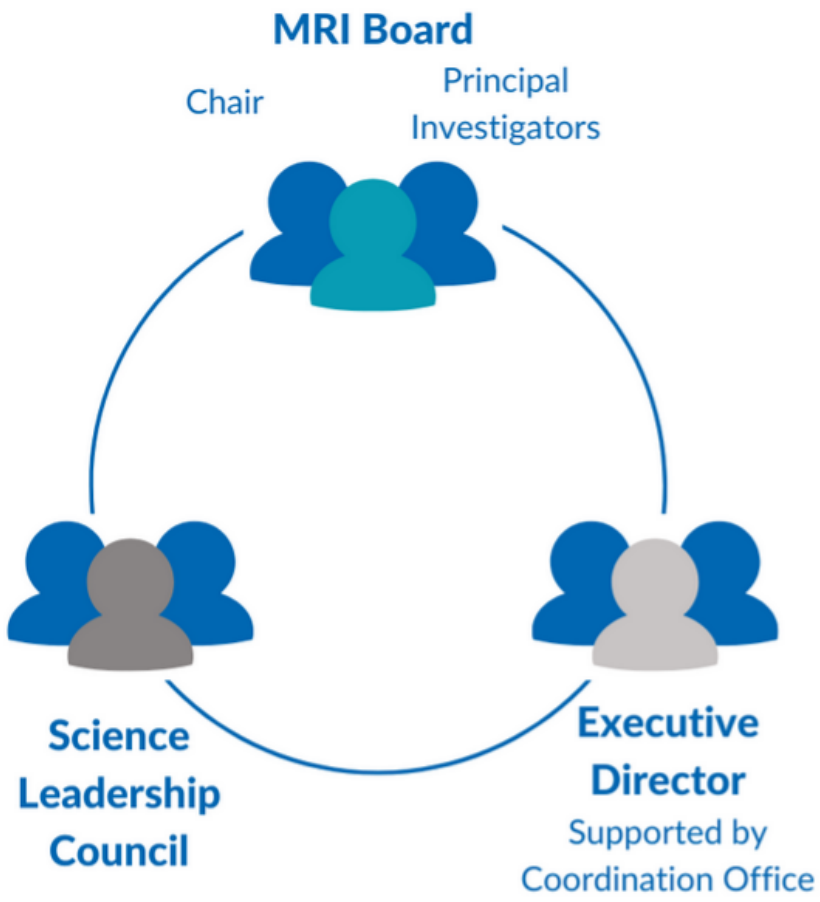
MRI Governance

The MRI Board (Chair and co-Principal Investigators), together with the MRI Science Leadership Council and the MRI Executive Director, constitute the MRI Governing Body. Together, they oversee and provide strategic direction and advice to support the implementation and development of both flagship and community-led activities, with the support of the Coordination Office staff.

As a team, we focus on supporting the mountain research community in making meaningful and fulfilling connections for our changing mountains and safeguarding them as social-ecological treasures for ecosystems and people worldwide.

Learn more about us:

- Principal Investigators
- Science Leadership Council
- Coordination Office



Adaptation at Altitude

Our Objective

The Adaptation at Altitude programme seeks to increase the resilience and adaptive capacity of mountain communities and ecosystems to climate change by:

- Improving the knowledge of appropriate climate change adaptation strategies in the mountains
- Transferring that knowledge through science–policy platforms to inform decision-making in national, regional and global policy processes

Adaptation at Altitude will foster exchange among the mountain areas of the world and seek short and long-term solutions to the problems arising from climate change.

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Who is missing?

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